

Professional Pilot Paper – Options module

Advanced Financial Management

Time allowed

Reading and planning:	15 minutes
Writing:	3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

ACCA

Section A: BOTH questions are compulsory and MUST be attempted

- 1 You are the chief financial officer of Fly4000 a large company in the airline and travel business whose principal market base is in Europe and the Middle East. Its principal hub is a major Northern European airport and Fly4000 has a small holiday business through its partnership with a number of independent tour operators. It has a good reputation as a business carrier within its European market, earned through very high standards of punctuality and service. Following the recent disinvestment of associated interests and a joint venture, it has cash reserves of \$860 million.

FliHi is a smaller airline which also has its centre of operations at the same airport as Fly4000. It has, since it was founded in 1988, developed a strong transatlantic business as well as a substantial position in the long and medium haul holiday market. In the year to 31 December 2005 its reported turnover was in \$1.7 billion and its profit after tax for the financial year was \$50 million. The company's net assets are \$120 million and it has \$150 million of long term loans on its balance sheet. It has recently expanded its fleet of wide bodied jets suitable for its expanding holiday business and has orders placed for the new Airbus 380 super-Jumbo to supplement its long haul fleet. FliHi has route licenses to New York and six other major US cities.

FliHi's cash flow statement for the current and preceding year is as follows:

FliHi Consolidated Cash Flow Statement (extract) For the year ended 31 December 2005

	31 December 2005		31 December 2004	
	\$m	\$m	\$m	\$m
Net cash inflow from operating activities		210.0		95.0
Return on investment and servicing of finance				
Interest received	12.0		6.0	
Interest paid	(4.0)		(3.0)	
Interest element on finance leases	(6.5)		(4.0)	
		1.5		(1.0)
Taxation		(4.1)		(0.2)
Capital Expenditure		(120.2)		(75.0)
Acquisitions and disposals				
Proceeds from the sale of interest in joint ventures		10.0		15.0
Cash inflow before management of Liquid resources and financing		97.2		33.8
Management of liquid resources				
Decrease/(increase) in short term deposits		35.5		(32.2)
Financing				
Repayment of secured loans		(31.0)		(25.0)
Increase/(decrease) in cash for the year		101.7		(23.4)

There is no other airline of comparable size and business mix to Fly4000 although analysts regard Rover Airways as a useful comparator. The statement below contains market data relating to Rover Airways:

Key Fundamentals

Forward P/E*	11.00	Dividend Yield	0.00
Price to Book value of equity	1.25	1Yr Total Return (%)**	25.07
Price To Cash Flow	3.00	Beta**	2.00
1Yr Sales Growth	-1.67	1Yr EPS Growth	80.50

** Equity Market Cap £3bn

You also note the following:

The current risk-free rate is 4.5 per cent and the equity risk premium is estimated at 3.5 percent. The prevailing share price for Rover Airways is 290¢ per share and its P/E ratio is 10. The corporation tax rate for both companies is 30 per cent.

The gearing ratio for Rover Airways, expressed as total debt to total capital (debt plus equity), is 60 per cent and as total debt to equity is 150 per cent.

You may assume that:

- 1 FliHi has undertaken a consistent programme of reinvestment
- 2 The debt in both companies is not expected to be sensitive to market risk.

There has been considerable consolidation in the airline industry and you are advising your board of directors of Fly4000 on the value of FliHi as a potential target for acquisition. It is anticipated that over the longer term the domestic airline industry will settle down to a rate of growth in line with GDP growth in the European economy which stands at 4 per cent per annum (nominal). However, the current rates of growth for this company are likely to be sustained for the next five years before reverting to the GDP growth rate from the sixth year forward.

Required:

- (a) **Estimate the current cost of equity capital for FliHi using the Capital Asset Pricing Model, making notes on any assumptions that you have made.** (9 marks)
- (b) **Estimate the expected growth rate of Flihi using the current rate of retention of free cash flow and your estimate of the required rate of return on equity for each of the next six years. Make notes on any assumptions you have made.** (6 marks)
- (c) **Estimate the value of Flihi on the basis of its expected free cash flow to equity, explaining the limitations of the methods you have used.** (7 marks)
- (d) **Write a brief report outlining the considerations your colleagues on the board of Fly4000 might bear in mind when contemplating this acquisition.** (8 marks)

(30 marks)

- 2 You are the finance director of Sydonics Engineering and expect that a bid to build a new plant in Southern France may be accepted in three months time. If the contract is accepted, an immediate capital spend of €150million will be required in three months and the company will receive a €75million grant from the European Development Fund in nine months time. The current Euro/sterling exchange rate is EUR 0.6900 to the pound.

Three month and nine month Euro LIBOR is 2.76563 per cent and 3.05194 per cent respectively. The three and nine month sterling LIBOR is 4.62313 per cent and 4.73031 per cent respectively. You have decided to hedge the exchange rate risk by the purchase of EUR/STERLING at-the-money options which have a contract size of 100,000 Euros. The monthly volatility of the Euro against sterling is 6.35 per cent. At the current exchange rate, the project has a net present value of £25 million at the company's cost of capital of 8.5 per cent.

The board of directors are concerned about the use of derivatives in managing the firm's treasury operations. They argue that the diversity of the firm's interests in Europe, the UK and the United States means that such hedging transactions are unnecessary.

Required:

- (a) **Prepare a memorandum, to be considered at the next board meeting, which summarises the arguments for and against foreign currency risk hedging and recommends a general policy concerning the hedging of foreign exchange risk.** (10 marks)
(Including 2 professional marks)
- (b) **Prepare a short report justifying your use of derivatives to minimise the firm's exposure to foreign exchange risk. Your report should contain:**
- (i) **The likely option price for an at-the-money option, stating the circumstances in which the option would be exercised. You should use the Grabbe variant of the Black-Scholes model for both transactions, adjusted on the basis that deposits generate a rate of return of LIBOR.** (10 marks)
 - (ii) **A calculation of the number of contracts that would be required to eliminate the exchange rate risk and the cost of establishing a hedge to cover the likely foreign currency exposure.** (4 marks)
 - (iii) **A summary of the issues the board should bear in mind when reviewing a hedging proposal such as this, taking into account the limitations of the modelling methods employed and the balance of risk to which the firm will still be exposed to when the position is hedged.** (6 marks)

(30 marks)

Section B: TWO questions ONLY to be attempted

- 3 The board of directors of Jonas Chemical Systems Limited has used payback for many years as an initial selection tool to identify projects for subsequent and more detailed analysis by its financial investment team. The firm's capital projects are characterised by relatively long investment periods and even longer recovery phases. Unfortunately, for a variety of reasons, the cash flows towards the end of each project tend to be very low or indeed sometimes negative. As the company's new chief financial officer (CFO), you are concerned about the use of payback in this context and would favour a more thorough pre-evaluation of each capital investment proposal before it is submitted for detailed planning and approval. You recognise that many board members like the provision of a payback figure as this, they argue, gives them a clear idea as to when the project can be expected to recover its initial capital investment.

All capital projects must be submitted to the board for initial approval before the financial investment team begins its detailed review. At the initial stage the board sees the project's summarised cash flows, a supporting business case and an assessment of the project payback and accounting rate of return.

A recent capital investment proposal, which has passed to the implementation stage after much discussion at board level, had summarised cash flows and other information as follows:

Distillation Plant at the Gulf Refining Centre

	Investment Phase		Recovery Phase	
	Cash flow (tax adjusted, nominal)	Cumulative Cash Flow	Cash flow (tax adjusted, nominal)	Cumulative Cash Flow
	\$m	\$m	\$m	\$m
01 January 2006	(9.50)	(9.50)		
31 December 2006	(5.75)	(15.25)		
31 December 2007	(3.00)	(18.25)		
31 December 2008			4.5	(13.75)
31 December 2009			6.40	(7.35)
31 December 2010			7.25	(0.10)
31 December 2011			6.50	6.40
31 December 2012			5.50	11.90
31 December 2013			4.00	15.90
31 December 2014			(2.00)	13.90
31 December 2015			(5.00)	8.90
Cost of Capital		8%		
Expected net present value (\$m)		1.964		
Net present value volatility (\$m) – annualised		1.02		
Internal rate of return		11.0%		
Payback (years)		5.015		

The normal financial rules are that a project should only be considered if it has a payback period of less than five years. In this case the project was passed to detail review by the financial investment team who, on your instruction, have undertaken a financial simulation of the project's net present value to generate the expected value and volatility as shown above. The board minute of the discussion relating to the project's preliminary approval was as follows:

31 May 2005 Agenda Item 6

New capital projects – preliminary approvals

Outline consideration was given to the construction of a new distillation facility at the Gulf Refining Centre which is regarded as a key strategic component of the company's manufacturing capability. The cash flow projections had been prepared in accordance with existing guidelines and there was some uncertainty with respect to capital build and future profitability. Mrs Chua (chief financial officer) had given approval for the project to come to the board given its strategic importance and the closeness of the payback estimate to the company's barrier for long term capital investment of five years. Mr Lazar (non-executive director) suggested that they would need more information about the impact of risk upon the project's outcome before giving final approval. Mr Bright (operations director) agreed but asked why the board needed to consider capital proposals twice. The board was of the view that what was needed was clearer information about each proposal and the risks to which they were exposed. The chair requested the CFO to provide a review of

the company's capital approval procedures to include better assessment of the firm's financial exposure. The revised guidelines should include procedures for both the preliminary and final approval stages. Approved (Action CFO to report)

Required:

- (a) Prepare a paper for the next board meeting, recommending procedures for the assessment of capital investment projects. Your paper should make proposals about the involvement of the board at a preliminary stage and the information that should be provided to inform their decision. You should also provide an assessment of the alternative appraisal methods.** (8 marks)
- (b) Using the appraisal methods you have recommended in (a), prepare a paper outlining the case for the acceptance of the project to build a distillation facility at the Gulf plant with an assessment of the company's likely value at risk. You are not required to undertake an assessment of the impact of the project upon the firm's financial accounts.** (12 marks)

(Including 2 professional marks)

(20 marks)

- 4 You are the chief financial officer of a multinational company in the Do-It-Yourself (DIY) retail business based in the United States. Your company is considering a major expansion into the rapidly developing China market where one of your competitors has already established a presence with three stores, one in Beijing and two in Shanghai. After conducting local market research and a personal review, you are convinced that, although your competitor has successfully opened a new market in those cities, the demand is considerably greater than its ability to supply. Your overseas operations group report that they can open the appropriate supply chains and that, unlike the competition, you will be able to get a greater variety of goods onto the shelves and maintain supply at competitive prices.

Your assessment is that the company will need to raise the equivalent of \$380 million of new finance over 10 years for this venture, of which \$80 million could come from the company's existing liquid reserves. You have completed your review of the financial merits of the case and the project offers a rate of return in excess of 80 per cent. The company's current credit rating is assessed at AA-. Its total market capitalisation is \$3.5bn, which includes a ten year syndicated loan of \$0.5 billion due for retirement in three years. The balance of the firm's capital is in the form of common stock (ordinary shares) trading on the New York and Hong Kong markets.

You wish to undertake a preliminary review of the options for financing this project. Your assessment is that borrowing the money is a possibility but that the increase in gearing would drop your credit rating to A+. You believe that the likelihood of that happening is 60 per cent, with a further 40 per cent chance that the company's rating could fall to A. The company's existing weighted average cost of capital (tax adjusted at the company's average corporation tax rate of 30 per cent) is 6.8 per cent. The current nominal yield curve and credit spreads for the retail sector are shown below:

Exhibit 1: 30 year yield curve

Yield

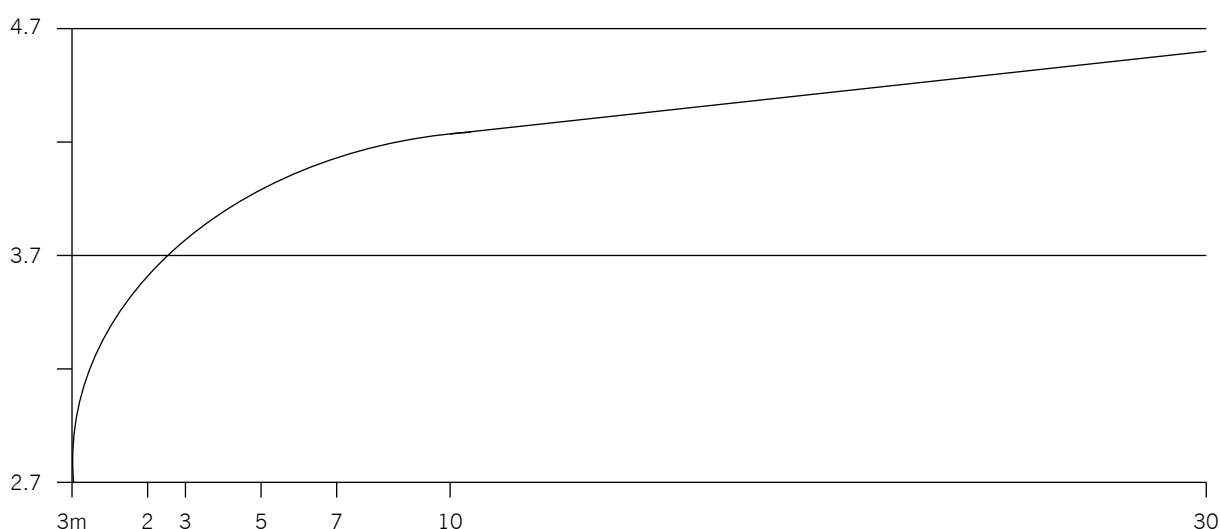


Exhibit 2: Yield spreads for retail sector (in basis points)

Rating	1yr	2yr	3yr	5yr	7yr	10yr	30yr
Aaa/AAA	4	8	12	18	20	30	50
Aa1/AA+	8	12	20	30	32	35	60
Aa2/AA	15	24	30	34	40	50	65
Aa3/AA-	24	35	40	45	54	56	78
A1/A+	28	37	44	55	60	70	82
A2/A	55	65	75	85	95	107	120

Required:

- (a) Estimate the expected cost of capital for this project on the assumption that the additional finance is raised through a bond issue in the US market. (10 marks)
- (b) Draft a brief report for the board which outlines the alternative sources of finance that are potentially available for this project. Include, in your report, a brief discussion of the advantages and disadvantages and the likely impact of each alternative source upon the firm's cost of capital. (10 marks)

(Including 2 professional marks)

(20 marks)

- 5 You have been appointed as the chief financial officer of a multimedia company which is financed by private equity. There is considerable public interest in the company and it continues a very rapid rate of growth under the leadership of its dynamic founder and chief executive officer, Martin Pickle. Martin Pickle owns over 30 per cent of the company's equity and has also loaned the business substantial sums to sustain its overseas development. The balance of the other investors consist of some small shareholdings held by current and past employees and the remainder is in the hands of a private equity company which is represented by two directors on the board.

You enjoy a substantial salary and package of other benefits. Your role description gives you overall responsibility to the board for the financial direction of the company, the management of its financial resources, direction and oversight of its internal control systems and responsibility for its risk management. After two months in the job you are called to a meeting with Martin Pickle and the company's non-executive chairman. In that time you have made significant progress in improving the financial controls of the business and the current year end, which is three weeks away, looks very promising. The company's underlying earnings growth promises to be in excess of 20 per cent and its cash generation is strong. The CEO tells you that he would like you to put together a plan to take the company to full listing as a first step to him undertaking a substantial reduction in his financial stake in the business. He tells you that this discussion must be confidential, as he expects that the market would react adversely to the news. However, he would like to see what could be done to make sure that the year end figures are as strong as possible. Given your performance, he also tell you that they would like to offer you a substantial incentive in the form of share options.

Required:

- (a) **Prepare a board paper, describing the procedure for obtaining a listing on an international stock exchange such as the London or New York Stock Exchange.** (6 marks)
- (b) **Prepare a briefing note, itemising the advantages and disadvantages of such a step for a medium-sized company.** (6 marks)
(Including 2 professional marks)
- (c) **Discuss any ethical considerations or concerns you may have concerning this proposed course of action.** (8 marks)

(20 marks)

Formula Sheet

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2 w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + b_i(E(r_m) - R_f)$$

The asset beta formula

$$b_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} b_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} b_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

The Black Scholes Option Pricing Model	The FOREX modified Black and Scholes option pricing model
$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$ Where: $d_1 = \frac{\ln(P_a / P_e) + (r + 0.5 s^2) t}{s \sqrt{t}}$ $d_2 = d_1 - s \sqrt{t}$	$c = e^{-rt} [F_0 N(d_1) - X N(d_2)]$ Or $p = e^{-rt} [X N(-d_2) - F_0 N(-d_1)]$ Where: $d_1 = \frac{\ln(F_0 / X) + s^2 T / 2}{s \sqrt{T}}$ and $d_2 = d_1 - s \sqrt{T}$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

		Discount rate (r)									
Periods (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		<i>Discount rate (r)</i>									
<i>Periods</i>		1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
(n)											
1		0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909
2		1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736
3		2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487
4		3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170
5		4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791
6		5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355
7		6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868
8		7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335
9		8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759
10		9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145
11		10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495
12		11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814
13		12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103
14		13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367
15		13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606
(n)		11%	12%	13%	14%	15%	16%	17%	18%	19%	20%
1		0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833
2		1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528
3		2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106
4		3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589
5		3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991
6		4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326
7		4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605
8		5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837
9		5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031
10		5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192
11		6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327
12		6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439
13		6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533
14		6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611
15		7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675

Standard normal distribution table

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2703	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990

This table can be used to calculate $N(d_i)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0.5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0.5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Thursday 6 December 2007

Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae tables are on pages 9–13.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper



Section A – BOTH questions are compulsory and MUST be attempted

- 1 You are the chief financial officer of International Enterprises, a multinational company with interests in Europe and the Far East. You are concerned about certain aspects of the company's financial management. The company has enjoyed a high rate of growth over the last three years as a result of a single product's development. This product has had a big impact in the fast moving mobile communications industry. However, the company does not have any new products in development and is relying on expanding its market share and developing upgraded versions of the current product.

As part of your preparation for the board meeting to discuss the 2007 draft accounts, you have prepared a projected income statement and balance sheet for the year ending 31 December 2008. These projections are based upon a number of agreed assumptions taken from the company's strategic plan. As part of the agenda, the board will also consider its dividend target for the forthcoming year.

International Enterprises

Income statement for the year ended 31 December	2008 (projected) \$m	2007 (draft) \$m	2006 (actual) \$m
Revenue	288.1	261.9	220.0
Cost of sales	143.2	132.6	104.0
Gross profit	144.9	129.3	116.0
/less other operating costs	36.1	27.0	24.0
Operating profit	108.8	102.3	92.0
Finance costs	1.8	2.3	2.3
Profit before tax	107.0	100.0	89.7
Income tax expense (at 30%)	32.1	30.0	26.9
Profit for the period	74.9	70.0	62.8

Balance sheet as at 31 December	2008 (projected) \$m	2007 (draft) \$m	2006 (actual) \$m
Non-current assets (see note)			
Buildings, plant and machinery	168.0	116.0	96.0
Current assets			
Inventories	3.2	3.7	2.3
Receivables	25.6	29.1	19.6
Cash	151.8	155.8	121.7
Total current assets	180.6	188.6	143.6
Total assets	348.6	304.6	239.6

Equity and liabilities

Paid up share capital			
Ordinary shares (25c)	25.0	25.0	20.0
Other reserves	12.0	12.0	10.0
Retained earnings	216.9	170.0	120.0
/less dividends payable	0.0	-28.0	-20.0
	216.9	142.0	100.0
Total equity	253.9	179.0	130.0

Balance sheet continued on the next page.

Balance sheet continued

	2008 (projected) \$m	2007 (draft) \$m	2006 (actual) \$m
Current liabilities			
Trade payables	8.8	7.7	6.4
Tax payable	28.5	25.6	23.3
Dividends payable	0.0	28.0	20.0
Interest payable	1.8	2.3	2.3
Total current liabilities	39.1	63.6	52.0
Non-current liabilities			
Loans	35.0	45.0	45.0
Provisions (deferred tax)	20.6	17.0	12.6
Total non-current liabilities	55.6	62.0	57.6
Total liabilities	94.7	125.6	109.6
Total equity and liabilities	348.6	304.6	239.6
Note	2008	2007	2006
	\$m	\$m	\$m
Non-current assets	280.0	200.0	160.0
less accumulated depreciation	112.0	84.0	64.0
Net book value of non-current assets	168.0	116.0	96.0

The projected figures assume:

- (i) \$10 million of the existing loans will be repaid during the year.
- (ii) Capital investment in plant and equipment of \$80 million will be undertaken.

The company is quoted on an international stock exchange and its beta value (based upon three years of monthly return data) is 1.40. The current risk free rate is 3% and the equity risk premium is 5%. The current share price is \$16.20 and the sector price/earnings ratio is 24. The company's cost of debt capital remains at its current rate of 5%. You may assume that the current cost of equity capital remains unchanged over the term of the projection.

Required:

- (a) **Prepare a cash flow forecast for the year ended 31 December 2008.**

Note: the format does not need to comply with accounting standards.

(6 marks)

- (b) **Estimate the company's maximum dividend capacity after the target level of capital reinvestment is undertaken and making any working capital adjustments you deem necessary.**

(6 marks)

- (c) **Draft a brief report for senior management reviewing the potential performance of the business in the year ended 31 December 2008 if the expectations contained within the strategic plan are fulfilled. You should use the Economic Value Added (EVA™) and any other performance measures you think appropriate.**

Note: requirement (c) includes 2 professional marks.

(18 marks)

(30 marks)

- 2 Burcolene is a large European-based petrochemical manufacturer, with a wide range of basic bulk chemicals in its product range and with strong markets in Europe and the Pacific region. In recent years, margins have fallen as a result of competition from China and, more importantly, Eastern European countries that have favourable access to the Russian petrochemical industry. However, the company has managed to sustain a 5% growth rate in earnings through aggressive management of its cost base, the management of its risk and careful attention to its value base.

As part of its strategic development, Burcolene is considering a leveraged (debt-financed) acquisition of PetroFrancais, a large petrochemical business that has engaged in a number of high quality alliances with oil drilling and extraction companies in the newly opened Russian Arctic fields. However, the growth of the company has not been particularly strong in recent years, although Burcolene believes that an expected long term growth of 4% per annum is realistic under its current management.

Preliminary discussions with its banks have led Burcolene to the conclusion that an acquisition of 100% of the equity of PetroFrancais, financed via a bond issue, would not have a significant impact upon the company's existing credit rating. The key issues, according to the company's advisors, are the terms of the deal and the likely effect of the acquisition on the company's value and its financial leverage.

Both companies are quoted on an international stock exchange and below are relevant data relating to each company:

Financial data as at 30 November 2007

	Burcolene	PetroFrancais
Market value of debt in issue (\$bn)	3.30	5.80
Market value of equity in issue (\$bn)	9.90	6.70
Number of shares in issue (million)	340.00	440.00
Share options outstanding (million)	25.40	–
Exercise price of options (\$ per share)	22.00	–
Company tax rate (%)	30.00	25.00
Equity beta	1.85	0.95
Default risk premium	1.6%	3.0%
Net operating profit after tax and net reinvestment (\$ million)	450.00	205.00
Current EPS (\$ per share)	1.19	0.44

The global equity risk premium is 4.0% and the most appropriate risk free rate derived from the returns on government stock is 3.0%.

Burcolene has a share option scheme as part of its executive remuneration package. In accordance with the accounting standards, the company has expensed its share options at fair value. The share options held by the employees of Burcolene were granted on 1 January 2004. The vesting date is 30 November 2009 and the exercise date is 30 November 2010. Currently, the company has a 5% attrition rate as members leave the company and, of those remaining at the vesting date, 20% are expected not to have achieved the standard of performance required. Your estimate is that the options have a time value of \$7.31.

PetroFrancais operates a defined benefits pension scheme which, at its current actuarial valuation, shows a deficit of \$430 million.

You have been appointed to advise the senior management team of Burcolene on the validity of the free cash flow to equity model as a basis for valuing both firms and on the financial implications of this acquisition for Burcolene. Following your initial discussions with management, you decide that the following points are relevant:

1. The free cash flow to all classes of capital invested can be reliably approximated as net operating profit after tax (NOPAT) less net reinvestment.
2. Given the rumours in the market concerning a potential acquisition, the existing market valuations may not fully reflect each company's value.
3. The acquisition would be financed by a new debt issue by Burcolene.

Required:

- (a) Estimate the weighted average cost of capital and the current entity value for each business, taking into account the impact of the share option scheme and the pension fund deficit on the value of each company. (16 marks)

- (b) Write a briefing paper for management, advising them on:

- (i) The validity of the free cash flow model, given the growth rate assumptions made by management for both firms;
- (ii) The most appropriate method of deriving a bid price; and
- (iii) The implications of an acquisition such as this for Burcolene's gearing and cost of capital.

Note: requirement (b) includes 2 professional marks.

(14 marks)

(30 marks)

Section B – TWO questions ONLY to be attempted

- 3** Digunder, a property development company, has gained planning permission for the development of a housing complex at Newtown which will be developed over a three year period. The resulting property sales less building costs have an expected net present value of \$4 million at a cost of capital of 10% per annum. Digunder has an option to acquire the land in Newtown, at an agreed price of \$24 million, which must be exercised within the next two years. Immediate building of the housing complex would be risky as the project has a volatility attaching to its net present value of 25%.

One source of risk is the potential for development of Newtown as a regional commercial centre for the large number of professional firms leaving the capital, Bigcity, because of high rents and local business taxes. Within the next two years, an announcement by the government will be made about the development of transport links into Newtown from outlying districts including the area where Digunder hold the land option concerned. The risk free rate of interest is 5% per annum.

Required:

- (a) Estimate the value of the option to delay the start of the project for two years using the Black and Scholes option pricing model and comment upon your findings. Assume that the government will make its announcement about the potential transport link at the end of the two-year period.** (12 marks)
- (b) On the basis of your valuation of the option to delay, estimate the overall value of the project, giving a concise rationale for the valuation method you have used.** (4 marks)
- (c) Describe the limitations of the valuation method you used in (a) above and describe how you would value the option if the government were to make the announcement at ANY time over the next two years.** (4 marks)

(20 marks)

- 4 The chairman of your company has become concerned about the accumulation of cash in hand and in the deposit accounts shown in the company's balance sheet. The company is in the manufacturing sector, supplying aerospace components to the civil aviation markets in the UK and Europe. For the last 20 years the company has grown predominantly by acquisition and has not invested significantly in research and development on its own account. The acquisitions have given the company the technology that it has required and have all tended to be small, relative to the company's total market capitalisation. The company has a healthy current asset ratio of 1.3, although its working capital cycle has an average of 24 unfunded days.

The company has not systematically embraced new manufacturing technologies nor has it sought to reduce costs as a way of rebuilding profitability. Managerial and structural problems within divisions have led to a number of substantial projects overrunning and losses being incurred as a result. It has also proven difficult to ensure the accountability of managers promoting projects – many of which have not subsequently earned the cash flows originally promised. At the corporate level, much of the company's accounting is on a contracts basis and over the years it has tended to be cautious in its revenue recognition practices. This has meant that earnings growth has lagged behind cash flow.

Over the last 12 months the company has come under strong competitive pressure on the dominant defence side of its business which, coupled with the slow-down in spending in this area across the major western economies, has slowed the rate of growth of its earnings. The company's gearing ratio is very low at 12% of total market capitalisation and borrowing has invariably been obtained in the European fixed interest market and used to support capital investment in its European production facility. In the current year, investment plans are at the lowest they have been in real terms since the company was founded in the 1930s.

In discussion, the chairman comments upon the poor nature of the company's buildings and its poor levels of pay which could, in his view, be improved to reflect standards across the industry. Directors' pay, he reminds you, is some 15% below industry benchmarks and there is very little equity participation by the board of directors. He also points out that the company's environmental performance has not been good. Last year the company was fined for an untreated discharge into a local river. There are, he says, many useful things the company could do with the money to help improve the long-term health of the business. However, he does admit some pessimism that business opportunities will ever again be the same as in previous years and he would like a free and frank discussion at the next board meeting about the options for the company. The company has a very open culture where ideas are encouraged and freely debated.

The chairman asks if you, as the newly appointed chief financial officer, would lead the discussion at the next board.

Required:

- (a) In preparation for a board paper entitled 'Agenda for Change', write brief notes which identify the strategic financial issues the company faces and the alternatives it might pursue. (10 marks)
- (b) Identify and discuss any ethical issues you believe are in the above case and how the various alternatives you have identified in (a) may lead to their resolution. (10 marks)

(20 marks)

- 5** Your company, which is in the airline business, is considering raising new capital of \$400 million in the bond market for the acquisition of new aircraft. The debt would have a term to maturity of four years. The market capitalisation of the company's equity is \$1.2 billion and it has a 25% market gearing ratio (market value of debt to total market value of the company). This new issue would be ranked for payment, in the event of default, equally with the company's other long-term debt and the latest credit risk assessment places the company at AA. Interest would be paid to holders annually. The company's current debt carries an average coupon of 4% and has three years to maturity. The company's effective rate of tax is 30%.

The current yield curve suggests that, at three years, government treasuries yield 3.5% and at four years they yield 5.1%. The current credit risk spread is estimated to be 50 basis points at AA. If the issue proceeds, the company's investment bankers suggest that a 90 basis point spread will need to be offered to guarantee take up by its institutional clients.

Required:

- (a) Advise on the coupon rate that should be applied to the new debt issue to ensure that it is fully subscribed.**
(4 marks)
- (b) Estimate the current and revised market valuation of the company's debt and the increase in the company's effective cost of debt capital.**
(8 marks)
- (c) Discuss the relative advantages and disadvantages of this mode of capital financing in the context of the company's financial objectives.**
(8 marks)

(20 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model	The FOREX modified Black-Scholes option pricing model
$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$ Where: $d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$ $d_2 = d_1 - s\sqrt{t}$	$c = e^{-rt} F_0 N(d_1) - XN(d_2)$ Or $p = e^{-rt} (XN(-d_2) - F_0 N(-d_1))$ Where: $d_1 = \frac{\ln(F_0 / X) + s^2 T / 2}{s\sqrt{T}}$ and $d_2 = d_1 - s\sqrt{T}$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.941	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.305	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Thursday 5 June 2008

Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae tables are on pages 11–15.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

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The question paper begins on page 3.**

Section A – BOTH questions are compulsory and MUST be attempted

- 1 Mercury Training was established in 1999 and since that time it has developed rapidly. The directors are considering either a flotation or an outright sale of the company.

The company provides training for companies in the computer and telecommunications sectors. It offers a variety of courses ranging from short intensive courses in office software to high level risk management courses using advanced modelling techniques. Mercury employs a number of in-house experts who provide technical materials and other support for the teams that service individual client requirements. In recent years, Mercury has diversified into the financial services sector and now also provides computer simulation systems to companies for valuing acquisitions. This business now accounts for one third of the company's total revenue.

Mercury currently has 10 million, 50c shares in issue. Jupiter is one of the few competitors in Mercury's line of business. However, Jupiter is only involved in the training business. Jupiter is listed on a small company investment market and has an estimated beta of 1.5. Jupiter has 50 million shares in issue with a market price of 580c. The average beta for the financial services sector is 0.9. Average market gearing (debt to total market value) in the financial services sector is estimated at 25%.

Other summary statistics for both companies for the year ended 31 December 2007 are as follows:

	Mercury	Jupiter
Net assets at book value (\$million)	65	45
Earnings per share (c)	100	50
Dividend per share (c)	25	25
Gearing (debt to total market value)	30%	12%
Five year historic earnings growth (annual)	12%	8%

Analysts forecast revenue growth in the training side of Mercury's business to be 6% per annum, but the financial services sector is expected to grow at just 4%.

Background information:

The equity risk premium is 3.5% and the rate of return on short-dated government stock is 4.5%.

Both companies can raise debt at 2.5% above the risk free rate.

Tax on corporate profits is 40%.

Required:

- (a) Estimate the cost of equity capital and the weighted average cost of capital for Mercury Training. Explain the circumstances where each of the two rates would be used. (10 marks)
- (b) Advise the owners of Mercury Training on a range of likely issue prices for the company. (10 marks)
- (c) Discuss the advantages and disadvantages, to the directors of Mercury Training, of a public listing versus private equity finance as a means of disposing of their interest in the company. (8 marks)

(28 marks)

- 2 Venus Systems, a publicly quoted company, is a specialist manufacturer of mechanical control units for both the defence and civil aviation industries. Its principal customers are the defence procurement agencies of a number of western governments and European Aerospace Co, an aeroplane manufacturer. Over recent years the company has suffered a collapse in profitability and has attempted to respond by reducing its defence related business and focusing on its civil aviation business.

On the civil side, long delays at European Aerospace Co in the development of a new large-bodied passenger aeroplane, the European Aircoach, have severely impacted upon suppliers such as Venus Systems. As a result Venus's share price has declined over the last three years, in line with movements in the sector index. However, market valuations have not followed the general decline in earnings across the sector. Market analysts attribute this to the high level of advance orders by airlines for the Aircoach and the confident expectation that full production will commence in mid 2009.

Orders for defence components have fallen rapidly over the last three years and the company has taken the decision to scale down its defence division and switch production resources to the civil side of the business. Wherever possible the company has redeployed and retrained its workforce. Indeed, its commitment to its workforce has helped maintain good industrial relations and the redeployment has been successfully matched against its natural labour turnover in the civil division.

During 2007 the company commenced the decommissioning of its defence manufacturing site and began rationalising both defence and civil production to one site.

The summary financial statements for the company over the last three years are as follows:

Venus Systems

Accounts for the year ended 30 April:

	2008 \$m	2007 \$m	2006 \$m
Income statements			
Revenue	72.6	88.2	90.0
Cost of Sales	44.0	50.5	58.0
Gross Profit	28.6	37.7	32.0
/less other operating costs	27.0	25.2	24.0
Operating profit	1.6	12.5	8.0
Finance Costs	1.4	1.4	2.5
Profit before tax	0.2	11.1	5.5
Income tax expense (at 30%)	0.1	3.3	1.7
Profit for the period	0.1	7.8	3.8

	2008	2007	2006
	\$m	\$m	\$m
Statements of financial position (Balance Sheets)			
Non-current assets			
Buildings, plant and machinery	17·0	42·0	84·0
Current Assets			
Inventories	11·7	6·7	8·1
Receivables	36·3	22·1	17·5
Cash	56·1	54·8	26·6
Total Current Assets	<u>104·1</u>	<u>83·6</u>	<u>52·2</u>
Total Assets	<u><u>121·1</u></u>	<u><u>125·6</u></u>	<u><u>136·2</u></u>
Equity and Liabilities			
Paid up share capital			
Ordinary shares (25¢)	25·0	25·0	20·0
Other reserves	12·0	12·0	10·0
Retained earnings	39·7	39·6	36·8
less dividends payable	0·0	0·0	–5·0
	<u>39·7</u>	<u>39·6</u>	<u>31·8</u>
Total equity	<u>76·7</u>	<u>76·6</u>	<u>61·8</u>
Current Liabilities			
Trade payables	2·7	2·9	3·5
Deferred income and accrued expenses	6·1	7·3	7·8
Tax payable	3·2	1·6	6·1
Dividends payable	0·0	0·0	5·0
Interest payable	1·4	1·4	2·5
Total current liabilities	<u>13·4</u>	<u>13·2</u>	<u>24·9</u>
Non-current liabilities			
Loans	25·0	25·0	45·0
Provisions (deferred tax)	6·0	10·8	4·5
Total non-current liabilities	<u>31·0</u>	<u>35·8</u>	<u>49·5</u>
Total liabilities	<u>44·4</u>	<u>49·0</u>	<u>74·4</u>
Total Equity and Liabilities	<u><u>121·1</u></u>	<u><u>125·6</u></u>	<u><u>136·2</u></u>
Notes	2008	2007	2006
	\$m	\$m	\$m
Non-current assets at start of year	170·0	220·0	190·0
additions	0·0	0·0	30·0
less disposals	–20·0	–50·0	0·0
	<u>150·0</u>	<u>170·0</u>	<u>220·0</u>
Accumulated depreciation at start of the year	128·0	136·0	114·0
less depreciation on disposals	–10·0	–25·0	0·0
depreciation charge for the year	15·0	17·0	22·0
Accumulated depreciation at the end of the year	<u>133·0</u>	<u>128·0</u>	<u>136·0</u>
Net book value of non-current assets	<u><u>17·0</u></u>	<u><u>42·0</u></u>	<u><u>84·0</u></u>

Summary statements of cash flows	2008	2007	2006
	\$m	\$m	\$m
Operating cash flow	-4.0	25.4	35.0
/less interest	-1.4	-2.5	-2.5
/less taxation	-3.3	-1.7	-6.5
Free cash flow before reinvestment	-8.7	21.2	26.0
Dividend paid	0.0	-5.0	-12.0
Capital expenditure	10.0	25.0	-30.0
Financing	0.0	-13.0	0.0
Net cash flow	1.3	28.2	-16.0
Market value of equity (\$ million)	40.0	97.0	105.0

The board believes that it can reverse the decline in the company's profitability with the following turnaround strategy:

1. By cost saving and other measures, it can return to its 2007 level of operating profit on its existing business.
2. By investment in new fabrication equipment at a cost of \$35 million, it can increase the value of the company by \$20.2 million.
3. Through the introduction of just-in-time manufacturing systems, it anticipates that it can reduce its inventories by 10 days in each of the next three years. Its current receivables can be brought back to the 2007 level by improvements in its credit control but, because of the nature of its business, it does not anticipate being able to improve beyond that.
4. The company is also considering a scheme whereby 10% of the equity shares will be repurchased at the current price of 40c per share plus a premium of 10c.

The non-executive directors have questioned the continuing viability of the company in the short to medium term and whether a more aggressive strategy than that proposed by management is necessary. As a result, you have been appointed as a financial advisor to the board. The board has requested that you provide a report evaluating the position thoroughly but not one that is 'full of figures'. The executive directors, whilst concerned about the company's position, are very keen to reassure the non-executive members of the board about the continuing viability of the business and have given you full access to the company's information and its planning processes. However, the non-executive directors are concerned about the company's prospect of survival in the short to medium term. As part of your review of the risk of failure, you believe that it will be necessary to calculate a current Z-score for the company.

In the calculation of ratios using figures in the statement of financial position, you may use year-end figures throughout.

Retail price inflation stands at 3% per annum and has been at that rate for the last three years.

Note:

The Altman Z-score is calculated as follows:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

Where:

X_1 = working capital/total assets

X_2 = retained earnings/total assets

X_3 = earnings before interest and tax/total assets

X_4 = market value of equity/book value of total debt

X_5 = sales/total assets

In Altman's model, a Z-score of 3 or more indicates a high likelihood of non-failure, 1.8 or less indicates a high likelihood of failure.

Required:

Prepare a report for the board of directors containing an analysis of the company's position. Your report should include:

- (a) An introduction outlining the principal causes of financial distress in any business. (5 marks)**
- (b) An appraisal of the impact of the proposed turnaround strategy on the company's performance, efficiency, risk and liquidity. Your appraisal of the company's exposure to risk should include an assessment of the risk of failure using Altman's Z-score and any other risk measures you consider appropriate. (17 marks)**
- (c) Advice on any corporate governance and ethical issues that the company should consider when reviewing its turnaround strategy. (6 marks)**

Appropriateness of the format and presentation of the report and the effectiveness with which its advice is communicated. (4 marks)

(32 marks)

Section B – TWO questions ONLY to be attempted

- 3** Asteroid Systems is a German-based company with a subsidiary in Switzerland. The company's financial manager expects the Swiss business will remit the equivalent of Euros 1·5 million in two months. Her expectations of the future remittance are based upon the current SFr/Euro forward rate.

The current spot and forward rates for Swiss francs against the Euro are extracted from the Financial Times and are shown in the table below.

	Closing mid-point	Change on day	Bid/offer spread	Days mid high low	One month Rate	One month annual %	Three month Rate	Three month annual %
Switzerland (SFr/€)	1·6242	0·0107	239–244	1·6261 1·6147	1·6223	1·4	1·6176	1·6

In the Euro money market the company can make fixed interest deposits at LIBOR and can borrow at LIBOR plus 20 basis points for terms of greater than one month but up to six months. The company can borrow at fixed rates in the Swiss money market. LIBOR rates, as quoted in the Financial Times, are as follows:

	EUR	CHF
spot	3·56688	2·06000
1 week	3·57300	2·06000
2 week	3·58438	2·07000
1 month	3·60900	2·08000
2 month	3·72538	2·17000
3 month	3·78238	2·20000

The company's financial manager is keen to eliminate transaction risk. However, because of the margin requirements and their impact upon the firm's cash flow, she would prefer not to use exchange traded derivatives. Swiss franc borrowing or lending rates would need to be negotiated with the bank.

Required:

- Estimate the lowest acceptable Swiss borrowing or lending rate for a money market hedge maturing in two months.** (10 marks)
- Discuss the relative advantages and disadvantages of the use of a money market hedge compared with using exchange traded derivatives for hedging a foreign exchange exposure.** (6 marks)
- Discuss the extent to which currency hedging can reduce a firm's cost of capital.** (4 marks)

(20 marks)

- 4 Mr Moon is the CEO of Saturn Systems, a very large listed company in the telecommunications business. The company is in a very strong financial position, having developed rapidly in recent years through a strategy based upon growth by acquisition. Currently, earnings and earnings growth are at all time highs although the company's cash reserves are at a low level following a number of strategic investments in the last financial year. The previous evening Mr Moon gave a speech at a business dinner and during questions made some remarks that Pluto Limited was an attractive company with 'great assets' and that he would be a 'fool' if he did not consider the possibility 'like everyone else' of acquiring the company. Pluto is a long established supplier to Saturn Systems and if acquired would add substantially to the market capitalisation of the business.

Mr Moon's comments were widely reported in the following morning's financial newspapers and, by 10 am, the share price of Pluto had risen 15% in out-of-hours and early trading. The first that you, Saturn's chief financial officer, heard about the issue was when you received an urgent call from Mr Moon's office. You have just completed a background investigation of Pluto, along with three other potential targets instigated at Saturn's last board meeting in May. Following that investigation, you have now commenced a review of the steps required to raise the necessary debt finance for a bid and the procedure you would need to follow in setting up a due diligence investigation of each company.

On arriving at Mr Moon's office you are surprised to see the chairman of the board in attendance. Mr Moon has just put down the telephone and is clearly very agitated. They tell you about the remarks made by Mr Moon the previous evening and that the call just taken was from the Office of the Regulator for Public Companies. The regulator had wanted to know if a bid was to be made and what announcement the company intended to make. They had been very neutral in their response pending your advice but had promised to get back to the regulator within the hour. They knew that if they were forced to admit that a bid was imminent and then withdrew that they would not be able to bid again for another six months. Looking at you they ask as one: 'what do we do now?' After a short discussion you returned to your office and began to draft a memorandum with a recommendation about how to proceed.

Required:

- (a) **Assess the regulatory, financial and ethical issues in this case.** (15 marks)
- (b) **Propose a course of action that the company should now pursue, including a draft of any announcement that should be made, given that the board of Saturn Systems wishes to hold open the option of making a bid in the near future.** (5 marks)

(20 marks)

- 5 Neptune is a listed company in the telecommunications business. You are a senior financial management advisor employed by the company to review its capital investment appraisal procedures and to provide advice on the acceptability of a significant new capital project – the Galileo.

The project is a domestic project entailing immediate capital expenditure of \$800 million at 1 July 2008 and with projected revenues over five years as follows:

	30 June	30 June	30 June	30 June	30 June
Year ended	2009	2010	2011	2012	2013
Revenue (\$ million)	680·00	900·00	900·00	750·00	320·00

Direct costs are 60% of revenues and indirect, activity based costs are \$140 million for the first year of operations, growing at 5% per annum over the life of the project. In the first two years of operations, acceptance of this project will mean that other work making a net contribution before indirect costs of \$150 million for each of the first two years will not be able to proceed. The capital expenditure of \$800 million is to be paid immediately and the equipment will have a residual value after five years' operation of \$40 million. The company depreciates plant and equipment on a straight-line basis and, in this case, the annual charge will be allocated to the project as a further indirect charge. Preconstruction design and contracting costs incurred over the previous three years total \$50 million and will be charged to the project in the first year of operation.

The company pays tax at 30% on its taxable profits and can claim a 50% first year allowance on qualifying capital expenditure followed by a writing down allowance of 40% applied on a reducing balance basis. Given the timing of the company's tax payments, tax credits and charges will be paid or received twelve months after they arise. The company has sufficient other profits to absorb any capital allowances derived from this project.

The company currently has \$7,500 million of equity and \$2,500 million of debt in issue quoted at current market values. The current cost of its debt finance is \$LIBOR plus 180 basis points. \$LIBOR is currently 5·40%, which is 40 basis points above the one month Treasury bill rate. The equity risk premium is 3·5% and the company's beta is 1·40. The company wishes to raise the additional finance for this project by a new bond issue. Its advisors do not believe that this will alter the company's bond rating. The new issue will incur transaction costs of 2% of the issue value at the date of issue.

Required:

- (a) Estimate the adjusted present value of the project resulting from the new investment and from the refinancing proposal and justify the use of this technique. (14 marks)
- (b) Estimate the modified internal rate of return generated by the project cash flows, excluding the effects of refinancing. (6 marks)

(20 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d (1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model	The FOREX modified Black-Scholes option pricing model
$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$ Where: $d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$ $d_2 = d_1 - s\sqrt{t}$	$c = e^{-rt} F_0 N(d_1) - XN(d_2)$ Or $p = e^{-rt} (XN(-d_2) - F_0 N(-d_1))$ Where: $d_1 = \frac{\ln(F_0 / X) + s^2 T / 2}{s\sqrt{T}}$ and $d_2 = d_1 - s\sqrt{T}$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.941	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.305	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·49987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Thursday 4 December 2008

Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 6–10.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – BOTH questions are compulsory and MUST be attempted

- 1 It is now 1 December 2008. You have been hired as a financial consultant to the Blipton International Entertainment Group which is evaluating a proposal from its hotel division to build a 400 bedroom hotel in the East End of London. This area has developed rapidly over the last 15 years and the prospects have been further enhanced by the announcement that London is to host the 2012 Olympics. Blipton is based in Dubai and both reports and accounts for all its transactions in dollars. The current dollar/sterling spot rate is \$1.4925/£. The operating costs for the hotel are expected to be £30 per occupied room per day (variable) and a fixed cost of £1.7 million per annum expressed in current prices. The proportion of bedrooms occupied, on the basis of opening for 365 days a year, is expected to be as follows:

Year ended	occupancy
31 December 2009	construction
31 December 2010	40%
31 December 2011	50%
31 December 2012	90%
31 December 2013	60%
31 December 2014	60%

UK inflation is currently projected by the Bank of England as 2.5% per annum and inflation in the United States is 4.8% per annum. These rates are expected to be constant over the term of the project. Blipton's real cost of capital is 4.2%. UK hotel property values within the London area are expected to rise in real terms by 8% per annum.

The construction cost for this hotel is estimated to be £6.2 million and it will be built over the 12 months to 31 December 2009. As part of the UK's Olympic Development Plan, a 50% first year capital allowance is available for tax purposes on building projects related to the Games. The balance of the capital expenditure can be claimed in equal instalments over the following three years. UK profit tax is 30% and is levied and paid on profits in the year they arise. There is no additional tax liability on remittance to or from Dubai. The company has sufficient UK profits on its other activities to absorb the capital allowances on this project.

In making investment decisions of this type the company operates the following procedure:

1. All cash flows including construction costs are assumed to arise at the end of the year concerned and are to be projected in nominal (money) terms over the six year period.
2. The residual value of the investment at the end of six years is assumed to be the open market value of the property less a charge for repairs and renewals.
3. The charge for repairs and renewals is expected to be £1.2 million in current prices payable on disposal.
4. The net present value of the project should be based upon a 100% remittance of net cash flows to Dubai and should be calculated in dollars.
5. Average room rates are set at the level required to recover variable cost plus 100%.

Required:

Prepare a report for management to include the following:

- (a) A six year nominal dollar projection of the after tax cash flow for this project distinguishing between cash flows arising from its investment phase and those arising from its return phase. (12 marks)
- (b) An estimate of the project's dollar net present value and the modified internal rate of return.

Note: you may use the following formula
$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

Where PV_R is the present value of the return phase of the project, PV_I is the present value of the investment phase and r_e is the firm's cost of capital. (8 marks)

- (c) An assessment of the viability of the project with a summary of the relative advantages and disadvantages of the net present value and modified internal rate of return methods in investment appraisal. (8 marks)

Quality and presentation of the report. (2 marks)

(30 marks)

- 2 Rosa Nelson, the Chief Financial Officer (CFO) of Jupiter Co, has been in discussion with the firm's advisors about refinancing the capital of the firm. She is considering a scheme to repay current borrowings of \$800 million and raise new capital through a bond issue of \$2,400 million. The current debt consists of several small loans raised in the Euro market with differing maturities and carrying an average rate of interest of 5.6%. The average term to maturity of the existing debt is four years. The new debt would be in the form of 10 year, fixed interest bonds with half being raised in the Yen and half in the Euro market. The yield to maturity of an appropriate government bond and the relevant credit risk premium for a company of Jupiter's credit rating in the Japanese and the European market is shown below:

	Yield to maturity		Credit risk premium (basis points)	
	4 years	10 years	4 years	10 years
Japanese Government Bonds	1.00%	1.80%	35	50
European Government Bonds	4.20%	4.60%	45	85

Jupiter's current beta is 1.50. The current risk free rate is 4.0% and the equity risk premium is 3.0%.

The company currently earns a free cash flow to equity of \$400 million after interest, tax and net reinvestment. The company consistently reinvests 30% of that free cash flow within the firm and makes the balance available for distribution to investors. The free cash flow to equity model has provided a reasonable estimate of the company's equity market valuation in the past. The current share price, based upon 500 million fully paid, 25¢ equity shares, is 1,380¢. The current rate of tax on corporate profits is 25%. Management is of the view that the additional borrowing will lead to the company being able to increase its earnings growth rate to 4%.

You may assume:

1. Interest on the firm's debt is paid annually.
2. The debt in issue and proposed has a zero beta.
3. The firm's share price will be unaffected by the alteration in gearing.
4. Foreign exchange risk may be ignored.

Required:

As Deputy Chief Financial Officer prepare a briefing note for Rosa Nelson. Your note should include:

- (a) An assessment of the firm's current cost of debt, cost of equity and weighted average cost of capital.**
(6 marks)
 - (b) An assessment of the firm's expected cost of debt, cost of equity and weighted average cost of capital after the redemption of the existing debt and the issue of the new bonds.**
(8 marks)
 - (c) An assessment of the minimum rate of return that the company needs to earn on the new debt capital before interest is paid.**
(6 marks)
 - (d) A comparison of the proposed method of raising capital for investment purposes with alternative means of raising the debt finance required.**
(8 marks)
- Quality and presentation of the briefing note.** (2 marks)
- (30 marks)**

Section B – TWO questions ONLY to be attempted

- 3 Aston Co, a medium-sized internet trading company, has conducted a review of its monthly operating cash flow. Its monthly average operating cash flow has been \$14,400 with an observed monthly volatility over the previous five years of trading of 13%. There has been no growth in the trend of operating cash flow over the last 12 months and there is great concern among the small group of shareholders about the credit risk to which the company is exposed. The company has \$1.5 million of borrowing repayable in five years and pays an effective interest rate of 8% per annum on the debt. Interest is paid monthly. The company has \$8,500 of cash in hand or on deposit. The bank has first call upon the assets of the firm. The company estimates the proportion of its loan that would be recoverable in the event of default to be 90% of the outstanding debt value including accrued interest. The inter-bank offered rate is 5.5% per annum and the bank normally seeks to recover a risk adjustment of 34 basis points above the spread required to compensate for the expected loss on the loan.

Required:

- (a) On the basis of the annualised operating cash flow after interest and its volatility, estimate the probability of default within 12 months on the assumption that the company has no other lines of credit available.

(10 marks)

- (b) Using the probability of default estimated in (a) discuss the issues that the bank would consider when making a loan of this type and demonstrate how an annual charge of 8% on this loan would be justified.

(10 marks)

(20 marks)

- 4 Solar Supermarkets, a listed company, is reviewing the approach that it should take to remunerating its executive directors and other senior managers. Over the years, the company's share price has performed well although there is now concern that price and cost competition from overseas entrants into the domestic market will have a significant impact on the firm's profitability. As a result, the directors believe that large investment in new technologies and markets will be required over the next five years. Traditionally, management has been rewarded by salary, a generous system of benefits, and a bonus scheme that has taken up to 4% of turnover. The directors are considering introducing a generous share option scheme with a five year vesting period.

There is also a view, expressed by some of the company's principal equity investors, that the company should consider returning cash to them through the sale of its property holdings. The company has over 200 stores nationally and 15 overseas, of which all except five are owned by the company. In the domestic economy, growth in the value of commercial property has averaged 8% per annum in recent years whilst retail growth has remained static at 5.5%. A sale and leaseback, or the flotation of a separate property company that would rent the stores to Solar Supermarkets at commercial rates, are two suggestions that have been made at investor meetings. Either approach, it is suggested, would return value to investors and create a supply of capital for further expansion. There have been press rumours, possibly fed from sources within the investor community, that the company may be a target for a private equity acquisition. However, no formal approach has been made to the company.

The only other area of controversy to emerge about the company which has concerned the directors followed an announcement about the company pension scheme. Although the scheme is well funded the directors took the decision to close the current final salary scheme to new employees and to replace it with a money purchase scheme. Current employees would not be affected.

Required:

Discuss the strategic, financial and ethical issues that this case presents and the merits of the proposed share option and sale and leaseback schemes.

(20 marks)

- 5 Following a collapse in credit confidence in the banking sector globally, there have been high levels of volatility in the financial markets around the world. Phobos Co is a UK listed company and has a borrowing requirement of £30 million arising in two months' time on 1 March and expects to be able to make repayment of the full amount six months from now. The governor of the central bank has suggested that interest rates are now at their peak and could fall over the next quarter. However, the chairman of the Federal Reserve in the United States has suggested that monetary conditions may need to be tightened, which could lead to interest rate rises throughout the major economies. In your judgement there is now an equal likelihood that rates will rise or fall by as much as 100 basis points depending upon economic conditions over the next quarter.

LIBOR is currently 6.00% and Phobos can borrow at a fixed rate of LIBOR plus 50 basis points on the short term money market but the company treasurer would like to keep the maximum borrowing rate at or below 6.6%.

Short term sterling index futures have a contract size of £500,000 and a tick size of £12.50. The open and settlement prices of three month futures contracts are shown below (settlement at the end of the month):

	Open	Settlement
March	93.800	93.880
June	93.870	93.940
September	93.890	93.970

You may assume that basis diminishes to zero at contract maturity at a constant rate over time and that time intervals can be counted in months.

Options on short sterling futures have a contract size of £500,000 and the premiums (shown as an annual percentage) available against a range of exercise prices are as follows:

Exercise	calls			puts		
	March	June	September	March	June	September
93750	0.155	0.260	0.320	0.045	0.070	0.100
94000	0.038	0.110	0.175	0.168	0.170	0.205
94250	0.010	0.040	0.080	0.300	0.350	0.360

Required:

- (a) Estimate the effective interest rate cost if the anticipated interest rate exposure is hedged:

- (i) using the sterling interest rate futures; and
- (ii) the options on short sterling futures.

(14 marks)

- (b) Outline the benefits and dangers to Phobos of using derivative agreements in the management of interest rate risk.

(6 marks)

(20 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model	The FOREX modified Black-Scholes option pricing model
$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$ Where: $d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$ $d_2 = d_1 - s\sqrt{t}$	$c = e^{-rt} [F_0 N(d_1) - X N(d_2)]$ Or $p = e^{-rt} [X N(-d_2) - F_0 N(-d_1)]$ Where: $d_1 = \frac{\ln(F_0 / X) + s^2 T / 2}{s\sqrt{T}}$ and $d_2 = d_1 - s\sqrt{T}$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

Discount rate (r)

Periods

(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.941	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.305	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·49987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Thursday 4 June 2009

Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 9–13.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

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The question paper begins on page 3.

Section A – BOTH questions are compulsory and MUST be attempted

- 1 You have been conducting a detailed review of an investment project proposed by one of the divisions of your business. Your review has two aims: first to correct the proposal for any errors of principle and second, to recommend a financial measure to replace payback as one of the criteria for acceptability when a project is presented to the company's board of directors for approval. The company's current weighted average cost of capital is 10% per annum.

The initial capital investment is for \$150 million followed by \$50 million one year later. The post tax cash flows, for this project, in \$million, including the estimated tax benefit from capital allowances for tax purposes, are as follows:

Year	0	1	2	3	4	5	6
Capital investment (plant and machinery):							
First phase	-127.50						
Second phase		-36.88					
Project post tax cash flow (\$ millions)			44.00	68.00	60.00	35.00	20.00

Company tax is charged at 30% and is paid/recovered in the year in which the liability is incurred. The company has sufficient profits elsewhere to recover capital allowances on this project, in full, in the year they are incurred. All the capital investment is eligible for a first year allowance for tax purposes of 50% followed by a writing down allowance of 25% per annum on a reducing balance basis.

You notice the following points when conducting your review:

1. An interest charge of 8% per annum on a proposed \$50 million loan has been included in the project's post tax cash flow before tax has been calculated.
2. Depreciation for the use of company shared assets of \$4 million per annum has been charged in calculating the project post tax cash flow.
3. Activity based allocations of company indirect costs of \$8 million have been included in the project's post tax cash flow. However, additional corporate infrastructure costs of \$4 million per annum have been ignored which you discover would only be incurred if the project proceeds.
4. It is expected that the capital equipment will be written off and disposed of at the end of year six. The proceeds of the sale of the capital equipment are expected to be \$7 million which have been included in the forecast of the project's post tax cash flow. You also notice that an estimate for site clearance of \$5 million has not been included nor any tax saving recognised on the unclaimed writing down allowance on the disposal of the capital equipment.

Required:

- (a) **Prepare a corrected project evaluation using the net present value technique supported by a separate assessment of the sensitivity of the project to a \$1million change in the initial capital expenditure.**
(14 marks)
- (b) **Estimate the discounted payback period and the duration for this project commenting on the relative advantages and disadvantages of each method.**
(5 marks)
- (c) **Draft a brief report for presentation to the board of directors with a recommendation on the acceptability of this project and on the techniques that the board should consider when reviewing capital investment projects in future.**
(8 marks)

Professional marks will be awarded in part (c) for the clarity, presentation and persuasiveness of the report.
(1 mark)

(28 marks)

- 2 BBS Stores, a publicly quoted limited company, is considering unbundling a section of its property portfolio. The company believes that it should use the proceeds to reduce the company's medium-term borrowing and to reinvest the balance in the business (option 1). However, the company's investors have argued strongly that a sale and rental scheme would release substantial cash to investors (option 2). You are a financial consultant and have been given the task of assessing the likely impact of these alternative proposals on the company's financial performance, cost of capital and market value.

Attached is the summarised BBS Stores' statement of financial position. The company owns all its stores.

	As at year end 2008 \$m	As at year end 2007 \$m
ASSETS		
Non-current assets		
Intangible assets	190	160
Property, plant and equipment	4,050	3,600
Other assets	500	530
	<u>4,740</u>	<u>4,290</u>
Current assets	840	1,160
	<u>5,580</u>	<u>5,450</u>
LIABILITIES		
Current liabilities	1,600	2,020
Non-current liabilities		
Medium-term loan notes	1,130	1,130
Other non-financial liabilities	890	900
	<u>2,020</u>	<u>2,030</u>
Total liabilities	<u>3,620</u>	<u>4,050</u>
Net assets	<u>1,960</u>	<u>1,400</u>
EQUITY		
Called up share capital – equity	425	420
Retained earnings	1,535	980
	<u>1,960</u>	<u>1,400</u>

The company's profitability has improved significantly in recent years and earnings for 2008 were \$670 million (2007: \$540 million).

The company's property, plant and equipment within non-current assets for 2008 are as follows:

	Land and buildings \$m	Fixtures, fittings & equipment \$m	Assets under construction \$m	Total \$m
Year end 2008				
At revaluation	2,297	4,038	165	6,500
Accumulated depreciation		(2,450)		(2,450)
Net book value	<u>2,297</u>	<u>1,588</u>	<u>165</u>	<u>4,050</u>

The property portfolio was revalued at the year end 2008. The assets under construction are valued at a market value of \$165 million and relate to new building.

In recent years commercial property values have risen in real terms by 4% per annum. Current inflation is 2.5% per annum. Property rentals currently earn an 8% return.

The proposal is that 50% of the property portfolio (land and buildings) and 50% of the assets under construction would be sold to a newly established property holding company called RPH that would issue bonds backed by the assured rental income stream from BBS Stores. BBS Stores would not hold any equity interest in the newly formed company nor would they take any part in its management.

BBS Stores is currently financed by equity in the form of 25c fully paid ordinary shares with a current market value of 400c per share. The capital debt for the company consists of medium-term loan notes of which \$360 million are repayable at the end of two years and \$770 million are repayable at the end of six years. Both issues of medium-term notes carry a floating rate of LIBOR plus 70 basis points. The interest liability on the six year notes has been swapped at a fixed rate of 5.5% in exchange for LIBOR which is also currently 5.5%. The reduction in the firm's gearing implied by option 1 would improve the firm's credit rating and reduce its current credit spread by 30 basis points. The change in gearing resulting from the second option is not expected to have any impact upon the firm's credit rating. There has been no alteration in the rating of the company since the earliest debt was issued.

The BBS Stores equity beta is currently 1.824. A representative portfolio of commercial property companies has an equity beta of 1.25 and an average market gearing (adjusted for tax) of 50%. The risk free rate of return is 5% and the equity risk premium is 3%. The company's current accounting rate of return on new investment is 13% before tax. You may assume that debt betas are zero throughout.

The effective rate of company tax is 35%.

Required:

On the assumption that the property unbundling proceeds, prepare a report for consideration by senior management which should include the following:

- (a) A comparative statement showing the impact upon the statement of financial position and on the earnings per share on the assumption that the cash proceeds of the property sale are used:**
 - (i) To repay the debt, repayable in two years, in full and for reinvestment in non-current assets;**
 - (ii) To repay the debt, repayable in two years, in full and to finance a share repurchase at the current share price with the balance of the proceeds.** (13 marks)
- (b) An estimate of the weighted average cost of capital for the remaining business under both options on the assumption that the share price remains unchanged.** (10 marks)
- (c) An evaluation of the potential impact of each alternative on the market value of the firm (you are not required to calculate a revised market value for the firm).** (6 marks)

Professional marks will be awarded in question 2 for the clarity, presentation and persuasiveness of the report. (3 marks)

(32 marks)

Section B – TWO questions ONLY to be attempted

- 3** You are an importer of stone chippings for building purposes and you have entered into a fixed price contract for the delivery of 10,000 metric tonnes per month for the next six months.

The first delivery is due in one month's time.

Each tonne costs €220 under the fixed price contract and will be paid in Euros at the end of the month in question. Your domestic currency is the dollar and your supplier is in the Euro area. The current rate of exchange is Euro 0.8333 to the dollar. The quoted forward rates and the risk free interest rates in the dollar zone are as follows:

	1	2	3	4	5	6
Forward Rates (Euros per \$)	0.8326	0.8314	0.8302	0.8289	0.8278	0.8267
\$ short zero coupon yield curve	3.25%	3.45%	3.50%	3.52%	3.52%	3.52%

Required:

- (a) Estimate the forward exchange rate that would be fixed for a six month currency swap with monthly deliveries against the current order of 10,000 metric tonnes per month. (12 marks)
- (b) Outline the advantages and disadvantages of using a plain vanilla currency swap with monthly delivery compared with a strip of forward contracts. (8 marks)

(20 marks)

- 4 You have been appointed as deputy Chief Financial Officer to a large multinational pharmaceutical company with trading interests in 24 countries in sub-Saharan Africa, South America and the Indian sub-continent. Your company also has important trading links with the United States, Malaysia and Singapore. There have been a number of issues arising in the previous six months which have impacted upon the company's business interests.
- (i) Following an investigation you discover that commissions were paid to a senior official in one country to ensure that the local drug licensing agency concerned facilitated the acceptance of one of your principal revenue earning drugs for use within its national health service.
 - (ii) You have discovered that an agent of your firm, aware that the licensing agreement might be forthcoming, purchased several call option contracts on your company's equity.
 - (iii) A senior member of the firm's treasury team has been taking substantial positions in currency futures in order to protect the risk of loss on the translation of dollar assets into the domestic currency. Over the last 12 months significant profits have been made but the trades do not appear to have been properly authorised. You discover that a long position in 50, \$250,000 contracts is currently held but over the last four weeks the dollar has depreciated by 10% and all the signs are that it will depreciate considerably more over the next two months.
 - (iv) One drug company has managed to copy a novel drug that you have just released for the treatment of various forms of skin cancer. You have patent protection in the country concerned but your company has not been able to initiate proceedings through the local courts. Contacts with the trade officials at your embassy in the country concerned suggest that the government has made sure that the proceedings have not been allowed to proceed.

The company's chief financial officer has asked you to look into these issues and, with respect to (iv), any World Trade Organisation (WTO) agreements that might be relevant, and to advise her on how the company should proceed in each case.

Required:

Prepare a memorandum advising the Chief Financial Officer on the issues involved and recommending how she should, in each case and in the circumstances, proceed.

(20 marks)

5 Slow Fashions Ltd is considering the following series of investments for the current financial year 2009:

Project bid proposals (\$'000) for immediate investment with the first cash return assumed to follow in 12 months and at annual intervals thereafter.

Project	Now	2010	2011	2012	2013	2014	2015	NPV	IRR
P0801	-620	280	400	120				55	16%
P0802	-640	80	120	200	210	420	-30	69	13%
P0803	-240	120	120	60	10			20	15%
P0804	-1000	300	500	250	290			72	13%
P0805	-120	25	55	75	21			19	17%
P0806	-400	245	250					29	15%

There is no real option to delay any of these projects. All except project P0801, can be scaled down but not scaled up. P0801 is a potential fixed three-year contract to supply a supermarket chain and cannot be varied. The company has a limited capital budget of \$1.2 million and is concerned about the best way to allocate its capital to the projects listed. The company has a current cost of finance of 10% but it would take a year to establish further funding at that rate. Further funding for a short period could be arranged at a higher rate.

Required:

- (a) **Draft a capital investment plan with full supporting calculations justifying those projects which should be adopted giving:**
- (i) **The priorities for investment,**
 - (ii) **The net present value and internal rate of return of the plan; and**
 - (iii) **The net present value per dollar invested on the plan.** (12 marks)
- (b) **Estimate and advise upon the maximum interest rate which the company should be prepared to pay to finance investment in all of the remaining projects available to it.** (8 marks)

(20 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d (1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model	The FOREX modified Black-Scholes option pricing model
$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$ Where: $d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$ $d_2 = d_1 - s\sqrt{t}$	$c = e^{-rt} [F_0 N(d_1) - X N(d_2)]$ Or $p = e^{-rt} [X N(-d_2) - F_0 N(-d_1)]$ Where: $d_1 = \frac{\ln(F_0 / X) + s^2 T / 2}{s\sqrt{T}}$ and $d_2 = d_1 - s\sqrt{T}$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

Discount rate (r)

Periods

(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.941	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.305	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		<i>Discount rate (r)</i>									
<i>Periods</i>											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·49987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Thursday 10 December 2009

Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 9–13.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – BOTH questions are compulsory and MUST be attempted

- 1 Kodiak Company is a small software design business established four years ago. The company is owned by three directors who have relied upon external accounting services in the past. The company has grown quickly and the directors have appointed you as a financial consultant to advise on the value of the business under their ownership.

The directors have limited liability and the bank loan is secured against the general assets of the business. The directors have no outstanding guarantees on the company's debt.

The company's latest income statement and the extracted balances from the latest statement of financial position are as follows:

Income Statement	\$'000	Financial Position	\$000
Revenue	5,000	Opening non-current assets	1,200
Cost of Sales	3,000	Additions	66
Gross profit	2,000	Non-current assets (gross)	1,266
Other operating costs	1,877	Accumulated depreciation	367
Operating profit	123	Net book value	899
Interest on loan	74	Net current assets	270
Profit before tax	49	Loan	(990)
Income tax expense	15	Net Assets Employed	179
Profit for the period	34		

During the current year:

- (1) Depreciation is charged at 10% per annum on the year end non-current asset balance before accumulated depreciation, and is included in other operating costs in the income statement.
- (2) The investment in net working capital is expected to increase in line with the growth in gross profit.
- (3) Other operating costs consisted of:

	\$000
Variable component at 15% of sales	750
Fixed costs	1,000
Depreciation on non-current assets	127

- (4) Revenue and variable costs are projected to grow at 9% per annum and fixed costs are projected to grow at 6% per annum.
- (5) The company pays interest on its outstanding loan of 7.5% per annum and incurs tax on its profits at 30%, payable in the following year. The company does not pay dividends.
- (6) The net current assets reported in the statement of financial position contain \$50,000 of cash.

One of your first tasks is to prepare for the directors a forward cash flow projection for three years and to value the firm on the basis of its expected free cash flow to equity. In discussion with them you note the following:

- The company will not dispose of any of its non-current assets but will increase its investment in **new** non-current assets by 20% per annum. The company's depreciation policy matches the currently available tax write off for capital allowances. This straight-line write off policy is not likely to change.
- The directors will not take a dividend for the next three years but will then review the position taking into account the company's sustainable cash flow at that time.
- The level of the loan will be maintained at \$990,000 and, on the basis of the forward yield curve, interest rates are not expected to change.
- The directors have set a target rate of return on their equity of 10% per annum which they believe fairly represents the opportunity cost of their invested funds.

Required:

- (a) Prepare a three-year cash flow forecast for the business on the basis described above highlighting the free cash flow to equity in each year. (12 marks)
- (b) Estimate the value of the business based upon the expected free cash flow to equity and a terminal value based upon a sustainable growth rate of 3% per annum thereafter. (6 marks)
- (c) Advise the directors on the assumptions and the uncertainties within your valuation. (6 marks)
- (d) With reference to option pricing theory, advise the directors how limited liability may give a different value to the business from the value estimated in part (b) above. (4 marks)

(28 marks)

- 2 Anchorage Retail Company is a large high street and on-line retailer that has lost its position as the premier quality clothes, household goods and food chain in the European market. Five years previously there had been speculation that the company would be a takeover target for any one of a number of private equity firms. However, a newly appointed and flamboyant Chief Executive Officer, John Bear, initiated a major capital reconstruction and a highly aggressive turnaround strategy.

The reaction to that turnaround strategy was an improvement in the company's share price from \$3 to \$7 per share over the subsequent three years. The private equity firms who had been interested in acquiring the company were deterred for two principal reasons. First John Bear had a reputation for his aggressive style and his history of defending his companies against takeover. Second the share price of Anchorage had reached a record high.

In recent months a belief in the investment community had become widespread that the revival of the company's performance had more to do with the reorganisation of the firm's capital than the success of John Bear's turnaround strategy. John Bear insisted, however, that the improvements in the reported 'bottom line' reflected a sustainable improvement in the performance of the business. However, the recent recession in the European retail market following the 'credit crunch' led to a sharp reduction in Anchorage's share price reinforced by concerns in the financial markets that John Bear has become too dominant in the board of the company.

The most recent accounts for Anchorage Retail, in summary form, are as follows:

Anchorage Retail Company

	2009	2008		2009
	\$m	\$m		\$m
Income statement			Summary cash flow statement	
Sales Turnover	9,000	8,500		
Cost of Sales	5,500	5,250	Operating cash flow	1,610
Gross Profit	3,500	3,250	less interest	(110)
less other operating costs	2,250	2,220	less taxation	(270)
Operating profit	1,250	1,030	Free cash flow before reinvestment	1,230
Finance Costs	80	110	Dividend paid	(270)
Profit before tax	1,170	920	CAPEX	(740)
Income tax expense (at 30%)	310	270	Financing	(70)
Profit for the period	860	650	Net cash flow	150
	2009	2008		
	\$m	\$m		
Statement of financial position				
Assets				
Non-current assets	4,980	4,540		
Current assets	1,220	850		
Total assets	6,200	5,390		
Equity and Liabilities				
Ordinary share capital (25c)	400	425		
Share premium	230	200		
Capital redemption reserve	2,300	2,300		
Other reserves	(6,540)	(6,500)		
Retained earnings	5,990	5,400		
Dividends payable	(350)	(270)		
Total equity	2,030	1,555		
Non-current liabilities	1,900	1,865		
Current liabilities	2,270	1,970		
Total equity and liabilities	6,200	5,390		

The management of Polar Finance, a large private equity investment fund, has begun a review following the sale of a substantial part of its investment portfolio. It is now considering Anchorage as a potential target for acquisition. They have contacted you and asked if you would provide a brief report on the financial performance of Anchorage Retail and give an independent view on a bid the company is considering for the business. The suggested bid would be in the form of a cash offer of \$3.20 a share which would represent a 60¢ premium on the current share price. Reviewing the fund's existing business portfolio prior to acquisition you estimate that its asset beta is 0.285. Polar Finance has equity funds under management of \$1,125 million and a market based gearing ratio (debt as a proportion of total capital employed) of 0.85. This acquisition would be financed from additional cash resources and by additional borrowing of \$2.5 billion. It is expected that Anchorage's proportion of the total post-acquisition cash flows will be 20%. Polar Finance does not pay tax on its income.

During your investigations you discover the following:

1. The equity beta for Anchorage is 0.75. The current risk free rate is 5%. In order to estimate the rate of return on the market using the dividend growth model you note that the current dividend yield on a broadly based market index is 3.1% and the growth in GDP is 4% nominal. The growth of the firms in the index is fairly represented by growth in GDP.
2. Anchorage has a gearing ratio based upon market capitalisation of 24%. You estimate that its current cost of debt capital is 6.2%. You may assume that Anchorage's cost of finance has been constant over the last twelve months.

You may use year end financial position values when calculating performance ratios.

Required:

Prepare a report for Polar Finance:

- (a) **Outlining the principal risks that Polar Finance should consider when assessing an acquisition of this size.**
(6 marks)
- (b) **Summarising the performance of Anchorage in 2009 compared with 2008 on the basis of the EVA® for each year and using two other ratios you consider appropriate.**
(12 marks)
- (c) **Estimating the impact of this acquisition upon the required rate of return of equity investors in Polar Finance.**
(6 marks)
- (d) **Evaluating the argument that this company may have been systematically undervalued by the market and therefore a suitable target for acquisition.**
(4 marks)

Professional marks will be awarded for the appropriateness of the format and presentation of the report and the effectiveness with which its advice is communicated.
(4 marks)

(32 marks)

Section B – TWO questions ONLY to be attempted

- 3** Alaska Salvage is in discussion with potential lenders about financing an ambitious five-year project searching for lost gold in the central Atlantic. The company has had great success in the past with its various salvage operations and is now quoted on the London Alternative Investment Market. The company is currently financed by 120,000 equity shares trading at \$85 per share. It needs to borrow \$1.6 million and is concerned about the level of the fixed rates being suggested by the lenders. After lengthy discussions the lenders are prepared to offer finance against a mezzanine issue of fixed rate five-year notes with warrants attached. Each \$10,000 note, repayable at par, would carry a warrant for 100 equity shares at an exercise price of \$90 per share. The estimated volatility of the returns on the company's equity is 20% and the risk free rate of interest is 5%. The company does not pay dividends to its equity investors.

You may assume that the issue of these loan notes will not influence the current value of the firm's equity. The issue will be made at par.

Required:

- (a) Estimate, using Black-Scholes Option Pricing Model as appropriate, the current value of each warrant to the lender noting the assumptions that you have made in your valuation.** (10 marks)
- (b) Estimate the coupon rate that would be required by the lenders if they wanted a 13% rate of return on their investment.** (4 marks)
- (c) Discuss the advantages and disadvantages of issuing mezzanine debt in the situation outlined in the case.** (6 marks)

(20 marks)

- 4 You are the Chief Financial Officer of Moose Co. Moose Co is a manufacturer of cleaning equipment and has an international market for its products. Your company places a strong emphasis on innovation and design with patent protection across all its product range.

The company has two principal manufacturing centres, one in Europe which has been reduced in size in recent years because of high labour costs and the other in South East Asia. However, Moose Co's development has relied upon ready access to the debt market both in Europe and in South East Asia and the company is planning significant expansion with a new manufacturing and distribution centre in South America. Your company is highly profitable with strong cash flows although in the last two quarters there has been a downturn in sales in all markets as the global recession has begun to take effect.

Since August 2007, credit conditions have deteriorated across all of the major economies as banks have curtailed their lending following the down rating of US asset-backed securities. In 2008 and 2009 many banks recorded significant multibillion dollar losses as they attempted to sell off what had become known as 'toxic debt', leading to a further collapse in their value. In response many banks also attempted to repair their financial position by rights and other equity issues.

The founder and executive chairman of the company, Alan Bison, is planning a round of meetings with a number of investment banks in leading financial centres around the world to explore raising a \$350 million dollar loan for the new development. It has already been suggested that a loan of this size would need to be syndicated or alternatively raised through a bond issue.

In preparation for those meetings he has asked you to provide him with some briefing notes.

Required:

- (a) **Given conditions in the global debt market as described above, advise on the likely factors banks will consider in offering a loan of this size.** (7 marks)
- (b) **Assess the relative advantages of loan syndication versus a bond issue to Moose Co.** (7 marks)
- (c) **Assess the relative advantages and disadvantages of entering into a capital investment of this scale at this stage of the global economic cycle.** (6 marks)

(20 marks)

- 5 To finance capital investment in its domestic market, the Katmai Company raised \$150 million through the issue of 12-year floating rate notes at 120 basis points over LIBOR, interest payable at six month intervals. Following a review of the current yield curve, the company's Chief Financial Officer has become concerned about the potential impact of rising LIBOR on the firm's future cash flows. The loan now has 10 years to maturity. The CFO asks you, his deputy, to examine the choices that are now available to the firm and to recommend the best course of action. She comments that a swap is an obvious choice but that she would appreciate a briefing on the advantages and disadvantages of the alternative approaches to managing the company's interest rate risk and an estimate of the six monthly Value at Risk (VaR) if nothing is done. As part of your investigation you note that 10-year swap rates are quoted at 5.25–5.40. In estimating the VaR you note that the firm has a policy of 95% confidence level on its exposure to non-core risk and that the annual volatility of LIBOR is currently 150 basis points.

Required:

- (a) **Evaluate the alternative choices the company has for managing its interest rate exposure and recommend, with justification, the course of action the company should follow.** (9 marks)
- (b) **Estimate the six-monthly interest rate and the effective annual rate payable if a vanilla interest rate swap is agreed.** (5 marks)
- (c) **Estimate the six monthly Value at Risk on the interest rate exposure associated with this borrowing and comment upon the interpretation of the result.** (6 marks)

(20 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d (1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model	The FOREX modified Black-Scholes option pricing model
$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$ Where: $d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$ $d_2 = d_1 - s\sqrt{t}$	$c = e^{-rt} [F_0 N(d_1) - X N(d_2)]$ Or $p = e^{-rt} [X N(-d_2) - F_0 N(-d_1)]$ Where: $d_1 = \frac{\ln(F_0 / X) + s^2 T / 2}{s\sqrt{T}}$ and $d_2 = d_1 - s\sqrt{T}$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.941	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.305	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Thursday 10 June 2010

Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 6–10.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – BOTH questions are compulsory and MUST be attempted

- 1 The Seal Island Nuclear Power Company has received initial planning consent for an Advanced Boiling Water Reactor. This project is one of a number that has been commissioned by the Government of Roseland to help solve the energy needs of its expanding population of 60 million and meet its treaty obligations by cutting CO₂ emissions to 50% of their 2010 levels by 2030.

The project proposal is now moving to the detailed planning stage which will include a full investment appraisal within the financial plan. The financial plan so far developed has been based upon experience of this reactor design in Japan, the US and South Korea.

The core macro economic assumptions are that Roseland GDP will grow at an annual rate of 4% (nominal) and inflation will be maintained at the 2% target set by the Government.

The construction programme is expected to cost \$1 billion over three years, with construction commencing in January 2012. These capital expenditures have been projected, including expected future cost increases, as follows:

Year end	2012	2013	2014
Construction costs (\$ million)	300	600	100

Generation of electricity will commence in 2015 and the annual operating surplus in cash terms is expected to be \$100 million per annum (at 1 January 2015 price and cost levels). This value has been well validated by preliminary studies and includes the cost of fuel reprocessing, ongoing maintenance and systems replacement as well as the continuing operating costs of running the plant. The operating surplus is expected to rise in line with nominal GDP growth. The plant is expected to have an operating life of 30 years.

Decommissioning costs at the end of the project have been estimated at \$600 million at current (2012) costs. Decommissioning costs are expected to rise in line with nominal GDP growth.

The company's nominal cost of capital is 10% per annum. All estimates, unless otherwise stated, are at 1 January 2012 price and cost levels.

Required:

Produce a preliminary briefing note which, on the basis of the above information, includes:

- (i) An estimate of the net present value for this project as at the commencement of construction in 2012. (11 marks)
- (ii) A discussion of the principal uncertainties associated with this project. (7 marks)
- (iii) A sensitivity of the project's net present value (in percentage and in \$), to changes in the construction cost, the annual operating surplus and the decommissioning cost. (Assume that the increase in construction costs would be proportional to the initial investment for each year.) (6 marks)
- (iv) An explanation of how simulations, such as the Monte Carlo simulation, could be used to assess the volatility of the net present value of this project. (4 marks)

Note: the formula for an annuity discounted at an annual rate (i) and where cash flows are growing at an annual rate (g) is as follows:

$$A_n = \left[\frac{1 - \left(\frac{1+g}{1+i} \right)^n}{i-g} \right] (1+g)$$

(28 marks)

- 2 AggroChem Co is undertaking a due diligence investigation of LeverChem Co and is reviewing the potential bid price for an acquisition. You have been appointed as a consultant to advise the company's management on the financial aspects of the bid.

AggroChem is a fully listed company financed wholly by equity. LeverChem is listed on an alternative investment market. Both companies have been trading for over 10 years and have shown strong levels of profitability recently. However, both companies' shares are thinly traded. It is thought that the current market value of LeverChem's shares at $33\frac{1}{3}\%$ higher than the book value is accurate, but it is felt that AggroChem shares are not quoted accurately by the market.

The following information is taken from the financial statements of both companies at the start of the current year:

	AggroChem \$'000	LeverChem \$'000
Assets less current liabilities	4,400	4,200
Capital Employed		
Equity	4,400	1,200
5-year floating rate loan at yield rate plus 3%		3,000
Total capital employed	4,400	4,200
Net operating profit after tax (NOPAT)	580	430
Net amount retained for reinvestment in assets	180	150

It can be assumed that the retained earnings for both companies are equal to the net reinvestment in assets.

The assets of both companies are stated at fair value. Discussions with the AtReast Bank have led to an agreement that the floating rate loan to LeverChem can be transferred to the combined business on the same terms. The current yield rate is 5% and the current equity risk premium is 6%. It can be assumed that the risk free rate of return is equivalent to the yield rate. AggroChem's beta has been estimated to be 1.26.

AggroChem Co wants to use the Black-Scholes option pricing (BSOP) model to assess the value of the combined business and the maximum premium payable to LeverChem's shareholders. AggroChem has conducted a review of the volatility of the NOPAT values of both companies since both were formed and has estimated that the volatility of the combined business assets, if the acquisition were to go ahead, would be 35%. The exercise price should be calculated as the present value of a discount (zero-coupon) bond with an identical yield and term to maturity of the current bond.

Required:

Prepare a report for the management of AggroChem on the valuation of the combined business following acquisition and the maximum premium payable to the shareholders of LeverChem. Your report should:

- Using the free cash flow model, estimate the market value of equity for AggroChem Co, explaining any assumptions made.** (9 marks)
- Explain the circumstances in which the Black-Scholes option pricing (BSOP) model could be used to assess the value of a company, including the data required for the variables used in the model.** (5 marks)
- Using the BSOP methodology, estimate the maximum price and premium AggroChem may pay for LeverChem.** (9 marks)
- Discuss the appropriateness of the method used in part (iii) above, by considering whether the BSOP model can provide a meaningful value for a company.** (5 marks)

Professional marks will be awarded in question 2 for the clarity and presentation of the report. (4 marks)

(32 marks)

Section B – TWO questions ONLY to be attempted

- 3 The finance division of GoSlo Motor Corporation has made a number of loans to customers with a current pool value of \$200 million. The loans have an average term to maturity of four years. The loans generate a steady income to the business of 10·5% per annum. The company will use 95% of the loan's pool as collateral for a collateralised loan obligation structured as follows:

- 80% of the collateral value to support a tranche of A-rated floating rate loan notes offering investors LIBOR plus 140 basis points.
- 10% of the collateral value to support a tranche of B-rated fixed rate loan notes offering investors 11%.
- 10% of the collateral value to support a tranche as subordinated certificates (unrated).

In order to minimise interest rate risk, the company has decided to enter into a fixed for variable rate swap on the A-rated floating rate notes exchanging LIBOR for 8·5%.

Service charges of \$240,000 per annum will be charged for administering the income receivable from the loans.

You may ignore prepayment risk.

Required:

- (a) Calculate the expected returns of the investments in each of the three tranches described above. Estimate the sensitivity of the subordinated certificates to a reduction of 1% in the returns generated by the pool. (10 marks)
- (b) Explain the purpose and the methods of credit enhancement that can be employed on a securitisation such as this scheme. (4 marks)
- (c) Discuss the risks inherent to the investors in a scheme such as this. (6 marks)

(20 marks)

- 4 The MandM Company, a large listed company, has two divisions. The first, the MoneyMint division produces coins and notes for the national exchequer and generates 80% of the company's revenues. The second, the LunarMint division, manufactures a brand of sweets which are very popular with traders in the financial markets. The company is considering disposing of its LunarMint division. The LunarMint business is no longer viewed as part of the core business of the MandM Company. The Chief Executive Officer commented that he could never understand why the company entered into sweet-making in the first place. The LunarMint business is profitable and low risk, but has not been a high priority for investment.

Required:

Outline the issues that should be considered when disposing of the LunarMint division noting the risks that might be involved.

(20 marks)

- 5 You are the financial manager of Multidrop (Group) a European based company which has subsidiary businesses in North America, Europe, and Singapore. It also has foreign currency balances outstanding with two non-group companies in the UK and Malaysia. Last year the transaction costs of *ad-hoc* settlements both within the group and with non-group companies were significant and this year you have reached agreement with the non-group companies to enter into a netting agreement to clear indebtedness with the minimum of currency flows. It has been agreed that Multidrop (Europe) will be the principal in the netting arrangement and that all settlements will be made in Euros at the prevailing spot rate.

The summarised list of year end indebtedness is as follows:

Owed by:	Owed to:	
Multidrop (Europe)	Multidrop (US)	US\$6.4 million
Multidrop (Singapore)	Multidrop (Europe)	S\$16 million
Alposong (Malaysia)	Multidrop (US)	US\$5.4 million
Multidrop (US)	Multidrop (Europe)	€8.2 million
Multidrop (Singapore)	Multidrop (US)	US\$5.0 million
Multidrop (Singapore)	Alposong (Malaysia)	Rm25 million
Alposong (Malaysia)	NewRing (UK)	£2.2 million
NewRing (UK)	Multidrop (Singapore)	S\$4.0 million
Multidrop (Europe)	Alposong (Malaysia)	Rm8.3 million

Currency cross rates (mid-market) are as follows:

Currency		UK £	US \$	Euro	Sing \$	Rm
1 UK £	=	1.0000	1.4601	1.0653	2.1956	5.3128
1 US \$	=	0.6849	1.0000	0.7296	1.5088	3.6435
1 Euro	=	0.9387	1.3706	1.0000	2.0649	4.9901
1 Sing \$	=	0.4555	0.6628	0.4843	1.0000	2.4150
1 Rm	=	0.1882	0.2745	0.2004	0.4141	1.0000

You may assume settlement will be at the mid-market rates quoted.

Required:

- (a) Calculate the inter group and inter-company currency transfers that will be required for settlement by Multidrop (Europe). (12 marks)
- (b) Discuss the advantages and disadvantages of netting arrangements with both group and non-group companies. (8 marks)

(20 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d (1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model	The FOREX modified Black-Scholes option pricing model
$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$ Where: $d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$ $d_2 = d_1 - s\sqrt{t}$	$c = e^{-rt} [F_0 N(d_1) - X N(d_2)]$ Or $p = e^{-rt} [X N(-d_2) - F_0 N(-d_1)]$ Where: $d_1 = \frac{\ln(F_0 / X) + s^2 T / 2}{s\sqrt{T}}$ and $d_2 = d_1 - s\sqrt{T}$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.941	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.305	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

		Discount rate (r)									
Periods											
(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Thursday 9 December 2010

Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 9–13.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – BOTH questions are compulsory and MUST be attempted

- 1 Doric Co, a listed company, has two manufacturing divisions: parts and fridges. It has been manufacturing parts for domestic refrigeration and air conditioning systems for a number of years, which it sells to producers of fridges and air conditioners worldwide. It also sells around 30% of the parts it manufactures to its fridge production division. It started producing and selling its own brand of fridges a few years ago. After limited initial success, competition in the fridge market became very tough and revenue and profits have been declining. Without further investment there are currently few growth prospects in either the parts or the fridge divisions. Doric Co borrowed heavily to finance the development and launch of its fridges, and has now reached its maximum overdraft limit. The markets have taken a pessimistic view of the company and its share price has declined to 50c per share from a high of \$2.83 per share around three years ago.

Extracts from the most recent financial statements:

Financial position as at 30 November 2010

Assets

Non-Current Assets

Land and buildings

Machinery and equipment

\$m

70

50

120

Current Assets

Inventory

Receivables

180

40

220

Total Assets

340

Equity and Liabilities

Share capital (40c per share par value)

Reserves

40

20

60

Non-Current Liabilities

7% Unsecured bonds 2020

Other unsecured loans (currently $5\frac{1}{3}\%$ interest)

120

30

150

Current Liabilities

Payables

Bank overdraft (currently 10% interest)

70

60

130

Total Liabilities and capital

340

Income for the year ended 30 November 2010

Sales revenue: Parts division
Fridge division

Costs prior to depreciation, interest payments and tax:

Parts division
Fridge division

Tax allowable depreciation

Finance cost (interest)

Tax

Loss

\$m

170

340

(120)

(370)

(18)

(16)

Nil

(14)

A survey from the refrigeration and air conditioning parts market has indicated that there is potential for Doric Co to manufacture parts for mobile refrigeration units used in cargo planes and containers. If this venture goes ahead then the parts division before-tax profits are expected to grow by 5% per year. The proposed venture would need an initial one-off investment of \$50 million.

Suggested proposals

The Board of Directors has arranged for a meeting to discuss how to proceed and is considering each of the following proposals:

1. To cease trading and close down the company entirely.
2. To undertake corporate restructuring in order to reduce the level of debt and obtain the additional capital investment required to continue current operations.
3. To close the fridge division and continue the parts division through a leveraged management buy-out, involving some executive directors and managers from the parts division. The new company will then pursue its original parts business as well as the development of the parts for mobile refrigeration business, described above. All the current and long-term liabilities will be initially repaid using the proceeds from the sale of the fridge division. The finance raised from the management buy-out will pay for any remaining liabilities, the additional capital investment required to continue operations and re-purchase the shares at a premium of 20%.

The following information has been provided for each proposal:

Cease trading

Estimated realisable values of assets not sold as going concern are:

	\$ million
Land and buildings	60
Machinery and equipment	40
Inventory	90
Receivables	20

Corporate restructuring

The existing ordinary shares will be cancelled and ordinary shareholders will be issued with 40 million new \$1 ordinary shares in exchange for a cash payment at par. The existing unsecured bonds will be cancelled and replaced with 270 million of \$1 ordinary shares. The bond holders will contribute \$90 million in cash. All the shares will be listed and traded. The bank overdraft will be converted into a secured ten-year loan with a fixed annual interest rate of 7%. The other unsecured loans will be repaid. In addition to this, the directors of the restructured company will get 4 million \$1 share options for an exercise price of \$1.10, which will expire in four years.

An additional one-off capital investment of \$80 million in machinery and equipment is necessary to increase sales revenue for both divisions by 7%, with no change to the costs. After the one-off 7% growth, sales will continue at the new level for the foreseeable future.

It is expected that the Doric's cost of capital rate will reduce by 550 basis points following the restructuring from the current rate.

Management buy-out

The parts division is half the size of the fridge division in terms of the assets and liabilities attributable to it. If the management buy-out proposal is chosen, a pro rata additional capital investment will be made to machinery and equipment on a one-off basis to increase sales revenue of the parts division by 7%. Sales revenue will then continue at the new level for the foreseeable future.

All liabilities categories have equal claim for repayment against the company's assets.

It is expected that Doric's cost of capital rate will decrease by 100 basis points following the management buy-out from the current rate.

The following additional information has been provided:

Redundancy and other costs will be approximately \$54 million if the whole company is closed, and pro rata for individual divisions that are closed. These costs have priority for payment before any other liabilities in case of closure. The taxation effects relating to this may be ignored.

Corporation tax on profits is 20% and losses cannot be carried forward for tax purposes. Assume that tax is payable in the year incurred.

All the non-current assets, including land and buildings, are eligible for tax allowable depreciation of 15% annually on the book values. The annual reinvestment needed to keep operations at their current levels is roughly equivalent to the tax allowable depreciation. The \$50 million investment in the mobile refrigeration business is not eligible for any tax allowable depreciation.

Doric's current cost of capital is 12%.

Required:

Prepare a report for the Board of Directors, evaluating the financial and non-financial impact of all the three proposals to Doric Co's main stakeholder groups, that includes:

- (i) An estimate of the return the debt holders and shareholders would receive in the event that Doric Co ceases trading and is closed down.** (3 marks)
- (ii) An estimate of the income position and the value of Doric Co in the event that the restructuring proposal is selected. State any assumptions made.** (8 marks)
- (iii) An estimate of the amount of additional finance needed and the value of Doric Co if the management buy-out proposal is selected. State any assumptions made.** (8 marks)
- (iv) A discussion of the impact of each proposal on the existing shareholders, the unsecured bond holders, and the executive directors and managers involved in the management buy-out. Suggest which proposal is likely to be selected.** (12 marks)

Professional marks will be awarded in question 1 for the appropriateness and format of the report. (4 marks)

(35 marks)

- 2 Fubuki Co, an unlisted company based in Megaera, has been manufacturing electrical parts used in mobility vehicles for people with disabilities and the elderly, for many years. These parts are exported to various manufacturers worldwide but at present there are no local manufacturers of mobility vehicles in Megaera. Retailers in Megaera normally import mobility vehicles and sell them at an average price of \$4,000 each. Fubuki Co wants to manufacture mobility vehicles locally and believes that it can sell vehicles of equivalent quality locally at a discount of 37·5% to the current average retail price.

Although this is a completely new venture for Fubuki Co, it will be in addition to the company's core business. Fubuki Co's directors expect to develop the project for a period of four years and then sell it for \$16 million to a private equity firm. Megaera's government has been positive about the venture and has offered Fubuki Co a subsidised loan of up to 80% of the investment funds required, at a rate of 200 basis points below Fubuki Co's borrowing rate. Currently Fubuki Co can borrow at 300 basis points above the five-year government debt yield rate.

A feasibility study commissioned by the directors, at a cost of \$250,000, has produced the following information.

1. Initial cost of acquiring suitable premises will be \$11 million, and plant and machinery used in the manufacture will cost \$3 million. Acquiring the premises and installing the machinery is a quick process and manufacturing can commence almost immediately.
2. It is expected that in the first year 1,300 units will be manufactured and sold. Unit sales will grow by 40% in each of the next two years before falling to an annual growth rate of 5% for the final year. After the first year the selling price per unit is expected to increase by 3% per year.
3. In the first year, it is estimated that the total direct material, labour and variable overheads costs will be \$1,200 per unit produced. After the first year, the direct costs are expected to increase by an annual inflation rate of 8%.
4. Annual fixed overhead costs would be \$2·5 million of which 60% are centrally allocated overheads. The fixed overhead costs will increase by 5% per year after the first year.
5. Fubuki Co will need to make working capital available of 15% of the anticipated sales revenue for the year, at the beginning of each year. The working capital is expected to be released at the end of the fourth year when the project is sold.

Fubuki Co's tax rate is 25% per year on taxable profits. Tax is payable in the same year as when the profits are earned. Tax allowable depreciation is available on the plant and machinery on a straight-line basis. It is anticipated that the value attributable to the plant and machinery after four years is \$400,000 of the price at which the project is sold. No tax allowable depreciation is available on the premises.

Fubuki Co uses 8% as its discount rate for new projects but feels that this rate may not be appropriate for this new type of investment. It intends to raise the full amount of funds through debt finance and take advantage of the government's offer of a subsidised loan. Issue costs are 4% of the gross finance required. It can be assumed that the debt capacity available to the company is equivalent to the actual amount of debt finance raised for the project.

Although no other companies produce mobility vehicles in Megaera, Haizum Co, a listed company, produces electrical-powered vehicles using similar technology to that required for the mobility vehicles. Haizum Co's cost of equity is estimated to be 14% and it pays tax at 28%. Haizum Co has 15 million shares in issue trading at \$2·53 each and \$40 million bonds trading at \$94·88 per \$100. The five-year government debt yield is currently estimated at 4·5% and the market risk premium at 4%.

Required:

(a) Evaluate, on financial grounds, whether Fubuki Co should proceed with the project. (17 marks)

(b) Discuss the appropriateness of the evaluation method used and explain any assumptions made in part (a) above. (8 marks)

(25 marks)

Section B – TWO questions ONLY to be attempted

- 3 The treasury division of Marengo Co, a large quoted company, holds equity investments in various companies around the world. One of the investments is in Arion Co, in which Marengo holds 200,000 shares, which is around 2% of the total number of Arion Co's shares traded on the stock market. Over the past year, due to the general strength in the equity markets following optimistic predictions of the performance of world economies, Marengo's investments have performed well. However, there is some concern that the share price of Arion Co may fall in the coming two months due to uncertainty in its markets. It is expected that any fall in share prices will be reversed following this period of uncertainty.

The treasury division managers in Marengo, Wenyu, Lola and Sam, held a meeting to discuss what to do with the investment in Arion Co and they each made a different suggestion as follows:

1. Wenyu was of the opinion that Marengo's shareholders would benefit most if no action were taken. He argued that the courses of action proposed by Lola and Sam, below, would result in extra costs and possibly increase the risk to Marengo Co.
2. Lola proposed that Arion Co's shares should be sold in order to eliminate the risk of a fall in the share price.
3. Sam suggested that the investment should be hedged using an appropriate derivative product.

Although no exchange-traded derivative products exist on Arion Co's shares, a bank has offered over-the-counter (OTC) option contracts at an exercise price of 350 cents per share in a contract size of 1,000 shares each, for the appropriate time period. Arion Co's current share price is 340 cents per share, although the volatility of the share prices could be as high as 40%.

It can be assumed that Arion Co will not pay any dividends in the coming few months and that the appropriate inter-bank lending rate will be 4% over that period.

Required:

- (a) Estimate the number of OTC put option contracts that Marengo Co will need to hedge against any adverse movement in Arion Co's share price. Provide a brief explanation of your answer.

Note: You may assume that the delta of a put option is equivalent to $N(-d_1)$ (7 marks)

- (b) Discuss possible reasons for the suggestions made by each of the three managers. (13 marks)

(20 marks)

- 4 Lamri Co (Lamri), a listed company, is expecting sales revenue to grow to \$80 million next year, which is an increase of 20% from the current year. The operating profit margin for next year is forecast to be the same as this year at 30% of sales revenue. In addition to these profits, Lamri receives 75% of the after-tax profits from one of its wholly owned foreign subsidiaries – Magnolia Co (Magnolia), as dividends. However, its second wholly owned foreign subsidiary – Strymon Co (Strymon) does not pay dividends.

Lamri is due to pay dividends of \$7.5 million shortly and has maintained a steady 8% annual growth rate in dividends over the past few years. The company has grown rapidly in the last few years as a result of investment in key projects and this is likely to continue.

For the coming year it is expected that Lamri will require the following capital investment.

1. An investment equivalent to the amount of depreciation to keep its non-current asset base at the present productive capacity. Lamri charges depreciation of 25% on a straight-line basis on its non-current assets of \$15 million. This charge has been included when calculating the operating profit amount.
2. A 25% investment in additional non-current assets for every \$1 increase in sales revenue.
3. \$4.5 million additional investment in non-current assets for a new project.

Lamri also requires a 15% investment in working capital for every \$1 increase in sales revenue.

Strymon produces specialist components solely for Magnolia to assemble into finished goods. Strymon will produce 300,000 specialist components at \$12 variable cost per unit and will incur fixed costs of \$2.1 million for the coming year. It will then transfer the components to Magnolia at full cost price, where they will be assembled at a cost of \$8 per unit and sold for \$50 per unit. Magnolia will incur additional fixed costs of \$1.5 million in the assembly process.

Tax-Ethic (TE) is a charitable organisation devoted to reducing tax avoidance schemes by companies operating in poor countries around the world. TE has petitioned Lamri's Board of Directors to reconsider Strymon's policy of transferring goods at full cost. TE suggests that the policy could be changed to cost plus 40% mark-up. If Lamri changes Strymon's policy, it is expected that Strymon would be asked to remit 75% of its after-tax profits as dividends to Lamri.

Other Information

1. Lamri's outstanding non-current liabilities of \$35 million, on which it pays interest of 8% per year, and its 30 million \$1 issued equity capital will not change for the coming year.
2. Lamri's, Magnolia's and Strymon's profits are taxed at 28%, 22% and 42% respectively. A withholding tax of 10% is deducted from any dividends remitted from Strymon.
3. The tax authorities where Lamri is based charge tax on profits made by subsidiary companies but give full credit for tax already paid by overseas subsidiaries.
4. All costs and revenues are in \$ equivalent amounts and exchange rate fluctuations can be ignored.

Required:

- (a) Calculate Lamri's dividend capacity for the coming year prior to implementing TE's proposal and after implementing the proposal. (14 marks)
- (b) Comment on the impact of implementing TE's proposal and suggest possible actions Lamri may take as a result. (6 marks)

(20 marks)

- 5 Prospice Mentis University (PMU) is a prestigious private institution and a member of the Holly League, which is made up of universities based in Rosinante and renowned worldwide as being of the highest quality. Universities in Rosinante have benefited particularly from students coming from Kantaka, and PMU has been no exception. However, PMU has recognised that Kantaka has a large population of able students who cannot afford to study overseas. Therefore it wants to investigate how it can offer some of its most popular degree programmes in Kantaka, where students will be able to study at a significantly lower cost. It is considering whether to enter into a joint venture with a local institution or to independently set up its own university site in Kantaka.

Offering courses overseas would be a first from a Holly League institution and indeed from any academic institution based in Rosinante. However, there have been less renowned academic institutions from other countries which have formed joint ventures with small private institutions in Kantaka to deliver degree programmes. These have been of low quality and are not held in high regard by the population or the government of Kantaka.

In Kantaka, government run universities and a handful of large private academic institutions, none of which have entered into joint ventures, are held in high regard. However, the demand for places in these institutions far outstrips the supply of places and many students are forced to go to the smaller private institutions or to study overseas if they can afford it.

After an initial investigation the following points have come to light:

1. The Kantaka government is keen to attract foreign direct investment (FDI) and offer tax concessions to businesses which bring investment funds into the country. It is likely that PMU would need to borrow a substantial amount of money if it were to set up independently. However, the investment funds required would be considerably smaller if it went into a joint venture.
2. Given the past experiences of poor quality education offered by joint ventures between small local private institutions and overseas institutions, the Kantaka government has been reluctant to approve degrees from such institutions. Also the government has not allowed graduates from these institutions to work in national or local government, or in nationalised organisations.
3. Over the past two years the Kantaka currency has depreciated against other currencies, but economic commentators believe that this may not continue for much longer.
4. A large proportion of PMU's academic success is due to innovative teaching and learning methods, and high quality research. The teaching and learning methods used in Kantaka's educational institutions are very different. Apart from the larger private and government run universities, little academic research is undertaken elsewhere in Kantaka's education sector.

Required:

Discuss the benefits and disadvantages of PMU entering into a joint venture instead of setting up independently in Kantaka. As part of your discussion, consider how the disadvantages can be mitigated and the additional information PMU needs in order to make its decision.

(20 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d (1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model	The FOREX modified Black-Scholes option pricing model
$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$ Where: $d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$ $d_2 = d_1 - s\sqrt{t}$	$c = e^{-rt} [F_0 N(d_1) - X N(d_2)]$ Or $p = e^{-rt} [X N(-d_2) - F_0 N(-d_1)]$ Where: $d_1 = \frac{\ln(F_0 / X) + s^2 T / 2}{s\sqrt{T}}$ and $d_2 = d_1 - s\sqrt{T}$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

Discount rate (r)

Periods

(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.941	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.305	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Thursday 9 June 2011

Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 8–12.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – BOTH questions are compulsory and MUST be attempted

- 1 Pursuit Co, a listed company which manufactures electronic components, is interested in acquiring Fodder Co, an unlisted company involved in the development of sophisticated but high risk electronic products. The owners of Fodder Co are a consortium of private equity investors who have been looking for a suitable buyer for their company for some time. Pursuit Co estimates that a payment of the equity value plus a 25% premium would be sufficient to secure the purchase of Fodder Co. Pursuit Co would also pay off any outstanding debt that Fodder Co owed. Pursuit Co wishes to acquire Fodder Co using a combination of debt finance and its cash reserves of \$20 million, such that the capital structure of the combined company remains at Pursuit Co's current capital structure level.

Information on Pursuit Co and Fodder Co

Pursuit Co

Pursuit Co has a market debt to equity ratio of 50:50 and an equity beta of 1.18. Currently Pursuit Co has a total firm value (market value of debt and equity combined) of \$140 million.

Fodder Co, Income Statement Extracts

Year Ended	31 May 2011	31 May 2010	31 May 2009	31 May 2008
All amounts are in \$'000				
Sales revenue	16,146	15,229	14,491	13,559
Operating profit (after operating costs and tax allowable depreciation)	5,169	5,074	4,243	4,530
Net interest costs	489	473	462	458
Profit before tax	4,680	4,601	3,781	4,072
Taxation (28%)	1,310	1,288	1,059	1,140
After tax profit	3,370	3,313	2,722	2,932
Dividends	123	115	108	101
Retained earnings	3,247	3,198	2,614	2,831

Fodder Co has a market debt to equity ratio of 10:90 and an estimated equity beta of 1.53. It can be assumed that its tax allowable depreciation is equivalent to the amount of investment needed to maintain current operational levels. However, Fodder Co will require an additional investment in assets of 22c per \$1 increase in sales revenue, for the next four years. It is anticipated that Fodder Co will pay interest at 9% on its future borrowings.

For the next four years, Fodder Co's sales revenue will grow at the same average rate as the previous years. After the forecasted four-year period, the growth rate of its free cash flows will be half the initial forecast sales revenue growth rate for the foreseeable future.

Information about the combined company

Following the acquisition, it is expected that the combined company's sales revenue will be \$51,952,000 in the first year, and its profit margin on sales will be 30% for the foreseeable future. After the first year the growth rate in sales revenue will be 5.8% per year for the following three years. Following the acquisition, it is expected that the combined company will pay annual interest at 6.4% on future borrowings.

The combined company will require additional investment in assets of \$513,000 in the first year and then 18c per \$1 increase in sales revenue for the next three years. It is anticipated that after the forecasted four-year period, its free cash flow growth rate will be half the sales revenue growth rate.

It can be assumed that the asset beta of the combined company is the weighted average of the individual companies' asset betas, weighted in proportion of the individual companies' market value.

Other information

The current annual government base rate is 4.5% and the market risk premium is estimated at 6% per year. The relevant annual tax rate applicable to all the companies is 28%.

SGF Co's interest in Pursuit Co

There have been rumours of a potential bid by SGF Co to acquire Pursuit Co. Some financial press reports have suggested that this is because Pursuit Co's share price has fallen recently. SGF Co is in a similar line of business as Pursuit Co and until a couple of years ago, SGF Co was the smaller company. However, a successful performance has resulted in its share price rising, and SGF Co is now the larger company.

The rumours of SGF Co's interest have raised doubts about Pursuit Co's ability to acquire Fodder Co. Although SGF Co has made no formal bid yet, Pursuit Co's board is keen to reduce the possibility of such a bid. The Chief Financial Officer has suggested that the most effective way to reduce the possibility of a takeover would be to distribute the \$20 million in its cash reserves to its shareholders in the form of a special dividend. Fodder Co would then be purchased using debt finance. He conceded that this would increase Pursuit Co's gearing level but suggested it may increase the company's share price and make Pursuit Co less appealing to SGF Co.

Required:

Prepare a report to the Board of Directors of Pursuit Co that

- (i) Evaluates whether the acquisition of Fodder Co would be beneficial to Pursuit Co and its shareholders. The free cash flow to firm method should be used to estimate the values of Fodder Co and the combined company assuming that the combined company's capital structure stays the same as that of Pursuit Co's current capital structure. Include all relevant calculations;** (16 marks)
- (ii) Discusses the limitations of the estimated valuations in part (i) above;** (4 marks)
- (iii) Estimates the amount of debt finance needed, in addition to the cash reserves, to acquire Fodder Co and concludes whether Pursuit Co's current capital structure can be maintained;** (3 marks)
- (iv) Explains the implications of a change in the capital structure of the combined company, to the valuation method used in part (i) and how the issue can be resolved;** (4 marks)
- (v) Assesses whether the Chief Financial Officer's recommendation would provide a suitable defence against a bid from SGF Co and would be a viable option for Pursuit Co.** (5 marks)

Professional marks will be awarded in question 1 for the format, structure and presentation of the report.

(4 marks)

(36 marks)

- 2 Casasophia Co, based in a European country that uses the Euro (€), constructs and maintains advanced energy efficient commercial properties around the world. It has just completed a major project in the USA and is due to receive the final payment of US\$20 million in four months.

Casasophia Co is planning to commence a major construction and maintenance project in Mazabia, a small African country, in six months' time. This government-owned project is expected to last for three years during which time Casasophia Co will complete the construction of state-of-the-art energy efficient properties and provide training to a local Mazabian company in maintaining the properties. The carbon-neutral status of the building project has attracted some grant funding from the European Union and these funds will be provided to the Mazabian government in Mazabian Shillings (MShs).

Casasophia Co intends to finance the project using the US\$20 million it is due to receive and borrow the rest through a € loan. It is intended that the US\$ receipts will be converted into € and invested in short-dated treasury bills until they are required. These funds plus the loan will be converted into MShs on the date required, at the spot rate at that time.

Mazabia's government requires Casasophia Co to deposit the MShs2.64 billion it needs for the project, with Mazabia's central bank, at the commencement of the project. In return, Casasophia Co will receive a fixed sum of MShs1.5 billion after tax, at the end of each year for a period of three years. Neither of these amounts is subject to inflationary increases. The relevant risk adjusted discount rate for the project is assumed to be 12%.

Financial Information

Exchange Rates available to Casasophia

	Per €1	Per €1
Spot	US\$1.3585–US\$1.3618	MShs116–MShs128
4-month forward	US\$1.3588–US\$1.3623	Not available

Currency Futures (Contract size €125,000, Quotation: US\$ per €1)

2-month expiry	1.3633
5-month expiry	1.3698

Currency Options (Contract size €125,000, Exercise price quotation: US\$ per €1, cents per Euro)

Exercise price	Calls		Puts	
	2-month expiry	5-month expiry	2-month expiry	5-month expiry
1.36	2.35	2.80	2.47	2.98
1.38	1.88	2.23	4.23	4.64

Casasophia Co Local Government Base Rate	2.20%
Mazabia Government Base Rate	10.80%
Yield on short-dated Euro Treasury Bills (assume 360-day year)	1.80%

Mazabia's current annual inflation rate is 9.7% and is expected to remain at this level for the next six months. However, after that, there is considerable uncertainty about the future and the annual level of inflation could be anywhere between 5% and 15% for the next few years. The country where Casasophia Co is based is expected to have a stable level of inflation at 1.2% per year for the foreseeable future. A local bank in Mazabia has offered Casasophia Co the opportunity to swap the annual income of MShs1.5 billion receivable in each of the next three years for Euros, at the estimated annual MShs/€ forward rates based on the current government base rates.

Required:

- (a) Advise Casasophia Co on, and recommend, an appropriate hedging strategy for the US\$ income it is due to receive in four months. Include all relevant calculations. (15 marks)
- (b) Provide a reasoned estimate of the additional amount of loan finance Casasophia Co needs to obtain to undertake the project in Mazabia in six months. (5 marks)
- (c) Given that Casasophia Co agrees to the local bank's offer of the swap, calculate the net present value of the project, in six months' time, in €. Discuss whether the swap would be beneficial to Casasophia Co. (10 marks)

(30 marks)

Section B – TWO questions ONLY to be attempted

- 3 GNT Co is considering an investment in one of two corporate bonds. Both bonds have a par value of \$1,000 and pay coupon interest on an annual basis. The market price of the first bond is \$1,079.68. Its coupon rate is 6% and it is due to be redeemed at par in five years. The second bond is about to be issued with a coupon rate of 4% and will also be redeemable at par in five years. Both bonds are expected to have the same gross redemption yields (yields to maturity).

GNT Co considers duration of the bond to be a key factor when making decisions on which bond to invest.

Required:

- (a) Estimate the Macaulay duration of the two bonds GNT Co is considering for investment. (9 marks)

- (b) Discuss how useful duration is as a measure of the sensitivity of a bond price to changes in interest rates. (8 marks)

(17 marks)

- 4 MesmerMagic Co (MMC) is considering whether to undertake the development of a new computer game based on an adventure film due to be released in 22 months. It is expected that the game will be available to buy two months after the film's release, by which time it will be possible to judge the popularity of the film with a high degree of certainty. However, at present, there is considerable uncertainty about whether the film, and therefore the game, is likely to be successful. Although MMC would pay for the exclusive rights to develop and sell the game now, the directors are of the opinion that they should delay the decision to produce and market the game until the film has been released and the game is available for sale.

MMC has forecast the following end of year cash flows for the four-year sales period of the game.

Year	1	2	3	4
Cash flows (\$ million)	25	18	10	5

MMC will spend \$7 million at the start of each of the next two years to develop the game, the gaming platform, and to pay for the exclusive rights to develop and sell the game. Following this, the company will require \$35 million for production, distribution and marketing costs at the start of the four-year sales period of the game.

It can be assumed that all the costs and revenues include inflation. The relevant cost of capital for this project is 11% and the risk free rate is 3.5%. MMC has estimated the likely volatility of the cash flows at a standard deviation of 30%.

Required:

- (a) Estimate the financial impact of the directors' decision to delay the production and marketing of the game. The Black-Scholes Option Pricing model may be used, where appropriate. All relevant calculations should be shown. (12 marks)

- (b) Briefly discuss the implications of the answer obtained in part (a) above. (5 marks)

(17 marks)

- 5 Mezza Co is a large food manufacturing and wholesale company. It imports fruit and vegetables from countries in South America, Africa and Asia, and packages them in steel cans, plastic tubs and as frozen foods, for sale to supermarkets around Europe. Its suppliers range from individual farmers to Government run cooperatives, and farms run by its own subsidiary companies. In the past, Mezza Co has been very successful in its activities, and has an excellent corporate image with its customers, suppliers and employees. Indeed Mezza Co prides itself on how it has supported local farming communities around the world and has consistently highlighted these activities in its annual reports.

However, in spite of buoyant stock markets over the last couple of years, Mezza Co's share price has remained static. It is thought that this is because there is little scope for future growth in its products. As a result the company's directors are considering diversifying into new areas. One possibility is to commercialise a product developed by a recently acquired subsidiary company. The subsidiary company is engaged in researching solutions to carbon emissions and global warming, and has developed a high carbon absorbing variety of plant that can be grown in warm, shallow sea water. The plant would then be harvested into carbon-neutral bio-fuel. This fuel, if widely used, is expected to lower carbon production levels.

Currently there is a lot of interest among the world's governments in finding solutions to climate change. Mezza Co's directors feel that this venture could enhance its reputation and result in a rise in its share price. They believe that the company's expertise would be ideally suited to commercialising the product. On a personal level, they feel that the venture's success would enhance their generous remuneration package which includes share options. It is hoped that the resulting increase in the share price would enable the options to be exercised in the future.

Mezza Co has identified the coast of Maienar, a small country in Asia, as an ideal location, as it has a large area of warm, shallow waters. Mezza Co has been operating in Maienar for many years and as a result, has a well developed infrastructure to enable it to plant, monitor and harvest the crop. Mezza Co's directors have strong ties with senior government officials in Maienar and the country's politicians are keen to develop new industries, especially ones with a long-term future.

The area identified by Mezza Co is a rich fishing ground for local fishermen, who have been fishing there for many generations. However, the fishermen are poor and have little political influence. The general perception is that the fishermen contribute little to Maienar's economic development. The coastal area, although naturally beautiful, has not been well developed for tourism. It is thought that the high carbon absorbing plant, if grown on a commercial scale, may have a negative impact on fish stocks and other wildlife in the area. The resulting decline in fish stocks may make it impossible for the fishermen to continue with their traditional way of life.

Required:

Discuss the key issues that the directors of Mezza Co should consider when making the decision about whether or not to commercialise the new product, and suggest how these issues may be mitigated or resolved.

(17 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.941	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.305	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Tuesday 6 December 2011

Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 8–12.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – BOTH questions are compulsory and MUST be attempted

- 1 Tramont Co is a listed company based in the USA and manufactures electronic devices. One of its devices, the *X-IT*, is produced exclusively for the American market. Tramont Co is considering ceasing the production of the *X-IT* gradually over a period of four years because it needs the manufacturing facilities used to make the *X-IT* for other products.

The government of Gamala, a country based in south-east Asia, is keen to develop its manufacturing industry and has offered Tramont Co first rights to produce the *X-IT* in Gamala and sell it to the USA market for a period of four years. At the end of the four-year period, the full production rights will be sold to a government-backed company for Gamalan Rupiahs (GR) 450 million after tax (this amount is not subject to inflationary increases). Tramont Co has to decide whether to continue production of the *X-IT* in the USA for the next four years or to move the production to Gamala immediately.

Currently each *X-IT* unit sold makes a unit contribution of \$20. This unit contribution is not expected to be subject to any inflationary increase in the next four years. Next year's production and sales estimated at 40,000 units will fall by 20% each year for the following three years. It is anticipated that after four years the production of the *X-IT* will stop. It is expected that the financial impact of the gradual closure over the four years will be cost neutral (the revenue from sale of assets will equal the closure costs). If production is stopped immediately, the excess assets would be sold for \$2.3 million and the costs of closure, including redundancy costs of excess labour, would be \$1.7 million.

The following information relates to the production of the *X-IT* moving to Gamala.

The Gamalan project will require an initial investment of GR 230 million, to pay for the cost of land and buildings (GR 150 million) and machinery (GR 80 million). The cost of machinery is tax allowable and will be depreciated on a straight-line basis over the next four years, at the end of which it will have a negligible value.

Tramont Co will also need GR 40 million for working capital immediately. It is expected that the working capital requirement will increase in line with the annual inflation rate in Gamala. When the project is sold, the working capital will not form part of the sale price and will be released back to Tramont Co.

Production and sales of the device are expected to be 12,000 units in the first year, rising to 22,000 units, 47,000 units and 60,000 units in the next three years respectively.

The following revenues and costs apply to the first year of operation:

- Each unit will be sold for \$70;
- The variable cost per unit comprising of locally sourced materials and labour will be GR 1,350, and;
- In addition to the variable cost above, each unit will require a component bought from Tramont Co for \$7, on which Tramont Co makes \$4 contribution per unit;
- Total fixed costs for the first year will be GR 30 million.

The costs are expected to increase by their countries' respective rates of inflation, but the selling price will remain fixed at \$70 per unit for the four-year period.

The annual corporation tax rate in Gamala is 20% and Tramont Co currently pays corporation tax at a rate of 30% per year. Both countries' corporation taxes are payable in the year that the tax liability arises. A bi-lateral tax treaty exists between the USA and Gamala, which permits offset of overseas tax against any USA tax liability on overseas earnings. The USA and Gamalan tax authorities allow losses to be carried forward and written off against future profits for taxation purposes.

Tramont Co has decided to finance the project by borrowing the funds required in Gamala. The commercial borrowing rate is 13% but the Gamalan government has offered Tramont Co a 6% subsidised loan for the entire amount of the initial funds required. The Gamalan government has agreed that it will not ask for the loan to be repaid as long as Tramont Co fulfils its contract to undertake the project for the four years. Tramont Co can borrow dollar funds at an interest rate of 5%.

Tramont Co's financing consists of 25 million shares currently trading at \$2.40 each and \$40 million 7% bonds trading at \$1,428 per \$1,000. Tramont Co's quoted beta is 1.17. The current risk free rate of return is estimated at 3% and the market risk premium is 6%. Due to the nature of the project, it is estimated that the beta applicable to the project if it is all-equity financed will be 0.4 more than the current all-equity financed beta of Tramont Co. If the Gamalan project is undertaken, the cost of capital applicable to the cash flows in the USA is expected to be 7%.

The spot exchange rate between the dollar and the Gamalan Rupiah is GR 55 per \$1. The annual inflation rates are currently 3% in the USA and 9% in Gamala. It can be assumed that these inflation rates will not change for the foreseeable future. All net cash flows arising from the project will be remitted back to Tramont Co at the end of each year.

There are two main political parties in Gamala: the Gamala Liberal (GL) Party and the Gamala Republican (GR) Party. Gamala is currently governed by the GL Party but general elections are due to be held soon. If the GR Party wins the election, it promises to increase taxes of international companies operating in Gamala and review any commercial benefits given to these businesses by the previous government.

Required:

Prepare a report for the Board of Directors of Tramont Co that

- (i) **Evaluates whether or not Tramont Co should undertake the project to produce the X-IT in Gamala and cease its production in the USA immediately. In the evaluation, include all relevant calculations in the form of a financial assessment and explain any assumptions made;**

Note: it is suggested that the financial assessment should be based on present value of the operating cash flows from the Gamalan project, discounted by an appropriate all-equity rate, and adjusted by the present value of all other relevant cash flows. (27 marks)

- (ii) **Discusses the potential change in government and other business factors that Tramont Co should consider before making a final decision.** (8 marks)

Professional marks will be awarded in question 1 for the format, structure and presentation of the answer.

(4 marks)

(39 marks)

- 2 Alecto Co, a large listed company based in Europe, is expecting to borrow €22,000,000 in four months' time on 1 May 2012. It expects to make a full repayment of the borrowed amount nine months from now. Currently there is some uncertainty in the markets, with higher than normal rates of inflation, but an expectation that the inflation level may soon come down. This has led some economists to predict a rise in interest rates and others suggesting an unchanged outlook or maybe even a small fall in interest rates over the next six months.

Although Alecto Co is of the opinion that it is equally likely that interest rates could increase or fall by 0.5% in four months, it wishes to protect itself from interest rate fluctuations by using derivatives. The company can borrow at LIBOR plus 80 basis points and LIBOR is currently 3.3%. The company is considering using interest rate futures, options on interest rate futures or interest rate collars as possible hedging choices.

The following information and quotes from an appropriate exchange are provided on Euro futures and options. Margin requirements may be ignored.

Three month Euro futures, €1,000,000 contract, tick size 0.01% and tick value €25

March 96.27

June 96.16

September 95.90

Options on three month Euro futures, €1,000,000 contract, tick size 0.01% and tick value €25. Option premiums are in annual %.

Calls			Strike	Puts		
March	June	September		March	June	September
0.279	0.391	0.446	96.00	0.006	0.163	0.276
0.012	0.090	0.263	96.50	0.196	0.581	0.754

It can be assumed that settlement for both the futures and options contracts is at the end of the month. It can also be assumed that basis diminishes to zero at contract maturity at a constant rate and that time intervals can be counted in months.

Required:

- (a) Briefly discuss the main advantage and disadvantage of hedging interest rate risk using an interest rate collar instead of options. (4 marks)
- (b) Based on the three hedging choices Alecto Co is considering and assuming that the company does not face any basis risk, recommend a hedging strategy for the €22,000,000 loan. Support your recommendation with appropriate comments and relevant calculations in €. (17 marks)
- (c) Explain what is meant by basis risk and how it would affect the recommendation made in part (b) above. (4 marks)

(25 marks)

Section B – TWO questions ONLY to be attempted

- 3** Levante Co has identified a new project for which it will need to increase its long-term borrowings from \$250 million to \$400 million. This amount will cover a significant proportion of the total cost of the project and the rest of the funds will come from cash held by the company.

The current \$250 million borrowing is in the form of a 4% bond which is trading at \$98.71 per \$100 and is due to be redeemed at par in three years. The issued bond has a credit rating of AA. The new borrowing will also be raised in the form of a traded bond with a par value of \$100 per unit. It is anticipated that the new project will generate sufficient cash flows to be able to redeem the new bond at \$100 par value per unit in five years. It can be assumed that coupons on both bonds are paid annually.

Both bonds would be ranked equally for payment in the event of default and the directors expect that as a result of the new issue, the credit rating for both bonds will fall to A. The directors are considering the following two alternative options when issuing the new bond:

- (i) Issue the new bond at a fixed coupon of 5% but at a premium or discount, whichever is appropriate to ensure full take up of the bond; or
- (ii) Issue the new bond at a coupon rate where the issue price of the new bond will be \$100 per unit and equal to its par value.

The following extracts are provided on the current government bond yield curve and yield spreads for the sector in which Levante Co operates:

Current Government Bond Yield Curve

Years	1	2	3	4	5
	3.2%	3.7%	4.2%	4.8%	5.0%

Yield spreads (in basis points)

Bond Rating	1 year	2 years	3 years	4 years	5 years
AAA	5	9	14	19	25
AA	16	22	30	40	47
A	65	76	87	100	112
BBB	102	121	142	167	193

Required:

- (a) Calculate the expected percentage fall in the market value of the existing bond if Levante Co's bond credit rating falls from AA to A. (3 marks)
- (b) Advise the directors on the financial implications of choosing each of the two options when issuing the new bond. Support the advice with appropriate calculations. (7 marks)
- (c) Among the criteria used by credit agencies for establishing a company's credit rating are the following: industry risk, earnings protection, financial flexibility and evaluation of the company's management.

Briefly explain each criterion and suggest factors that could be used to assess it. (8 marks)

(18 marks)

- 4 Proteus Co, a large listed company, has a number of subsidiaries in different industries but its main line of business is developing surveillance systems and intruder alarms. It has decided to sell a number of companies that it considers are peripheral to its core activities. One of these subsidiary companies is Tyche Co, a company involved in managing the congestion monitoring and charging systems that have been developed by Proteus Co. Tyche Co is a profitable business and it is anticipated that its revenues and costs will continue to increase at their current rate of 8% per year for the foreseeable future.

Tyche Co's managers and some employees want to buy the company through a leveraged management buy-out. An independent assessment estimates Tyche Co's market value at \$81 million if Proteus Co agrees to cancel its current loan to Tyche Co. The managers and employees involved in the buy-out will invest \$12 million for 75% of the equity in the company, with another \$4 million coming from a venture capitalist for the remaining 25% equity.

Palaemon Bank has agreed to lend the balance of the required funds in the form of a 9% loan. The interest is payable at the end of the year, on the loan amount outstanding at the start of each year. A covenant on the loan states that the following debt-equity ratios should not be exceeded at the end of each year for the next five years:

Year	1	2	3	4	5
Debt / Equity (%)	350%	250%	200%	150%	125%

Shown below is an extract of the latest annual income statement for Tyche Co:

	\$'000
Sales Revenue	60,000
Materials and consumables	12,000
Labour costs	22,000
Other costs	4,000
Allocated overhead charge payable to Proteus Co	14,000
Interest paid	2,000
Taxable Profit	6,000
Taxation	1,500
Retained Earnings	4,500

As part of the management buy-out agreement, it is expected that Proteus Co will provide management services costing \$12 million for the first year of the management buy-out, increasing by 8% per year thereafter.

The current tax rate is 25% on profits and it is expected that 25% of the after-tax profits will be payable as dividends every year. The remaining profits will be allocated to reserves. It is expected that Tyche Co will repay \$3 million of the outstanding loan at the end of each of the next five years from the cash flows generated from its business activity.

Required:

- Briefly discuss the possible benefits to Proteus Co of disposing Tyche Co through a management buy-out.** (4 marks)
- Calculate whether the debt-equity covenant imposed by Palaemon Bank on Tyche Co will be breached over the five-year period.** (9 marks)
- Discuss briefly the implications of the results obtained in part (b) and outline two possible actions Tyche Co may take if the covenant is in danger of being breached.** (5 marks)

(18 marks)

- 5** The Chairman and the Chief Executive Officer (CEO) of Kengai Co are discussing whether or not the company should adopt a triple bottom line (TBL) reporting system in order to demonstrate Kengai Co's level of sustainable development. Kengai Co's competitors are increasingly adopting TBL reporting and the Chairman feels that it would be beneficial to follow suit. The CEO, on the other hand, feels that pursuing TBL reporting would be expensive and is not necessary.

Required:

- (a) Explain what TBL reporting involves and how it would help demonstrate Kengai Co's sustainable development. Support your explanation by including examples of proxies that can be used to indicate the impact of the factors that would be included in a TBL report. (8 marks)
- (b) Discuss how producing a TBL report may help Kengai Co's management focus on improving the financial position of the company. Illustrate the discussion with examples where appropriate. (10 marks)

(18 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

Discount rate (r)

Periods

(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.941	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.305	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.37	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.26	10.58	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.13	11.35	10.63	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.00	12.11	11.30	10.56	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.87	12.85	11.94	11.12	10.38	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Tuesday 12 June 2012



Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 8–12.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – BOTH questions are compulsory and MUST be attempted

- 1 Nente Co, an unlisted company, designs and develops tools and parts for specialist machinery. The company was formed four years ago by three friends, who own 20% of the equity capital in total, and a consortium of five business angel organisations, who own the remaining 80%, in roughly equal proportions. Nente Co also has a large amount of debt finance in the form of variable rate loans. Initially the amount of annual interest payable on these loans was low and allowed Nente Co to invest internally generated funds to expand its business. Recently though, due to a rapid increase in interest rates, there has been limited scope for future expansion and no new product development.

The Board of Directors, consisting of the three friends and a representative from each business angel organisation, met recently to discuss how to secure the company's future prospects. Two proposals were put forward, as follows:

Proposal 1

To accept a takeover offer from Mije Co, a listed company, which develops and manufactures specialist machinery tools and parts. The takeover offer is for \$2.95 cash per share or a share-for-share exchange where two Mije Co shares would be offered for three Nente Co shares. Mije Co would need to get the final approval from its shareholders if either offer is accepted;

Proposal 2

To pursue an opportunity to develop a small prototype product that just breaks even financially, but gives the company exclusive rights to produce a follow-on product within two years.

The meeting concluded without agreement on which proposal to pursue.

After the meeting, Mije Co was consulted about the exclusive rights. Mije Co's directors indicated that they had not considered the rights in their computations and were willing to continue with the takeover offer on the same terms without them.

Currently, Mije Co has 10 million shares in issue and these are trading for \$4.80 each. Mije Co's price to earnings (P/E) ratio is 15. It has sufficient cash to pay for Nente Co's equity and a substantial proportion of its debt, and believes that this will enable Nente Co to operate on a P/E level of 15 as well. In addition to this, Mije Co believes that it can find cost-based synergies of \$150,000 after tax per year for the foreseeable future. Mije Co's current profit after tax is \$3,200,000.

The following financial information relates to Nente Co and to the development of the new product.

Nente Co financial information

Extract from the most recent income statement

	\$'000
Sales revenue	8,780
Profit before interest and tax	1,230
Interest	(455)
Tax	(155)
Profit after tax	620
Dividends	Nil

Extract from the most recent statement of financial position

	\$'000
Net non-current assets	10,060
Current assets	690
Total Assets	10,750
Share capital (40c per share par value)	960
Reserves	1,400
Non-current liabilities: Variable rate loans	6,500
Current liabilities	1,890
Total liabilities and capital	10,750

In arriving at the profit after tax amount, Nente Co deducted tax allowable depreciation and other non-cash expenses totalling \$1,206,000. It requires an annual cash investment of \$1,010,000 in non-current assets and working capital to continue its operations.

Nente Co's profits before interest and tax in its first year of operation were \$970,000 and have been growing steadily in each of the following three years, to their current level. Nente Co's cash flows grew at the same rate as well, but it is likely that this growth rate will reduce to 25% of the original rate for the foreseeable future.

Nente Co currently pays interest of 7% per year on its loans, which is 380 basis points over the government base rate, and corporation tax of 20% on profits after interest. It is estimated that an overall cost of capital of 11% is reasonable compensation for the risk undertaken on an investment of this nature.

New product development (Proposal 2)

Developing the new follow-on product will require an investment of \$2,500,000 initially. The total expected cash flows and present values of the product over its five-year life, with a volatility of 42% standard deviation, are as follows:

Year(s)	Now	1	2	3 to 7 (in total)
Cash flows (\$'000)	–	–	(2,500)	3,950
Present values (\$'000)	–	–	(2,029)	2,434

Required:

Prepare a report for the Board of Directors of Nente Co that:

- (i) Estimates the current value of a Nente Co share, using the free cash flow to firm methodology;** (7 marks)
- (ii) Estimates the percentage gain in value to a Nente Co share and a Mije Co share under each payment offer;** (8 marks)
- (iii) Estimates the percentage gain in the value of the follow-on product to a Nente Co share, based on its cash flows and on the assumption that the production can be delayed following acquisition of the exclusive rights of production;** (8 marks)
- (iv) Discusses the likely reaction of Nente Co and Mije Co shareholders to the takeover offer, including the assumptions made in the estimates above and how the follow-on product's value can be utilised by Nente Co.** (8 marks)

Professional marks will be awarded in question 1 for the presentation, structure and clarity of the answer.

(4 marks)

(35 marks)

- 2 Three proposals were put forward for further consideration after a meeting of the executive directors of Ennea Co to discuss the future investment and financing strategy of the business. Ennea Co is a listed company operating in the haulage and shipping industry.

Proposal 1

To increase the company's level of debt by borrowing a further \$20 million and use the funds raised to buy back share capital.

Proposal 2

To increase the company's level of debt by borrowing a further \$20 million and use these funds to invest in additional non-current assets in the haulage strategic business unit.

Proposal 3

To sell excess non-current haulage assets with a net book value of \$25 million for \$27 million and focus on offering more services to the shipping strategic business unit. This business unit will require no additional investment in non-current assets. All the funds raised from the sale of the non-current assets will be used to reduce the company's debt.

Ennea Co financial information

Extracts from the forecast financial position for the coming year

	\$m
Non-current assets	282
Current assets	66
Total assets	348
Equity and liabilities	
Share capital (40c per share par value)	48
Retained earnings	123
Total equity	171
Non-current liabilities	140
Current liabilities	37
Total liabilities	177
Total liabilities and capital	348

Ennea Co's forecast after tax profit for the coming year is expected to be \$26 million and its current share price is \$3.20 per share. The non-current liabilities consist solely of a 6% medium term loan redeemable within seven years. The terms of the loan contract stipulates that an increase in borrowing will result in an increase in the coupon payable of 25 basis points on the total amount borrowed, while a reduction in borrowing will lower the coupon payable by 15 basis points on the total amount borrowed.

Ennea Co's effective tax rate is 20%. The company's estimated after tax rate of return on investment is expected to be 15% on any new investment. It is expected that any reduction in investment would suffer the same rate of return.

Required:

- (a) **Estimate and discuss the impact of each of the three proposals on the forecast statement of financial position, the earnings and earnings per share, and gearing of Ennea Co.** (20 marks)
- (b) An alternative suggestion to proposal three was made where the non-current assets could be leased to other companies instead of being sold. The lease receipts would then be converted into an asset through securitisation. The proceeds from the sale of the securitised lease receipts asset would be used to reduce the outstanding loan borrowings.

Required:

Explain what the securitisation process would involve and what would be the key barriers to Ennea Co undertaking the process. (5 marks)

(25 marks)

Section B – TWO questions ONLY to be attempted

- 3** Sembilan Co, a listed company, recently issued debt finance to acquire assets in order to increase its activity levels. This debt finance is in the form of a floating rate bond, with a face value of \$320 million, redeemable in four years. The bond interest, payable annually, is based on the spot yield curve plus 60 basis points. The next annual payment is due at the end of year one.

Sembilan Co is concerned that the expected rise in interest rates over the coming few years would make it increasingly difficult to pay the interest due. It is therefore proposing to either swap the floating rate interest payment to a fixed rate payment, or to raise new equity capital and use that to pay off the floating rate bond. The new equity capital would either be issued as rights to the existing shareholders or as shares to new shareholders.

Ratus Bank has offered Sembilan Co an interest rate swap, whereby Sembilan Co would pay Ratus Bank interest based on an equivalent fixed annual rate of $3.76\frac{1}{4}\%$ in exchange for receiving a variable amount based on the current yield curve rate. Payments and receipts will be made at the end of each year, for the next four years. Ratus Bank will charge an annual fee of 20 basis points if the swap is agreed.

The current annual spot yield curve rates are as follows:

Year	One	Two	Three	Four
Rate	2.5%	3.1%	3.5%	3.8%

The current annual forward rates for years two, three and four are as follows:

Year	Two	Three	Four
Rate	3.7%	4.3%	4.7%

Required:

- (a) Based on the above information, calculate the amounts Sembilan Co expects to pay or receive every year on the swap (excluding the fee of 20 basis points). Explain why the fixed annual rate of interest of $3.76\frac{1}{4}\%$ is less than the four-year yield curve rate of 3.8%. (6 marks)
- (b) Demonstrate that Sembilan Co's interest payment liability does not change, after it has undertaken the swap, whether the interest rates increase or decrease. (5 marks)
- (c) Discuss the factors that Sembilan Co should consider when deciding whether it should raise equity capital to pay off the floating rate debt. (9 marks)

(20 marks)

- 4 Tisa Co is considering an opportunity to produce an innovative component which, when fitted into motor vehicle engines, will enable them to utilise fuel more efficiently. The component can be manufactured using either process Omega or process Zeta. Although this is an entirely new line of business for Tisa Co, it is of the opinion that developing either process over a period of four years and then selling the productions rights at the end of four years to another company may prove lucrative.

The annual after-tax cash flows for each process are as follows:

Process Omega

Year	0	1	2	3	4
After-tax cash flows (\$000)	(3,800)	1,220	1,153	1,386	3,829

Process Zeta

Year	0	1	2	3	4
After-tax cash flows (\$000)	(3,800)	643	546	1,055	5,990

Tisa Co has 10 million 50c shares trading at 180c each. Its loans have a current value of \$3.6 million and an average after-tax cost of debt of 4.50%. Tisa Co's capital structure is unlikely to change significantly following the investment in either process.

Elfu Co manufactures electronic parts for cars including the production of a component similar to the one being considered by Tisa Co. Elfu Co's equity beta is 1.40, and it is estimated that the equivalent equity beta for its other activities, excluding the component production, is 1.25. Elfu Co has 400 million 25c shares in issue trading at 120c each. Its debt finance consists of variable rate loans redeemable in seven years. The loans paying interest at base rate plus 120 basis points have a current value of \$96 million. It can be assumed that 80% of Elfu Co's debt finance and 75% of Elfu Co's equity finance can be attributed to other activities excluding the component production.

Both companies pay annual corporation tax at a rate of 25%. The current base rate is 3.5% and the market risk premium is estimated at 5.8%.

Required:

- Provide a reasoned estimate of the cost of capital that Tisa Co should use to calculate the net present value of the two processes. Include all relevant calculations.** (8 marks)
- Calculate the internal rate of return (IRR) and the modified internal rate of return (MIRR) for Process Omega. Given that the IRR and MIRR of Process Zeta are 26.6% and 23.3% respectively, recommend which process, if any, Tisa Co should proceed with and explain your recommendation.** (8 marks)
- Elfu Co has estimated an annual standard deviation of \$800,000 on one of its other projects, based on a normal distribution of returns. The average annual return on this project is \$2,200,000.

Required:

Estimate the project's Value at Risk (VAR) at a 99% confidence level for one year and over the project's life of five years. Explain what is meant by the answers obtained. (4 marks)

(20 marks)

- 5 Kilenc Co, a large listed company based in the UK, produces pharmaceutical products which are exported around the world. It is reviewing a proposal to set up a subsidiary company to manufacture a range of body and facial creams in Lanosia. These products will be sold to local retailers and to retailers in nearby countries.

Lanosia has a small but growing manufacturing industry in pharmaceutical products, although it remains largely reliant on imports. The Lanosian government has been keen to promote the pharmaceutical manufacturing industry through purchasing local pharmaceutical products, providing government grants and reducing the industry's corporate tax rate. It also imposes large duties on imported pharmaceutical products which compete with the ones produced locally.

Although politically stable, the recent worldwide financial crisis has had a significant negative impact on Lanosia. The country's national debt has grown substantially following a bailout of its banks and it has had to introduce economic measures which are hampering the country's ability to recover from a deep recession. Growth in real wages has been negative over the past three years, the economy has shrunk in the past year and inflation has remained higher than normal during this time.

On the other hand, corporate investment in capital assets, research and development, and education and training, has grown recently and interest rates remain low. This has led some economists to suggest that the economy should start to recover soon. Employment levels remain high in spite of low nominal wage growth.

Lanosian corporate governance regulations stipulate that at least 40% of equity share capital must be held by the local population. In addition at least 50% of members on the Board of Directors, including the Chairman, must be from Lanosia. Kilenc Co wants to finance the subsidiary company using a mixture of debt and equity. It wants to raise additional equity and debt finance in Lanosia in order to minimise exchange rate exposure. The small size of the subsidiary will have minimal impact on Kilenc Co's capital structure. Kilenc Co intends to raise the 40% equity through an initial public offering (IPO) in Lanosia and provide the remaining 60% of the equity funds from its own cash funds.

Required:

- (a) **Discuss the key risks and issues that Kilenc Co should consider when setting up a subsidiary company in Lanosia, and suggest how these may be mitigated.** (15 marks)
- (b) The directors of Kilenc Co have learnt that a sizeable number of equity trades in Lanosia are conducted using dark pool trading systems.

Required:

Explain what dark pool trading systems are and how Kilenc Co's proposed Initial Public Offering (IPO) may be affected by these. (5 marks)

(20 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_o = \frac{D_o(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Tuesday 4 December 2012



Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – BOTH questions are compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 11–15.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – BOTH questions are compulsory and MUST be attempted

- 1 Coeden Co is a listed company operating in the hospitality and leisure industry. Coeden Co's board of directors met recently to discuss a new strategy for the business. The proposal put forward was to sell all the hotel properties that Coeden Co owns and rent them back on a long-term rental agreement. Coeden Co would then focus solely on the provision of hotel services at these properties under its popular brand name. The proposal stated that the funds raised from the sale of the hotel properties would be used to pay off 70% of the outstanding non-current liabilities and the remaining funds would be retained for future investments.

The board of directors are of the opinion that reducing the level of debt in Coeden Co will reduce the company's risk and therefore its cost of capital. If the proposal is undertaken and Coeden Co focuses exclusively on the provision of hotel services, it can be assumed that the current market value of equity will remain unchanged after implementing the proposal.

Coeden Co Financial Information

Extract from the most recent Statement of Financial Position

	\$'000
Non-current assets (re-valued recently)	42,560
Current assets	26,840
Total assets	69,400
Share capital (25c per share par value)	3,250
Reserves	21,780
Non-current liabilities (5·2% redeemable bonds)	42,000
Current liabilities	2,370
Total capital and liabilities	69,400

Coeden Co's latest free cash flow to equity of \$2,600,000 was estimated after taking into account taxation, interest and reinvestment in assets to continue with the current level of business. It can be assumed that the annual reinvestment in assets required to continue with the current level of business is equivalent to the annual amount of depreciation. Over the past few years, Coeden Co has consistently used 40% of its free cash flow to equity on new investments while distributing the remaining 60%. The market value of equity calculated on the basis of the free cash flow to equity model provides a reasonable estimate of the current market value of Coeden Co.

The bonds are redeemable at par in three years and pay the coupon on an annual basis. Although the bonds are not traded, it is estimated that Coeden Co's current debt credit rating is BBB but would improve to A+ if the non-current liabilities are reduced by 70%.

Other Information

Coeden Co's current equity beta is 1·1 and it can be assumed that debt beta is 0. The risk free rate is estimated to be 4% and the market risk premium is estimated to be 6%.

There is no beta available for companies offering just hotel services, since most companies own their own buildings. The average asset beta for property companies has been estimated at 0·4. It has been estimated that the hotel services business accounts for approximately 60% of the current value of Coeden Co and the property company business accounts for the remaining 40%.

Coeden Co's corporation tax rate is 20%. The three-year borrowing credit spread on A+ rated bonds is 60 basis points and 90 basis points on BBB rated bonds, over the risk free rate of interest.

Required:

- (a) Calculate, and comment on, Coeden Co's cost of equity and weighted average cost of capital before and after implementing the proposal. Briefly explain any assumptions made.** (20 marks)
- (b) Discuss the validity of the assumption that the market value of equity will remain unchanged after the implementation of the proposal.** (5 marks)
- (c) As an alternative to selling the hotel properties, the board of directors is considering a demerger of the hotel services and a separate property company which would own the hotel properties. The property company would take over 70% of Coeden Co's long-term debt and pay Coeden Co cash for the balance of the property value.**

Required:

Explain what a demerger is, and the possible benefits and drawbacks of pursuing the demerger option as opposed to selling the hotel properties. (8 marks)

(33 marks)

- 2 Lignum Co, a large listed company, manufactures agricultural machines and equipment for different markets around the world. Although its main manufacturing base is in France and it uses the Euro (€) as its base currency, it also has a few subsidiary companies around the world. Lignum Co's treasury division is considering how to approach the following three cases of foreign exchange exposure that it faces.

Case One

Lignum Co regularly trades with companies based in Zuhait, a small country in South America whose currency is the Zupesos (ZP). It recently sold machinery for ZP140 million, which it is about to deliver to a company based there. It is expecting full payment for the machinery in four months. Although there are no exchange traded derivative products available for the Zupesos, Medes Bank has offered Lignum Co a choice of two over-the-counter derivative products.

The first derivative product is an over-the-counter forward rate determined on the basis of the Zuhait base rate of 8.5% plus 25 basis points and the French base rate of 2.2% less 30 basis points.

Alternatively, with the second derivative product Lignum Co can purchase either Euro call or put options from Medes Bank at an exercise price equivalent to the current spot exchange rate of ZP142 per €1. The option premiums offered are: ZP7 per €1 for the call option or ZP5 per €1 for the put option.

The premium cost is payable in full at the commencement of the option contract. Lignum Co can borrow money at the base rate plus 150 basis points and invest money at the base rate minus 100 basis points in France.

Case Two

Namel Co is Lignum Co's subsidiary company based in Maram, a small country in Asia, whose currency is the Maram Ringit (MR). The current pegged exchange rate between the Maram Ringit and the Euro is MR35 per €1. Due to economic difficulties in Maram over the last couple of years, it is very likely that the Maram Ringit will devalue by 20% imminently. Namel Co is concerned about the impact of the devaluation on its Statement of Financial Position.

Given below is an extract from the current Statement of Financial Position of Namel Co.

	MR '000
Non-current assets	179,574
Current assets	146,622
Total assets	326,196
Share capital and reserves	102,788
Non-current liabilities	132,237
Current liabilities	91,171
Total capital and liabilities	326,196

The current assets consist of inventories, receivables and cash. Receivables account for 40% of the current assets. All the receivables relate to sales made to Lignum Co in Euro. About 70% of the current liabilities consist of payables relating to raw material inventory purchased from Lignum Co and payable in Euro. 80% of the non-current liabilities consist of a Euro loan and the balance are borrowings sourced from financial institutions in Maram.

Case Three

Lignum Co manufactures a range of farming vehicles in France which it sells within the European Union to countries which use the Euro. Over the previous few years, it has found that its sales revenue from these products has been declining and the sales director is of the opinion that this is entirely due to the strength of the Euro. Lignum Co's biggest competitor in these products is based in the USA and US\$ rate has changed from almost parity with the Euro three years ago, to the current value of US\$1.47 for €1. The agreed opinion is that the US\$ will probably continue to depreciate against the Euro, but possibly at a slower rate, for the foreseeable future.

Required:

Prepare a report for Lignum Co's treasury division that:

- (i) Briefly explains the type of currency exposure Lignum Co faces for each of the above cases; (3 marks)**
- (ii) Recommends which of the two derivative products Lignum Co should use to manage its exposure in case one and advises on alternative hedging strategies that could be used. Show all relevant calculations; (9 marks)**
- (iii) Computes the gain or loss on Namel Co's Statement of Financial Position, due to the devaluation of the Maram Ringit in case two, and discusses whether and how this exposure should be managed; (8 marks)**
- (iv) Discusses how the exposure in case three can be managed. (3 marks)**

Professional marks will be awarded in question 2 for the structure and presentation of the report. (4 marks)

(27 marks)

Section B – TWO questions ONLY to be attempted

- 3 Sibra Co is a listed company producing confectionary products which it sells around the world. It wants to acquire Dentro Co, an unlisted company producing high quality, luxury chocolates. Sibra Co proposes to pay for the acquisition using one of the following three methods:

Method 1

A cash offer of \$5.00 per Dentro Co share; or

Method 2

An offer of three of its shares for two of Dentro Co's shares; or

Method 3

An offer of a 2% coupon bond in exchange for 16 Dentro Co's shares. The bond will be redeemed in three years at its par value of \$100.

Extracts from the latest financial statements of both companies are as follows:

	Sibra Co	Dentro Co
	\$'000	\$'000
Sales revenue	44,210	4,680
Profit before tax	6,190	780
Taxation	(1,240)	(155)
Profit after tax	4,950	625
Dividends	(2,700)	(275)
Retained earnings for the year	2,250	350
Non-current assets	22,450	3,350
Current assets	3,450	247
Non-current liabilities	9,700	873
Current liabilities	3,600	436
Share capital (40c per share)	4,400	500
Reserves	8,200	1,788

Sibra Co's current share price is \$3.60 per share and it has estimated that Dentro Co's price to earnings ratio is 12.5% higher than Sibra Co's current price to earnings ratio. Sibra Co's non-current liabilities include a 6% bond redeemable in three years at par which is currently trading at \$104 per \$100 par value.

Sibra Co estimates that it could achieve synergy savings of 30% of Dentro Co's estimated equity value by eliminating duplicated administrative functions, selling excess non-current assets and through reducing the workforce numbers, if the acquisition were successful.

Required:

- (a) **Estimate the percentage gain on a Dentro Co share under each of the above three payment methods. Comment on the answers obtained.** (16 marks)

- (b) In relation to the acquisition, the board of directors of Sibra Co are considering the following two proposals:

Proposal 1

Once Sibra Co has obtained agreement from a significant majority of the shareholders, it will enforce the remaining minority shareholders to sell their shares; and

Proposal 2

Sibra Co will offer an extra 3 cents per share, in addition to the bid price, to 30% of the shareholders of Dentro Co on a first-come, first-serve basis, as an added incentive to make the acquisition proceed more quickly.

Required:

With reference to the key aspects of the global regulatory framework for mergers and acquisitions, briefly discuss the above proposals. (4 marks)

(20 marks)

- 4 Arbore Co is a large listed company with many autonomous departments operating as investment centres. It sets investment limits for each department based on a three-year cycle. Projects selected by departments would have to fall within the investment limits set for each of the three years. All departments would be required to maintain a capital investment monitoring system, and report on their findings annually to Arbore Co's board of directors.

The Durvo department is considering the following five investment projects with three years of initial investment expenditure, followed by several years of positive cash inflows. The department's initial investment expenditure limits are \$9,000,000, \$6,000,000 and \$5,000,000 for years one, two and three respectively. None of the projects can be deferred and all projects can be scaled down but not scaled up.

Project	Investment required at start of year			Project net present value
	Year one (Immediately)	Year two	Year three	
PDur01	\$4,000,000	\$1,100,000	\$2,400,000	\$464,000
PDur02	\$800,000	\$2,800,000	\$3,200,000	\$244,000
PDur03	\$3,200,000	\$3,562,000	\$0	\$352,000
PDur04	\$3,900,000	\$0	\$200,000	\$320,000
PDur05	\$2,500,000	\$1,200,000	\$1,400,000	Not provided

PDur05 project's annual operating cash flows commence at the end of year four and last for a period of 15 years. The project generates annual sales of 300,000 units at a selling price of \$14 per unit and incurs total annual relevant costs of \$3,230,000. Although the costs and units sold of the project can be predicted with a fair degree of certainty, there is considerable uncertainty about the unit selling price. The department uses a required rate of return of 11% for its projects, and inflation can be ignored.

The Durvo department's managing director is of the opinion that all projects which return a positive net present value should be accepted and does not understand the reason(s) why Arbore Co imposes capital rationing on its departments. Furthermore, she is not sure why maintaining a capital investment monitoring system would be beneficial to the company.

Required:

- (a) Calculate the net present value of project PDur05. Calculate and comment on what percentage fall in the selling price would need to occur before the net present value falls to zero. (6 marks)
- (b) Formulate an appropriate capital rationing model, based on the above investment limits, that maximises the net present value for department Durvo. Finding a solution for the model is not required. (3 marks)
- (c) Assume the following output is produced when the capital rationing model in part (b) above is solved:

Category 1: Total Final Value

\$1,184,409

Category 2: Adjustable Final Values

Project PDur01: 0.958

Project PDur02: 0.407

Project PDur03: 0.732

Project PDur04: 0.000

Project PDur05: 1.000

Category 3:

Constraints Utilised

Year one: \$9,000,000

Year two: \$6,000,000

Year three: \$5,000,000

Slack

Year one: \$0

Year two: \$0

Year three: \$0

Required:

Explain the figures produced in each of the three output categories.

(5 marks)

(d) Provide a brief response to the managing director's opinions by:

(i) Explaining why Arbore Co may want to impose capital rationing on its departments;

(2 marks)

(ii) Explaining the features of a capital investment monitoring system and discussing the benefits of maintaining such a system.

(4 marks)

(20 marks)

- 5** Strom Co is a clothing retailer, with stores selling mid-price clothes and clothing accessories throughout Europe. It sells its own-brand items, which are produced by small manufacturers located in Africa, who work solely for Strom Co. The recent European sovereign debt crisis has affected a number of countries in the European Union (EU). Consequently, Strom Co has found trading conditions to be extremely difficult, putting pressure on profits and sales revenue.

The sovereign debt crisis in Europe resulted in countries finding it increasingly difficult and expensive to issue government bonds to raise funds. Two main reasons have been put forward to explain why the crisis took place: firstly, a number of countries continued to borrow excessive funds, because their expenditure exceeded taxation revenues; and secondly, a number of countries allocated significant sums of money to support their banks following the 'credit crunch' and the banking crisis.

In order to prevent countries defaulting on their debt obligations and being downgraded, the countries in the EU and the International Monetary Fund (IMF) established a fund to provide financial support to member states threatened by the risk of default, credit downgrades and excessive borrowing yields. Strict economic conditions known as austerity measures were imposed on these countries in exchange for receiving financial support.

The austerity measures have affected Strom Co negatively, and the years 2011 and 2012 have been particularly bad, with sales revenue declining by 15% and profits by 25% in 2011, and remaining at 2011 levels in 2012. On investigation, Strom Co noted that clothing retailers selling clothes at low prices and at high prices were not affected as badly as Strom Co or other mid-price retailers. Indeed, the retailers selling low-priced clothes had increased their profits, and retailers selling luxury, expensive clothes had maintained their profits over the last two to three years.

In order to improve profitability, Strom Co's board of directors expects to cut costs where possible. A significant fixed cost relates to quality control, which includes monitoring the working conditions of employees of Strom Co's clothing manufacturers, as part of its ethical commitment.

Required:

- (a) Explain the role and aims of the International Monetary Fund (IMF) and discuss possible reasons why the austerity measures imposed on European Union (EU) countries might have affected Strom Co negatively.**
(10 marks)
- (b) Suggest, giving reasons, why the austerity measures might not have affected clothing retailers at the high and low price range, as much as the mid-price range retailers like Strom Co.**
(4 marks)
- (c) Discuss the risks to Strom Co of reducing the costs relating to quality control and how the detrimental impact of such reductions in costs could be decreased.**
(6 marks)

(20 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

Two asset portfolio

$$s_p = \sqrt{w_a^2 s_a^2 + w_b^2 s_b^2 + 2w_a w_b r_{ab} s_a s_b}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i (E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = br_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d (1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Tuesday 4 June 2013



Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 8–12.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – This ONE question is compulsory and MUST be attempted

- 1 Mlima Co is a private company involved in aluminium mining. About eight years ago, the company was bought out by its management and employees through a leveraged buyout (LBO). Due to high metal prices worldwide, the company has been growing successfully since the LBO. However, because the company has significant debt borrowings with strict restrictive covenants and high interest levels, it has had to reject a number of profitable projects. The company has currently two bonds in issue, as follows:

A 16% secured bond with a nominal value of \$80m, which is redeemable at par in five years. An early redemption option is available on this bond, giving Mlima Co the option to redeem the bond at par immediately if it wants to; and

A 13% unsecured bond with a nominal value of \$40m, which is redeemable at par in ten years.

Mlima Co's Board of Directors (BoD) has been exploring the idea of redeeming both bonds to provide it with more flexibility when making future investment decisions. To do so, the BoD has decided to consider a public listing of the company on a major stock exchange. It is intended that a total of 100 million shares will be issued in the newly-listed company. From the total shares, 20% will be sold to the public, 10% will be offered to the holders of the unsecured bond in exchange for redeeming the bond through an equity-for-debt swap, and the remaining 70% of the equity will remain in the hands of the current owners. The secured bond would be paid out of the funds raised from the listing.

The details of the possible listing and the distribution of equity were published in national newspapers recently. As a result, potential investors suggested that due to the small proportion of shares offered to the public and for other reasons, the shares should be offered at a substantial discount of as much as 20% below the expected share price on the day of the listing.

Mlima Co, financial information

It is expected that after the listing, deployment of new strategies and greater financial flexibility will boost Mlima Co's future sales revenue and, for the next four years, the annual growth rate will be 120% of the previous two years' average growth rate. After the four years, the annual growth rate of the free cash flows to the company will be 3·5%, for the foreseeable future. Operating profit margins are expected to be maintained in the future. Although it can be assumed that the current tax-allowable depreciation is equivalent to the amount of investment needed to maintain the current level of operations, the company will require an additional investment in assets of 30c per \$1 increase in sales revenue for the next four years.

Extracts from Mlima Co's past three years' Statement of Profit or Loss

Year ended	31 May 2013 \$ million	31 May 2012 \$ million	31 May 2011 \$ million
Sales revenue	389·1	366·3	344·7
Operating profit	58·4	54·9	51·7
Net interest costs	17·5	17·7	18·0
Profit before tax	40·9	37·2	33·7
Taxation	10·2	9·3	8·4
Profit after tax	30·7	27·9	25·3

Once listed, Mlima Co will be able to borrow future debt at an interest rate of 7%, which is only 3% higher than the risk-free rate of return. It has no plans to raise any new debt after listing, but any future debt will carry considerably fewer restrictive covenants. However, these plans do not take into consideration the Bahari project (see below).

Bahari Project

Bahari is a small country with agriculture as its main economic activity. A recent geological survey concluded that there may be a rich deposit of copper available to be mined in the north-east of the country. This area is currently occupied by subsistence farmers, who would have to be relocated to other parts of the country. When the results of the survey were announced, some farmers protested that the proposed new farmland where they would be moved to was less fertile and that their communities were being broken up. However, the protesters were intimidated and violently put down by the government, and the state-controlled media stopped reporting about them. Soon afterwards, their protests were ignored and forgotten.

In a meeting between the Bahari government and Mlima Co's BoD, the Bahari government offered Mlima Co exclusive rights to mine the copper. It is expected that there are enough deposits to last at least 15 years. Initial estimates

suggest that the project will generate free cash flows of \$4 million in the first year, rising by 100% per year in each of the next two years, and then by 15% in each of the two years after that. The free cash flows are then expected to stabilise at the year-five level for the remaining 10 years.

The cost of the project, payable at the start, is expected to be \$150 million, comprising machinery, working capital and the mining rights fee payable to the Bahari government. None of these costs is expected to be recoverable at the end of the project's 15-year life.

The Bahari government has offered Mlima Co a subsidised loan over 15 years for the full \$150 million at an interest rate of 3% instead of Mlima Co's normal borrowing rate of 7%. The interest payable is allowable for taxation purposes. It can be assumed that Mlima Co's business risk is not expected to change as a result of undertaking the Bahari project.

At the conclusion of the meeting between the Bahari government and Mlima Co's BoD, the president of Bahari commented that working together would be like old times when he and Mlima Co's chief executive officer (CEO) used to run a business together.

Other Information

Mlima Co's closest competitor is Ziwa Co, a listed company which mines metals worldwide. Mlima Co's directors are of the opinion that after listing Mlima Co's cost of capital should be based on Ziwa Co's ungeared cost of equity. Ziwa Co's cost of capital is estimated at 9.4%, its geared cost of equity is estimated at 16.83% and its pre-tax cost of debt is estimated at 4.76%. These costs are based on a capital structure comprising of 200 million shares, trading at \$7 each, and \$1,700 million 5% irredeemable bonds, trading at \$105 per \$100. Both Ziwa Co and Mlima Co pay tax at an annual rate of 25% on their taxable profits.

It can be assumed that all cash flows will be in \$ instead of the Bahari currency and therefore Mlima Co does not have to take account of any foreign exchange exposure from this venture.

Required:

(a) Prepare a report for the Board of Directors (BoD) of Mlima Co that:

- (i) Explains why Mlima Co's directors are of the opinion that Mlima Co's cost of capital should be based on Ziwa Co's ungeared cost of equity and, showing relevant calculations, estimate an appropriate cost of capital for Mlima Co;** (7 marks)
- (ii) Estimates Mlima Co's value without undertaking the Bahari project and then with the Bahari project. The valuations should use the free cash flow methodology and the cost of capital calculated in part (i). Include relevant calculations;** (14 marks)
- (iii) Advises the BoD whether or not the unsecured bond holders are likely to accept the equity-for-debt swap offer. Include relevant calculations;** (5 marks)
- (iv) Advises the BoD on the listing and the possible share price range, if a total of 100 million shares are issued. The advice should also include:**
 - A discussion of the assumptions made in estimating the share price range;
 - In addition to the reasons mentioned in the scenario above, a brief explanation of other possible reasons for changing its status from a private company to a listed one; and
 - An assessment of the possible reasons for issuing the share price at a discount for the initial listing;(12 marks)

Professional marks will be awarded in part (a) for the format, structure and presentation of the report.

(4 marks)

(b) Discuss the possible impact on, and response of, Mlima Co to the following ethical issues, with respect to the Bahari project:

- (i) The relocation of the farmers; and**
- (ii) The relationship between the Bahari president and Mlima Co's chief executive officer.**

Note: The total marks will be split equally between each part.

(8 marks)

(50 marks)

Section B – TWO questions ONLY to be attempted

- 2 Hav Co is a publicly listed company involved in the production of highly technical and sophisticated electronic components for complex machinery. It has a number of diverse and popular products, an active research and development department, significant cash reserves and a highly talented management who are very good in getting products to market quickly.

A new industry that Hav Co is looking to venture into is biotechnology, which has been expanding rapidly and there are strong indications that this recent growth is set to continue. However, Hav Co has limited experience in this industry. Therefore it believes that the best and quickest way to expand would be through acquiring a company already operating in this industry sector.

Strand Co is a private company operating in the biotechnology industry and is owned by a consortium of business angels and company managers. The owner-managers are highly skilled scientists who have developed a number of technically complex products, but have found it difficult to commercialise them. They have also been increasingly constrained by the lack of funds to develop their innovative products further.

Discussions have taken place about the possibility of Strand Co being acquired by Hav Co. Strand Co's managers have indicated that the consortium of owners is happy for the negotiations to proceed. If Strand Co is acquired, it is expected that its managers would continue to run the Strand Co part of the larger combined company.

Strand Co is of the opinion that most of its value is in its intangible assets, comprising intellectual capital. Therefore, the premium payable on acquisition should be based on the present value to infinity of the after tax excess earnings the company has generated in the past three years, over the average return on capital employed of the biotechnological industry. However, Hav Co is of the opinion that the premium should be assessed on synergy benefits created by the acquisition and the changes in value, due to the changes in the price-to-earnings (PE) ratio before and after the acquisition.

Given below are extracts of financial information for Hav Co for 2013 and Strand Co for 2011, 2012 and 2013:

Year ended 30 April	Hav Co	Strand Co		
	2013	2013	2012	2011
	\$ million	\$ million	\$ million	\$ million
Earnings before tax	1,980	397	370	352
Non-current assets	3,965	882	838	801
Current assets	968	210	208	198
Share capital (25c/share)	600	300	300	300
Reserves	2,479	183	166	159
Non-current liabilities	1,500	400	400	400
Current liabilities	354	209	180	140

The current average PE ratio of the biotechnology industry is 16.4 times and it has been estimated that Strand Co's PE ratio is 10% higher than this. However, it is thought that the PE ratio of the combined company would fall to 14.5 times after the acquisition. The annual after tax earnings will increase by \$140 million due to synergy benefits resulting from combining the two companies.

Both companies pay tax at 20% per annum and Strand Co's annual cost of capital is estimated at 7%. Hav Co's current share price is \$9.24 per share. The biotechnology industry's pre-tax return on capital employed is currently estimated to be 20% per annum.

Hav Co has proposed to pay for the acquisition using one of the following three methods:

- A cash offer of \$5.72 for each Strand Co share; or
- A cash offer of \$1.33 for each Strand Co share plus one Hav Co share for every two Strand Co shares; or
- A cash offer of \$1.25 for each Strand Co share plus one \$100 3% convertible bond for every \$5 nominal value of Strand Co shares. In six years, the bond can be converted into 12 Hav Co shares or redeemed at par.

Required:

- (a) Distinguish between the different types of synergy and discuss possible sources of synergy based on the above scenario. (9 marks)
- (b) Based on the two different opinions expressed by Hav Co and Strand Co, calculate the maximum acquisition premium payable in each case. (6 marks)
- (c) Calculate the percentage premium per share that Strand Co's shareholders will receive under each acquisition payment method and justify, with explanations, which payment method would be most acceptable to them. (10 marks)

(25 marks)

- 3** Kenduri Co is a large multinational company based in the UK with a number of subsidiary companies around the world. Currently, foreign exchange exposure as a result of transactions between Kenduri Co and its subsidiary companies is managed by each company individually. Kenduri Co is considering whether or not to manage the foreign exchange exposure using multilateral netting from the UK, with the Sterling Pound (£) as the base currency. If multilateral netting is undertaken, spot mid-rates would be used.

The following cash flows are due in three months between Kenduri Co and three of its subsidiary companies. The subsidiary companies are Lakama Co, based in the United States (currency US\$), Jaia Co, based in Canada (currency CAD) and Gochiso Co, based in Japan (currency JPY).

Owed by	Owed to	Amount
Kenduri Co	Lakama Co	US\$ 4.5 million
Kenduri Co	Jaia Co	CAD 1.1 million
Gochiso Co	Jaia Co	CAD 3.2 million
Gochiso Co	Lakama Co	US\$ 1.4 million
Jaia Co	Lakama Co	US\$ 1.5 million
Jaia Co	Kenduri Co	CAD 3.4 million
Lakama Co	Gochiso Co	JPY 320 million
Lakama Co	Kenduri Co	US\$ 2.1 million

Exchange rates available to Kenduri Co

	US\$/£1	CAD/£1	JPY/£1
Spot	1.5938–1.5962	1.5690–1.5710	131.91–133.59
3-month forward	1.5996–1.6037	1.5652–1.5678	129.15–131.05

Currency options available to Kenduri Co

Contract size £62,500, Exercise price quotation: US\$/£1, Premium: cents per £1

Exercise price	Call Options		Put Options	
	3-month expiry	6-month expiry	3-month expiry	6-month expiry
1.60	1.55	2.25	2.08	2.23
1.62	0.98	1.58	3.42	3.73

It can be assumed that option contracts expire at the end of the relevant month

Annual interest rates available to Kenduri Co and subsidiaries

	Borrowing rate	Investing rate
UK	4.0%	2.8%
United States	4.8%	3.1%
Canada	3.4%	2.1%
Japan	2.2%	0.5%

Required:

- Advise Kenduri Co on, and recommend, an appropriate hedging strategy for the US\$ cash flows it is due to receive or pay in three months, from Lakama Co. Show all relevant calculations to support the advice given.** (12 marks)
- Calculate, using a tabular format (transactions matrix), the impact of undertaking multilateral netting by Kenduri Co and its three subsidiary companies for the cash flows due in three months. Briefly discuss why some governments allow companies to undertake multilateral netting, while others do not.** (10 marks)
- When examining different currency options and their risk factors, it was noticed that a long call option had a high gamma value. Explain the possible characteristics of a long call option with a high gamma value.** (3 marks)

(25 marks)

- 4 Limni Co is a large company manufacturing hand-held electronic devices such as mobile phones and tablet computers. The company has been growing rapidly over the last few years, but it also has high research and development expenditure. It is involved in a number of projects worldwide, developing new and innovative products and systems in a rapidly changing industry. Due to the nature of the industry, this significant growth in earnings has never been stable, but has depended largely on the success of the new innovations and competitor actions. However, in the last two years it seems that the rapid period of growth is slowing, with fewer products coming to market compared to previous years.

Limni Co has never paid dividends and has financed projects through internally generated funds and with occasional rights issues of new share capital. It currently has insignificant levels of debt. The retained cash reserves have recently grown because of a drop in the level of investment in new projects.

The company has an active treasury division which invests spare funds in traded equities, bonds and other financial instruments; and releases the funds when required for new projects. The division also manages cash flow risk using money and derivative markets. The treasury division is currently considering investing in three companies with the following profit after tax (PAT) and dividend history:

Year	Company Theta		Company Omega		Company Kappa	
	PAT	Dividends	PAT	Dividends	PAT	Dividends
	\$000	\$000	\$000	\$000	\$000	\$000
2013	57,100	22,840	93,300	60,560	162,400	44,100
2012	54,400	21,760	90,600	57,680	141,500	34,200
2011	52,800	21,120	88,000	54,840	108,900	26,300
2010	48,200	19,280	85,400	52,230	105,700	20,250
2009	45,500	18,200	82,900	49,740	78,300	15,700

All of the three companies' share capital has remained largely unchanged since 2009.

Recently, Limni Co's Board of Directors (BoD) came under pressure from the company's larger shareholders to start returning some of the funds, currently retained by the company, back to the shareholders. The BoD thinks that the shareholders have a strong case to ask for repayments. However, it is unsure whether to pay a special, one-off large dividend from its dividend capacity and retained funds, followed by small annual dividend payments; or to undertake a periodic share buyback scheme over the next few years.

Limni Co is due to prepare its statement of profit or loss shortly and estimates that the annual sales revenue will be \$600 million, on which its profit before tax is expected to be 23% of sales revenue. It charges depreciation of 25% on a straight-line basis on its non-current assets of \$220 million. It estimates that \$67 million investment in current and non-current assets was spent during the year. It is due to receive \$15 million in dividends from its subsidiary companies, on which annual tax of 20% on average has been paid. Limni Co itself pays annual tax at 26%, and the tax authorities where Limni Co is based charge tax on dividend remittances made by overseas subsidiary companies, but give full credit on tax already paid on those remittances. In order to fund the new policy of returning funds to shareholders, Limni Co's BoD wants to increase the current estimated dividend capacity by 10%, by asking the overseas subsidiary companies for higher repatriations.

Required:

- Discuss Limni Co's current dividend, financing and risk management policies, and suggest how the decision to return retained funds back to the shareholders will affect these policies.** (8 marks)
- Evaluate the dividend policies of each of the three companies that Limni Co is considering investing in, and discuss which company Limni Co might select.** (8 marks)
- Calculate, and briefly comment on, how much the dividends from overseas companies need to increase by, to increase Limni Co's dividend capacity by 10%.** (6 marks)
- Discuss the benefits to Limni Co's shareholders of receiving repayments through a share buyback scheme as opposed to the dividend scheme described above.** (3 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

Discount rate (r)

Periods

(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Tuesday 3 December 2013



Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 7–11.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – This ONE question is compulsory and MUST be attempted

- 1 Since becoming independent just over 20 years ago, the country of Mehgam has adopted protectionist measures which have made it difficult for multinational companies to trade there. However, recently, after discussions with the World Trade Organisation (WTO), it seems likely that Mehgam will reduce its protectionist measures significantly.

Encouraged by these discussions, Chmura Co, a company producing packaged foods, is considering a project to set up a manufacturing base in Mehgam to sell its goods there and in other regional countries nearby. An initial investigation costing \$500,000 established that Mehgam had appropriate manufacturing facilities, adequate transport links and a reasonably skilled but cheap work force. The investigation concluded that, if the protectionist measures were reduced, then the demand potential for Chmura Co's products looked promising. It is also felt that an early entry into Mehgam would give Chmura Co an advantage over its competitors for a period of five years, after which the current project will cease, due to the development of new advanced manufacturing processes.

Mehgam's currency, the Peso (MP), is currently trading at MP72 per \$1. Setting up the manufacturing base in Mehgam will require an initial investment of MP2,500 million immediately, to cover the cost of land and buildings (MP1,250 million) and machinery (MP1,250 million). Tax allowable depreciation is available on the machinery at an annual rate of 10% on cost on a straight-line basis. A balancing adjustment will be required at the end of year five, when it is expected that the machinery will be sold for MP500 million (after inflation). The market value of the land and buildings in five years' time is estimated to be 80% of the current value. These amounts are inclusive of any tax impact.

Chmura Co will require MP200 million for working capital immediately. It is not expected that any further injections of working capital will be required for the five years. When the project ceases at the end of the fifth year, the working capital will be released back to Chmura Co.

Production of the packaged foods will take place in batches of product mixes. These batches will then be sold to supermarket chains, wholesalers and distributors in Mehgam and its neighbouring countries, who will repackage them to their individual requirements. All sales will be in MP. The estimated average number of batches produced and sold each year is given below:

Year	1	2	3	4	5
Batches produced and sold	10,000	15,000	30,000	26,000	15,000

The current selling price for each batch is estimated to be MP115,200. The costs related to producing and selling each batch are currently estimated to be MP46,500. In addition to these costs, a number of products will need a special packaging material which Chmura Co will send to Mehgam. Currently the cost of the special packaging material is \$200 per batch. Training and development costs, related to the production of the batches, are estimated to be 80% of the production and selling costs (excluding the cost of the special packaging) in the first year, before falling to 20% of these costs (excluding the cost of the special packaging) in the second year, and then nil for the remaining years. It is expected that the costs relating to the production and sale of each batch will increase annually by 10% but the selling price and the special packaging costs will only increase by 5% every year.

The current annual corporation tax rate in Mehgam is 25% and Chmura Co pays annual corporation tax at a rate of 20% in the country where it is based. Both countries' taxes are payable in the year that the tax liability arises. A bi-lateral tax treaty exists between the two countries which permits offset of overseas tax against any tax liabilities Chmura Co incurs on overseas earnings.

The risk-adjusted cost of capital applicable to the project on \$-based cash flows is 12%, which is considerably higher than the return on short-dated \$ treasury bills of 4%. The current rate of inflation in Mehgam is 8%, and in the country where Chmura Co is based, it is 2%. It can be assumed that these inflation rates will not change for the foreseeable future. All net cash flows from the project will be remitted back to Chmura Co at the end of each year.

Chmura Co's finance director is of the opinion that there are many uncertainties surrounding the project and has assessed that the cash flows can vary by a standard deviation of as much as 35% because of these uncertainties.

Recently Bulud Co offered Chmura Co the option to sell the entire project to Bulud Co for \$28 million at the start of year three. Chmura Co will make the decision of whether or not to sell the project at the end of year two.

Required:

- (a) Discuss the role of the World Trade Organisation (WTO) and the possible benefits and drawbacks to Mehgam of reducing protectionist measures. (9 marks)
- (b) Prepare an evaluative report for the Board of Directors of Chmura Co which addresses the following parts and recommends an appropriate course of action:
- (i) An estimate of the value of the project before considering Bulud Co's offer. Show all relevant calculations; (14 marks)
 - (ii) An estimate of the value of the project taking into account Bulud Co's offer. Show all relevant calculations; (9 marks)
 - (iii) A discussion of the assumptions made in parts (i) and (ii) above and the additional business risks which Chmura Co should consider before it makes the final decision whether or not to undertake the project. (14 marks)

Professional marks will be awarded in part (b) for the format, structure and presentation of the report.

(4 marks)

(50 marks)

Section B – TWO questions ONLY to be attempted

- 2 Awan Co is expecting to receive \$48,000,000 on 1 February 2014, which will be invested until it is required for a large project on 1 June 2014. Due to uncertainty in the markets, the company is of the opinion that it is likely that interest rates will fluctuate significantly over the coming months, although it is difficult to predict whether they will increase or decrease.

Awan Co's treasury team want to hedge the company against adverse movements in interest rates using one of the following derivative products:

Forward rate agreements (FRAs);
Interest rate futures; or
Options on interest rate futures.

Awan Co can invest funds at the relevant inter-bank rate less 20 basis points. The current inter-bank rate is 4.09%. However, Awan Co is of the opinion that interest rates could increase or decrease by as much as 0.9% over the coming months.

The following information and quotes are provided from an appropriate exchange on \$ futures and options. Margin requirements can be ignored.

Three-month \$ futures, \$2,000,000 contract size
Prices are quoted in basis points at 100 – annual % yield

December 2013: 94.80
March 2014: 94.76
June 2014: 94.69

Options on three-month \$ futures, \$2,000,000 contract size, option premiums are in annual %

Calls			Strike	Puts		
December	March	June		December	March	June
0.342	0.432	0.523	94.50	0.090	0.119	0.271
0.097	0.121	0.289	95.00	0.312	0.417	0.520

Voblaka Bank has offered the following FRA rates to Awan Co:

1–7: 4.37%
3–4: 4.78%
3–7: 4.82%
4–7: 4.87%

It can be assumed that settlement for the futures and options contracts is at the end of the month and that basis diminishes to zero at contract maturity at a constant rate, based on monthly time intervals. Assume that it is 1 November 2013 now and that there is no basis risk.

Required:

- (a) Based on the three hedging choices Awan Co is considering, recommend a hedging strategy for the \$48,000,000 investment, if interest rates increase or decrease by 0.9%. Support your answer with appropriate calculations and discussion. (19 marks)
- (b) A member of Awan Co's treasury team has suggested that if option contracts are purchased to hedge against the interest rate movements, then the number of contracts purchased should be determined by a hedge ratio based on the delta value of the option.

Required:

Discuss how the delta value of an option could be used in determining the number of contracts purchased. (6 marks)

(25 marks)

- 3 Makonis Co, a listed company producing motor cars, wants to acquire Nuvola Co, an engineering company involved in producing innovative devices for cars. Makonis Co is keen to incorporate some of Nuvola Co's innovative devices into its cars and thereby boosting sales revenue.

The following financial information is provided for the two companies:

	Makonis Co	Nuvola Co
Current share price	\$5.80	\$2.40
Number of issued shares	210 million	200 million
Equity beta	1.2	1.2
Asset beta	0.9	1.2

It is thought that combining the two companies will result in several benefits. Free cash flows to firm of the combined company will be \$216 million in current value terms, but these will increase by an annual growth rate of 5% for the next four years, before reverting to an annual growth rate of 2.25% in perpetuity. In addition to this, combining the companies will result in cash synergy benefits of \$20 million per year, for the next four years. These synergy benefits are not subject to any inflationary increase and no synergy benefits will occur after the fourth year. The debt-to-equity ratio of the combined company will be 40:60 in market value terms and it is expected that the combined company's cost of debt will be 4.55%.

The corporation tax rate is 20%, the current risk free rate of return is 2% and the market risk premium is 7%. It can be assumed that the combined company's asset beta is the weighted average of Makonis Co's and Nuvola Co's asset betas, weighted by their current market values.

Makonis Co has offered to acquire Nuvola Co through a mixed offer of one of its shares for two Nuvola Co shares plus a cash payment, such that a 30% premium is paid for the acquisition. Nuvola Co's equity holders feel that a 50% premium would be more acceptable. Makonis Co has sufficient cash reserves if the premium is 30%, but not if it is 50%.

Required:

- (a) Estimate the additional equity value created by combining Nuvola Co and Makonis Co, based on the free cash flows to firm method. Comment on the results obtained and briefly discuss the assumptions made. (13 marks)
- (b) Estimate the impact on Makonis Co's equity holders if the premium paid is increased to 50% from 30%. (5 marks)
- (c) Estimate the additional funds required if a premium of 50% is paid instead of 30% and discuss how this premium could be financed. (7 marks)

(25 marks)

- 4 Nubo Co has divisions operating in two diverse sectors: production of aircraft parts and supermarkets. Whereas the aircraft parts production division has been growing rapidly, the supermarkets division's growth has been slower. The company is considering selling the supermarkets division and focusing solely on the aircraft parts production division.

Extracts from the Nubo Co's most recent financial statements are as follows:

Year ended 30 November	2013
	\$m
Profit after tax	166
Non-current assets	550
Current assets	122
Non-current liabilities	387
Current liabilities	95

About 70% of Nubo Co's non-current assets and current assets are attributable to the supermarkets division and the remainder to the aircraft parts production division. Each of the two divisions generates roughly half of the total profit after tax. The market value of the two divisions is thought to be equivalent to the price-to-earnings (PE) ratios of the two divisions' industries. The supermarket industry's PE ratio is 7 and the aircraft parts production industry's PE ratio is 12.

Nubo Co can either sell the supermarkets division as a going concern or sell the assets of the supermarkets division separately. If the assets are sold separately, Nubo Co believes that it can sell the non-current assets for 115% of the book value and the current assets for 80% of the book value. The funds raised from the sale of the supermarkets division will be used to pay for all the company's current and non-current liabilities.

Following the sale of the supermarkets division and paying off the liabilities, Nubo Co will raise additional finance for new projects in the form of debt. It will be able to borrow up to a maximum of 100% of the total asset value of the new downsized company.

One of the new projects which Nubo Co is considering is a joint venture with Pilvi Co to produce an innovative type of machinery which will be used in the production of light aircraft and private jets. Both companies will provide the expertise and funding required for the project equally. Representatives from both companies will make up the senior management team and decisions will be made jointly. Legal contracts will be drawn up once profit-sharing and other areas have been discussed by the companies and agreed on.

Pilvi Co has approached Ulap Bank for the finance it requires for the venture, based on Islamic finance principles. Ulap Bank has agreed to consider the request from Pilvi Co, but because the financing requirement will be for a long period of time and because of uncertainties surrounding the project, Ulap Bank wants to provide the finance based on the principles of a Musharaka contract, with Ulap Bank requiring representation on the venture's senior management team. Normally Ulap Bank provides funds based on the principles of a Mudaraba contract, which the bank provides for short-term, low-risk projects, where the responsibility for running a project rests solely with the borrower.

Required:

- (a) Advise Nubo Co whether it should sell the supermarkets division as a going concern or sell the assets separately and estimate the additional cash and debt funds which could be available to the new, downsized company. Show all relevant calculations. (7 marks)
- (b) An alternative to selling the supermarkets division would be to demerge both the divisions. In this case, all of Nubo Co's liabilities would be taken over by the demerged supermarkets division. Also, either of the demerged companies can borrow up to 100% of their respective total asset values.

Required:

Discuss whether a demerger of the supermarkets division may be more appropriate than a sale. (6 marks)

- (c) Discuss why Ulap Bank may want to consider providing the finance based on a Musharaka contract instead of a Mudaraba contract, and the key concerns Nubo Co may have from the arrangement between Pilvi Co and Ulap Bank. (12 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Tuesday 3 June 2014



Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 9–13.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – This ONE question is compulsory and MUST be attempted

- 1 Cocoa-Mocha-Chai (CMC) Co is a large listed company based in Switzerland and uses Swiss Francs as its currency. It imports tea, coffee and cocoa from countries around the world, and sells its blended products to supermarkets and large retailers worldwide. The company has production facilities located in two European ports where raw materials are brought for processing, and from where finished products are shipped out. All raw material purchases are paid for in US dollars (US\$), while all sales are invoiced in Swiss Francs (CHF).

Until recently CMC Co had no intention of hedging its foreign currency exposures, interest rate exposures or commodity price fluctuations, and stated this intent in its annual report. However, after consultations with senior and middle managers, the company's new Board of Directors (BoD) has been reviewing its risk management and operations strategies.

The following two proposals have been put forward by the BoD for further consideration:

Proposal one

Setting up a treasury function to manage the foreign currency and interest rate exposures (but not commodity price fluctuations) using derivative products. The treasury function would be headed by the finance director. The purchasing director, who initiated the idea of having a treasury function, was of the opinion that this would enable her management team to make better decisions. The finance director also supported the idea as he felt this would increase his influence on the BoD and strengthen his case for an increase in his remuneration.

In order to assist in the further consideration of this proposal, the BoD wants you to use the following upcoming foreign currency and interest rate exposures to demonstrate how they would be managed by the treasury function:

- (i) a payment of US\$5,060,000 which is due in four months' time; and
- (ii) a four-year CHF60,000,000 loan taken out to part-fund the setting up of four branches (see proposal two below). Interest will be payable on the loan at a fixed annual rate of 2.2% or a floating annual rate based on the yield curve rate plus 0.40%. The loan's principal amount will be repayable in full at the end of the fourth year.

Proposal two

This proposal suggested setting up four new branches in four different countries. Each branch would have its own production facilities and sales teams. As a consequence of this, one of the two European-based production facilities will be closed. Initial cost-benefit analysis indicated that this would reduce costs related to production, distribution and logistics, as these branches would be closer to the sources of raw materials and also to the customers. The operations and sales directors supported the proposal, as in addition to above, this would enable sales and marketing teams in the branches to respond to any changes in nearby markets more quickly. The branches would be controlled and staffed by the local population in those countries. However, some members of the BoD expressed concern that such a move would create agency issues between CMC Co's central management and the management controlling the branches. They suggested mitigation strategies would need to be established to minimise these issues.

Response from the non-executive directors

When the proposals were put to the non-executive directors, they indicated that they were broadly supportive of the second proposal if the financial benefits outweigh the costs of setting up and running the four branches. However, they felt that they could not support the first proposal, as this would reduce shareholder value because the costs related to undertaking the proposal are likely to outweigh the benefits.

Additional information relating to proposal one

The current spot rate is US\$1.0635 per CHF1. The current annual inflation rate in the USA is three times higher than Switzerland.

The following derivative products are available to CMC Co to manage the exposures of the US\$ payment and the interest on the loan:

Exchange-traded currency futures

Contract size CHF125,000 price quotation: US\$ per CHF1

3-month expiry	1.0647
6-month expiry	1.0659

Exchange-traded currency options

Contract size CHF125,000, exercise price quotation: US\$ per CHF1, premium: cents per CHF1

Exercise price	Call Options		Put Options	
	3-month expiry	6-month expiry	3-month expiry	6-month expiry
1·06	1·87	2·75	1·41	2·16
1·07	1·34	2·22	1·88	2·63

It can be assumed that futures and option contracts expire at the end of the month and transaction costs related to these can be ignored.

Over-the-counter products

In addition to the exchange-traded products, Pecunia Bank is willing to offer the following over-the-counter derivative products to CMC Co:

- (i) A forward rate between the US\$ and the CHF of US\$ 1·0677 per CHF1.
- (ii) An interest rate swap contract with a counterparty, where the counterparty can borrow at an annual floating rate based on the yield curve rate plus 0·8% or an annual fixed rate of 3·8%. Pecunia Bank would charge a fee of 20 basis points each to act as the intermediary of the swap. Both parties will benefit equally from the swap contract.

Required:

- (a) Advise CMC Co on an appropriate hedging strategy to manage the foreign exchange exposure of the US\$ payment in four months' time. Show all relevant calculations, including the number of contracts bought or sold in the exchange-traded derivative markets. (15 marks)
- (b) Demonstrate how CMC Co could benefit from the swap offered by Pecunia Bank. (6 marks)
- (c) As an alternative to paying the principal on the loan as one lump sum at the end of the fourth year, CMC Co could pay off the loan in equal annual amounts over the four years similar to an annuity. In this case, an annual interest rate of 2% would be payable, which is the same as the loan's gross redemption yield (yield to maturity).

Required:

Calculate the modified duration of the loan if it is repaid in equal amounts and explain how duration can be used to measure the sensitivity of the loan to changes in interest rates. (7 marks)

- (d) Prepare a memorandum for the Board of Directors (BoD) of CMC Co which:

- (i) Discusses proposal one in light of the concerns raised by the non-executive directors; and (9 marks)
- (ii) Discusses the agency issues related to proposal two and how these can be mitigated. (9 marks)

Professional marks will be awarded in part (d) for the presentation, structure, logical flow and clarity of the memorandum. (4 marks)

(50 marks)

Section B – TWO questions ONLY to be attempted

- 2 You have recently commenced working for Burung Co and are reviewing a four-year project which the company is considering for investment. The project is in a business activity which is very different from Burung Co's current line of business.

The following net present value estimate has been made for the project:

All figures are in \$ million

Year	0	1	2	3	4
Sales revenue		23·03	36·60	49·07	27·14
Direct project costs		(13·82)	(21·96)	(29·44)	(16·28)
Interest		(1·20)	(1·20)	(1·20)	(1·20)
Profit		8·01	13·44	18·43	9·66
Tax (20%)		(1·60)	(2·69)	(3·69)	(1·93)
Investment/sale	(38·00)				4·00
Cash flows	(38·00)	6·41	10·75	14·74	11·73
Discount factors (7%)	1	0·935	0·873	0·816	0·763
Present values	(38·00)	5·99	9·38	12·03	8·95

Net present value is negative \$1·65 million, and therefore the recommendation is that the project should not be accepted.

In calculating the net present value of the project, the following notes were made:

- Since the real cost of capital is used to discount cash flows, neither the sales revenue nor the direct project costs have been inflated. It is estimated that the inflation rate applicable to sales revenue is 8% per year and to the direct project costs is 4% per year.
- The project will require an initial investment of \$38 million. Of this, \$16 million relates to plant and machinery, which is expected to be sold for \$4 million when the project ceases, after taking any taxation and inflation impact into account.
- Tax allowable depreciation is available on the plant and machinery at 50% in the first year, followed by 25% per year thereafter on a reducing balance basis. A balancing adjustment is available in the year the plant and machinery is sold. Burung Co pays 20% tax on its annual taxable profits. No tax allowable depreciation is available on the remaining investment assets and they will have a nil value at the end of the project.
- Burung Co uses either a nominal cost of capital of 11% or a real cost of capital of 7% to discount all projects, given that the rate of inflation has been stable at 4% for a number of years.
- Interest is based on Burung Co's normal borrowing rate of 150 basis points over the 10-year government yield rate.
- At the beginning of each year, Burung Co will need to provide working capital of 20% of the anticipated sales revenue for the year. Any remaining working capital will be released at the end of the project.
- Working capital and depreciation have not been taken into account in the net present value calculation above, since depreciation is not a cash flow and all the working capital is returned at the end of the project.

It is anticipated that the project will be financed entirely by debt, 60% of which will be obtained from a subsidised loan scheme run by the government, which lends money at a rate of 100 basis points below the 10-year government debt yield rate of 2·5%. Issue costs related to raising the finance are 2% of the gross finance required. The remaining 40% will be funded from Burung Co's normal borrowing sources. It can be assumed that the debt capacity available to Burung Co is equal to the actual amount of debt finance raised for the project.

Burung Co has identified a company, Lintu Co, which operates in the same line of business as that of the project it is considering. Lintu Co is financed by 40 million shares trading at \$3·20 each and \$34 million debt trading at \$94 per \$100. Lintu Co's equity beta is estimated at 1·5. The current yield on government treasury bills is 2% and it is estimated that the market risk premium is 8%. Lintu Co pays tax at an annual rate of 20%.

Both Burung Co and Lintu Co pay tax in the same year as when profits are earned.

Required:

- (a) Calculate the adjusted present value (APV) for the project, correcting any errors made in the net present value estimate above, and conclude whether the project should be accepted or not. Show all relevant calculations. (15 marks)
- (b) Comment on the corrections made to the original net present value estimate and explain the APV approach taken in part (a), including any assumptions made. (10 marks)

(25 marks)

- 3 Vogel Co, a listed engineering company, manufactures large scale plant and machinery for industrial companies. Until ten years ago, Vogel Co pursued a strategy of organic growth. Since then, it has followed an aggressive policy of acquiring smaller engineering companies, which it feels have developed new technologies and methods, which could be used in its manufacturing processes. However, it is estimated that only between 30% and 40% of the acquisitions made in the last ten years have successfully increased the company's shareholder value.

Vogel Co is currently considering acquiring Tori Co, an unlisted company, which has three departments. Department A manufactures machinery for industrial companies, Department B produces electrical goods for the retail market, and the smaller Department C operates in the construction industry. Upon acquisition, Department A will become part of Vogel Co, as it contains the new technologies which Vogel Co is seeking, but Departments B and C will be unbundled, with the assets attached to Department C sold and Department B being spun off into a new company called Ndege Co.

Given below are extracts of financial information for the two companies for the year ended 30 April 2014.

	Vogel Co \$ million	Tori Co \$ million
Sales revenue	790.2	124.6
Profit before depreciation, interest and tax (PBDIT)	244.4	37.4
Interest	13.8	4.3
Depreciation	72.4	10.1
Pre-tax profit	158.2	23.0

	Vogel Co \$ million	Tori Co \$ million
Non-current assets	723.9	98.2
Current assets	142.6	46.5
7% unsecured bond	—	40.0
Other non-current and current liabilities	212.4	20.2
Share capital (50c/share)	190.0	20.0
Reserves	464.1	64.5

Share of current and non-current assets and profit of Tori Co's three departments:

	Department A	Department B	Department C
Share of current and non-current assets	40%	40%	20%
Share of PBDIT and pre-tax profit	50%	40%	10%

Other information

- It is estimated that for Department C, the realisable value of its non-current assets is 100% of their book value, but its current assets' realisable value is only 90% of their book value. The costs related to closing Department C are estimated to be \$3 million.
- The funds raised from the disposal of Department C will be used to pay off Tori Co's other non-current and current liabilities.
- The 7% unsecured bond will be taken over by Ndege Co. It can be assumed that the current market value of the bond is equal to its book value.
- At present, around 10% of Department B's PBDIT come from sales made to Department C.
- Ndege Co's cost of capital is estimated to be 10%. It is estimated that in the first year of operation Ndege Co's free cash flows to firm will grow by 20%, and then by 5.2% annually thereafter.
- The tax rate applicable to all the companies is 20%, and Ndege Co can claim 10% tax allowable depreciation on its non-current assets. It can be assumed that the amount of tax allowable depreciation is the same as the investment needed to maintain Ndege Co's operations.

- (vii) Vogel Co's current share price is \$3 per share and it is estimated that Tori Co's price-to-earnings (PE) ratio is 25% higher than Vogel Co's PE ratio. After the acquisition, when Department A becomes part of Vogel Co, it is estimated that Vogel Co's PE ratio will increase by 15%.
- (viii) It is estimated that the combined company's annual after-tax earnings will increase by \$7 million due to the synergy benefits resulting from combining Vogel Co and Department A.

Required:

- (a) **Discuss the possible reasons why Vogel Co may have switched its strategy of organic growth to one of growing by acquiring companies.** (4 marks)
- (b) **Discuss the possible actions Vogel Co could take to reduce the risk that the acquisition of Tori Co fails to increase shareholder value.** (7 marks)
- (c) **Estimate, showing all relevant calculations, the maximum premium Vogel Co could pay to acquire Tori Co, explaining the approach taken and any assumptions made.** (14 marks)

(25 marks)

- 4 The chief executive officer (CEO) of Faoilean Co has just returned from a discussion at a leading university on the 'application of options to investment decisions and corporate value'. She wants to understand how some of the ideas which were discussed can be applied to decisions made at Faoilean Co. She is still a little unclear about some of the discussion on options and their application, and wants further clarification on the following:
- (i) Faoilean Co is involved in the exploration and extraction of oil and gas. Recently there have been indications that there could be significant deposits of oil and gas just off the shores of Ireland. The government of Ireland has invited companies to submit bids for the rights to commence the initial exploration of the area to assess the likelihood and amount of oil and gas deposits, with further extraction rights to follow. Faoilean Co is considering putting in a bid for the rights. The speaker leading the discussion suggested that using options as an investment assessment tool would be particularly useful to Faoilean Co in this respect.
 - (ii) The speaker further suggested that options were useful in determining the value of equity and default risk, and suggested that this was why companies facing severe financial distress could still have a positive equity value.
 - (iii) Towards the end of the discussion, the speaker suggested that changes in the values of options can be measured in terms of a number of risk factors known as the 'greeks', such as the 'vega'. The CEO is unclear why option values are affected by so many different risk factors.

Required:

- (a) **With regard to (i) above, discuss how Faoilean Co may use the idea of options to help with the investment decision in bidding for the exploration rights, and explain the assumptions made when using the idea of options in making investment decisions.** (11 marks)
- (b) **With regard to (ii) above, discuss how options could be useful in determining the value of equity and default risk, and why companies facing severe financial distress still have positive equity values.** (9 marks)
- (c) **With regard to (iii) above, explain why changes in option values are determined by numerous different risk factors and what 'vega' determines.** (5 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

Discount rate (r)

Periods

(n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Tuesday 2 December 2014



Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 8–12.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – This ONE question is compulsory and MUST be attempted

1 Nahara Co and Fugae Co

Nahara Co is a private holding company owned by the government of a wealthy oil-rich country to invest its sovereign funds. Nahara Co has followed a strategy of risk diversification for a number of years by acquiring companies from around the world in many different sectors.

One of Nahara Co's acquisition strategies is to identify and purchase undervalued companies in the airline industry in Europe. A recent acquisition was Fugae Co, a company based in a country which is part of the European Union (EU). Fugae Co repairs and maintains aircraft engines.

A few weeks ago, Nahara Co stated its intention to pursue the acquisition of an airline company based in the same country as Fugae Co. The EU, concerned about this, asked Nahara Co to sell Fugae Co before pursuing any further acquisitions in the airline industry.

Avem Co's acquisition interest in Fugae Co

Avem Co, a UK-based company specialising in producing and servicing business jets, has approached Nahara Co with a proposal to acquire Fugae Co for \$1,200 million. Nahara Co expects to receive a premium of at least 30% on the estimated equity value of Fugae Co, if it is sold.

Given below are extracts from the most recent statements of financial position of both Avem Co and Fugae Co.

	Avem Co \$ million	Fugae Co \$ million
Share capital (50c/share)	800	100
Reserves	3,550	160
Non-current liabilities	2,200	380
Current liabilities	130	30
Total capital and liabilities	6,680	670

Each Avem Co share is currently trading at \$7.50, which is a multiple of 7.2 of its free cash flow to equity. Avem Co expects that the total free cash flows to equity of the combined company will increase by \$40 million due to synergy benefits. After adding the synergy benefits of \$40 million, Avem Co then expects the multiple of the total free cash flow of the combined company to increase to 7.5.

Fugae Co's free cash flow to equity is currently estimated at \$76.5 million and it is expected to generate a return on equity of 11%. Over the past few years, Fugae Co has returned 77.3% of its annual free cash flow to equity back to Nahara Co, while retaining the balance for new investments.

Fugae Co's non-current liabilities consist entirely of \$100 nominal value bonds which are redeemable in four years at the nominal value, on which the company pays a coupon of 5.4%. The debt is rated at B+ and the credit spread on B+ rated debt is 80 basis points above the risk-free rate of return.

Proposed luxury transport investment project by Fugae Co

In recent years, the country in which Fugae Co is based has been expanding its tourism industry and hopes that this industry will grow significantly in the near future. At present tourists normally travel using public transport and taxis, but there is a growing market for luxury travel. If the tourist industry does expand, then the demand for luxury travel is expected to grow rapidly. Fugae Co is considering entering this market through a four-year project. The project will cease after four years because of increasing competition.

The initial cost of the project is expected to be \$42,000,000 and it is expected to generate the following after-tax cash flows over its four-year life:

Year	1	2	3	4
Cash flows (\$000s)	3,277.6	16,134.3	36,504.7	35,683.6

The above figures are based on the tourism industry expanding as expected. However, it is estimated that there is a 25% probability that the tourism industry will not grow as expected in the first year. If this happens, then the present value of the project's cash flows will be 50% of the original estimates over its four-year life.

It is also estimated that if the tourism industry grows as expected in the first year, there is still a 20% probability that the expected growth will slow down in the second and subsequent years, and the present value of the project's cash flows would then be 40% of the original estimates in each of these years.

Lumi Co, a leisure travel company, has offered \$50 million to buy the project from Fugae Co at the start of the second year. Fugae Co is considering whether having this choice would add to the value of the project.

If Fugae Co is bought by Avem Co after the project has begun, it is thought that the project will not result in any additional synergy benefits and will not generate any additional value for the combined company, above any value the project has already generated for Fugae Co.

Although there is no beta for companies offering luxury forms of travel in the tourist industry, Reka Co, a listed company, offers passenger transportation services on coaches, trains and luxury vehicles. About 15% of its business is in the luxury transport market and Reka Co's equity beta is 1.6. It is estimated that the asset beta of the non-luxury transport industry is 0.80. Reka Co's shares are currently trading at \$4.50 per share and its debt is currently trading at \$105 per \$100. It has 80 million shares in issue and the book value of its debt is \$340 million. The debt beta is estimated to be zero.

General information

The corporation tax rate applicable to all companies is 20%. The risk-free rate is estimated to be 4% and the market risk premium is estimated to be 6%.

Required:

- (a) Discuss whether or not Nahara Co's acquisition strategies, of pursuing risk diversification and of purchasing undervalued companies, can be valid.** (7 marks)
- (b) Discuss why the European Union (EU) may be concerned about Nahara Co's stated intention and how selling Fugae Co could reduce this concern.** (4 marks)
- (c) Prepare a report for the Board of Directors of Avem Co, which:**
 - (i) Estimates the additional value created for Avem Co, if it acquires Fugae Co without considering the luxury transport project;** (10 marks)
 - (ii) Estimates the additional value of the luxury transport project to Fugae Co, both with and without the offer from Lumi Co;** (18 marks)
 - (iii) Evaluates the benefit attributable to Avem Co and Fugae Co from combining the two companies with and without the project, and concludes whether or not the acquisition is beneficial. The evaluation should include any assumptions made.** (7 marks)

Professional marks will be awarded in part (c) for the format, structure and presentation of the report.

(4 marks)

(50 marks)

Section B – TWO questions ONLY to be attempted

- 2 Keshi Co is a large multinational company with a number of international subsidiary companies. A centralised treasury department manages Keshi Co and its subsidiaries' borrowing requirements, cash surplus investment and financial risk management. Financial risk is normally managed using conventional derivative products such as forwards, futures, options and swaps.

Assume it is 1 December 2014 today and Keshi Co is expecting to borrow \$18,000,000 on 1 February 2015 for a period of seven months. It can either borrow the funds at a variable rate of LIBOR plus 40 basis points or a fixed rate of 5.5%. LIBOR is currently 3.8% but Keshi Co feels that this could increase or decrease by 0.5% over the coming months due to increasing uncertainty in the markets.

The treasury department is considering whether or not to hedge the \$18,000,000, using either exchange-traded March options or over-the-counter swaps offered by Rozu Bank.

The following information and quotes for \$ March options are provided from an appropriate exchange. The options are based on three-month \$ futures, \$1,000,000 contract size and option premiums are in annual %.

March calls	Strike price	March puts
0.882	95.50	0.662
0.648	96.00	0.902

Option prices are quoted in basis points at 100 minus the annual % yield and settlement of the options contracts is at the end of March 2015. The current basis on the March futures price is 44 points; and it is expected to be 33 points on 1 January 2015, 22 points on 1 February 2015 and 11 points on 1 March 2015.

Rozu Bank has offered Keshi Co a swap on a counterparty variable rate of LIBOR plus 30 basis points or a fixed rate of 4.6%, where Keshi Co receives 70% of any benefits accruing from undertaking the swap, prior to any bank charges. Rozu Bank will charge Keshi Co 10 basis points for the swap.

Keshi Co's chief executive officer believes that a centralised treasury department is necessary in order to increase shareholder value, but Keshi Co's new chief financial officer (CFO) thinks that having decentralised treasury departments operating across the subsidiary companies could be more beneficial. The CFO thinks that this is particularly relevant to the situation which Suisen Co, a company owned by Keshi Co, is facing.

Suisen Co operates in a country where most companies conduct business activities based on Islamic finance principles. It produces confectionery products including chocolates. It wants to use Salam contracts instead of commodity futures contracts to hedge its exposure to price fluctuations of cocoa. Salam contracts involve a commodity which is sold based on currently agreed prices, quantity and quality. Full payment is received by the seller immediately, for an agreed delivery to be made in the future.

Required:

- (a) Based on the two hedging choices Keshi Co is considering, recommend a hedging strategy for the \$18,000,000 borrowing. Support your answer with appropriate calculations and discussion. (15 marks)
- (b) Discuss how a centralised treasury department may increase value for Keshi Co and the possible reasons for decentralising the treasury department. (6 marks)
- (c) Discuss the key differences between a Salam contract, under Islamic finance principles, and futures contracts. (4 marks)

(25 marks)

- 3** Riviere Co is a small company based in the European Union (EU). It produces high quality frozen food which it exports to a small number of supermarket chains located within the EU as well. The EU is a free trade area for trade between its member countries.

Riviere Co finds it difficult to obtain bank finance and relies on a long-term strategy of using internally generated funds for new investment projects. This constraint means that it cannot accept every profitable project and often has to choose between them.

Riviere Co is currently considering investment in one of two mutually exclusive food production projects: Privi and Drugi. Privi will produce and sell a new range of frozen desserts exclusively within the EU. Drugi will produce and sell a new range of frozen desserts and savoury foods to supermarket chains based in countries outside the EU. Each project will last for five years and the following financial information refers to both projects.

Project Drugi, annual after-tax cash flows expected at the end of each year (€000s)

Year	Current	1	2	3	4	5
Cash flows (€000s)	(11,840)	1,230	1,680	4,350	10,240	2,200
		Privi		Drugi		
Net present value		€2,054,000		€2,293,000		
Internal rate of return		17·6%		Not provided		
Modified internal rate of return		13·4%		Not provided		
Value at risk (over the project's life)						
95% confidence level		€1,103,500		Not provided		
90% confidence level		€860,000		Not provided		

Both projects' net present value has been calculated based on Riviere Co's nominal cost of capital of 10%. It can be assumed that both projects' cash flow returns are normally distributed and the annual standard deviation of project Drugi's present value of after-tax cash flows is estimated to be €400,000. It can also be assumed that all sales are made in € (Euro) and therefore the company is not exposed to any foreign exchange exposure.

Notwithstanding how profitable project Drugi may appear to be, Riviere Co's board of directors is concerned about the possible legal risks if it invests in the project because they have never dealt with companies outside the EU before.

Required:

- Discuss the aims of a free trade area, such as the European Union (EU), and the possible benefits to Riviere Co of operating within the EU.** (5 marks)
- Calculate the figures which have not been provided for project Drugi and recommend which project should be accepted. Provide a justification for the recommendation and explain what the value at risk measures.** (13 marks)
- Discuss the possible legal risks of investing in project Drugi which Riviere Co may be concerned about and how these may be mitigated.** (7 marks)

(25 marks)

- 4 Kamala Co, a listed company, manufactures parts and machinery for the construction industry. About five years ago, Kamala Co started to manufacture parts and machinery for hospitals and companies engaged in biomedical research using largely the same manufacturing and processing systems it already had in place. In 2011, a young and ambitious chief executive officer (CEO) took over the running of the company.

With the publication of the latest financial statements for the year to 30 November 2014, the CEO made a brief statement and it includes the following two points:

- The CEO was very pleased with growth in the financial ratios provided and sales revenue from 2012 to 2014. More pleasing was growth in the share price, which increased even faster than the growth in the market index, suggesting that Kamala Co has been a successful company.
- The CEO expressed a desire to make Kamala Co the leading manufacturer of parts and machinery for the construction industry by acquiring a major rival manufacturer in 2015, and financing the acquisition through an issue of a new bond and a small rights issue.

An analyst, after examining the recent financial statements and the two points above, was less positive about Kamala Co's future prospects.

Given below are extracts from the recent financial statements, some ratios, and other financial information for Kamala Co.

Kamala Co
Year ending 30 November (all amounts in \$m)

	2012	2013	2014
Sales revenue	3,760	4,054	5,230
Operating profit	714	819	1,098
Finance costs	97	168	269
Profit before tax	617	651	829
Taxation	154	163	207
Profit for the year	463	488	622
Dividends	139	137	152

Kamala Co
Year ending 30 November (all amounts in \$m)

	2012	2013	2014
Total non-current assets	3,962	5,507	7,669
Total current assets	980	1,410	1,880
Total non-current and current assets	4,942	6,917	9,549
Equity			
Ordinary shares (\$0.25)	750	750	750
Reserves	1,476	1,827	2,297
Total equity	2,226	2,577	3,047
Non-current liabilities			
Bank loans	476	1,176	1,316
Bonds	1,008	1,008	2,218
Total non-current liabilities	1,484	2,184	3,534
Current liabilities			
Trade and other payables	1,232	1,540	2,016
Bank overdraft	–	616	952
Total current liabilities	1,232	2,156	2,968
Total non-current and current liabilities	2,716	4,340	6,502

Kamala Co: By activity
Year ending 30 November (all amounts in \$m)

	2012	2013	2014
Sales revenue			
Construction	2,420	2,644	3,660
Hospitals and biomedical	1,340	1,410	1,570
Operating profit			
Construction	460	489	693
Hospitals and biomedical	254	330	405

Ratios: Kamala Co

	2012	2013	2014
Operating profit margin	19.0%	20.2%	21.0%
Dividend cover	3.3	3.6	4.1
Earnings per share	15.4c	16.3c	20.7c
Gearing [(debt/debt + equity)]	40%	46%	54%

Other financial information

	30 November 2012	30 November 2013	30 November 2014
Kamala Co share price (\$)	1.69	2.01	2.69
Market index	4,539	5,447	6,550
Industry index	840	1,092	1,422
Industry average PE ratio	9.2:1	12.1:1	15.3:1

	2013 (\$m)	2014 (\$m)
Depreciation deducted to arrive at the operating profit (equivalent to tax allowable depreciation)	826	1,150
Economic depreciation	990	1,380
Non-cash expenses (excluding depreciation)	150	170

Kamala Co's cost of capital is estimated to be 10%. The company's corporation tax rate is 25%.

Required:

- (a) Discuss the advantages and drawbacks of using the economic value added (EVA™) technique to assess a company's performance.** (6 marks)
- (b) Estimate Kamala Co's EVA™ for the years ending 30 November 2013 and 30 November 2014.** (5 marks)
- (c) Evaluate Kamala Co's performance and conclude whether the analyst's opinion or the chief executive officer's opinion has the greater validity. Include any additional ratio and activity trends, and share price analysis, which are deemed to be relevant to the evaluation.** (14 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \qquad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

Tuesday 2 June 2015



Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 9–13.

Do NOT open this paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

Section A – This ONE question is compulsory and MUST be attempted

1 Yilandwe

Yilandwe, whose currency is the Yilandwe Rand (YR), has faced extremely difficult economic challenges in the past 25 years because of some questionable economic policies and political decisions made by its previous governments. Although Yilandwe's population is generally poor, its people are nevertheless well-educated and ambitious. Just over three years ago, a new government took office and since then it has imposed a number of strict monetary and fiscal controls, including an annual corporation tax rate of 40%, in an attempt to bring Yilandwe out of its difficulties. As a result, the annual rate of inflation has fallen rapidly from a high of 65% to its current level of 33%. These strict monetary and fiscal controls have made Yilandwe's government popular in the larger cities and towns, but less popular in the rural areas which seem to have suffered disproportionately from the strict monetary and fiscal controls.

It is expected that Yilandwe's annual inflation rate will continue to fall in the coming few years as follows:

Year	Inflation rate
1	22.0%
2	14.7%
3 onwards	9.8%

Yilandwe's government has decided to continue the progress made so far, by encouraging foreign direct investment into the country. Recently, government representatives held trade shows internationally and offered businesses a number of concessions, including:

- (i) zero corporation tax payable in the first two years of operation; and
- (ii) an opportunity to carry forward tax losses and write them off against future profits made after the first two years.

The government representatives also promised international companies investing in Yilandwe prime locations in towns and cities with good transport links.

Imoni Co

Imoni Co, a large listed company based in the USA with the US dollar (\$) as its currency, manufactures high tech diagnostic components for machinery, which it exports worldwide. After attending one of the trade shows, Imoni Co is considering setting up an assembly plant in Yilandwe where parts would be sent and assembled into a specific type of component, which is currently being assembled in the USA. Once assembled, the component will be exported directly to companies based in the European Union (EU). These exports will be invoiced in Euro (€).

Assembly plant in Yilandwe: financial and other data projections

It is initially assumed that the project will last for four years. The four-year project will require investments of YR21,000 million for land and buildings, YR18,000 million for machinery and YR9,600 million for working capital to be made immediately. The working capital will need to be increased annually at the start of each of the next three years by Yilandwe's inflation rate and it is assumed that this will be released at the end of the project's life.

It can be assumed that the assembly plant can be built very quickly and production started almost immediately. This is because the basic facilities and infrastructure are already in place as the plant will be built on the premises and grounds of a school. The school is ideally located, near the main highway and railway lines. As a result, the school will close and the children currently studying there will be relocated to other schools in the city. The government has kindly agreed to provide free buses to take the children to these schools for a period of six months to give parents time to arrange appropriate transport in the future for their children.

The current selling price of each component is €700 and this price is likely to increase by the average EU rate of inflation from year 1 onwards.

The number of components expected to be sold every year are as follows:

Year	1	2	3	4
Sales component units (000s)	150	480	730	360

The parts needed to assemble into the components in Yilandwe will be sent from the USA by Imoni Co at a cost of \$200 per component unit, from which Imoni Co would currently earn a pre-tax contribution of \$40 for each component unit. However, Imoni Co feels that it can negotiate with Yilandwe's government and increase the transfer price to \$280 per component unit. The variable costs related to assembling the components in Yilandwe are currently

YR15,960 per component unit. The current annual fixed costs of the assembly plant are YR4,600 million. All these costs, wherever incurred, are expected to increase by that country's annual inflation every year from year 1 onwards.

Imoni Co pays corporation tax on profits at an annual rate of 20% in the USA. The tax in both the USA and Yilandwe is payable in the year that the tax liability arises. A bilateral tax treaty exists between Yilandwe and the USA. Tax allowable depreciation is available at 25% per year on the machinery on a straight-line basis.

Imoni Co will expect annual royalties from the assembly plant to be made every year. The normal annual royalty fee is currently \$20 million, but Imoni Co feels that it can negotiate this with Yilandwe's government and increase the royalty fee by 80%. Once agreed, this fee will not be subject to any inflationary increase in the project's four-year period.

If Imoni Co does decide to invest in an assembly plant in Yilandwe, its exports from the USA to the EU will fall and it will incur redundancy costs. As a result, Imoni Co's after-tax cash flows will reduce by the following amounts:

Year	1	2	3	4
Redundancy and lost contribution	20,000	55,697	57,368	59,089

Imoni Co normally uses its cost of capital of 9% to assess new projects. However, the finance director suggests that Imoni Co should use a project specific discount rate of 12% instead.

Other financial information

Current spot rates

Euro per Dollar	€0.714/\$1
YR per Euro	YR142/€1
YR per Dollar	YR101.4/\$1

Forecast future rates based on expected inflation rate differentials

Year	1	2	3	4
YR/\$1	120.1	133.7	142.5	151.9

Year	1	2	3	4
YR/€1	165.0	180.2	190.2	200.8

Expected inflation rates

EU expected inflation rate: Next two years	5%
EU expected inflation rate: Year 3 onwards	4%
USA expected inflation rate: Year 1 onwards	3%

Required:

(a) Discuss the possible benefits and drawbacks to Imoni Co of setting up its own assembly plant in Yilandwe, compared to licensing a company based in Yilandwe to undertake the assembly on its behalf. (5 marks)

(b) Prepare a report which:

- (i) Evaluates the financial acceptability of the investment in the assembly plant in Yilandwe; (21 marks)
- (ii) Discusses the assumptions made in producing the estimates, and the other risks and issues which Imoni Co should consider before making the final decision; (17 marks)
- (iii) Provides a reasoned recommendation on whether or not Imoni Co should invest in the assembly plant in Yilandwe. (3 marks)

Professional marks will be awarded in part (b) for the format, structure and presentation of the report.

(4 marks)

(50 marks)

Section B – TWO questions ONLY to be attempted

- 2 The treasury department of Chawan Co, a listed company, aims to maintain a portfolio of around \$360 million consisting of equity shares, corporate bonds and government bonds, which it can turn into cash quickly for investment projects. Chawan Co is considering disposing 27 million shares, valued at \$2.15 each, which it has invested in Oden Co. The head of Chawan Co's treasury department is of the opinion that, should the decision be made to dispose of its equity stake in Oden Co, this should be sold through a dark pool network and not sold on the stock exchange where Oden Co's shares are listed. In the last few weeks, there have also been rumours that Oden Co may become subject to a takeover bid.

Oden Co operates in the travel and leisure (T&L) sector, and the poor weather conditions in recent years, coupled with a continuing recession, has meant that the T&L sector is under-performing. Over the past three years, sales revenue fell by an average of 8% per year in the T&L sector. However, there are signs that the economy is starting to recover, but this is by no means certain.

Given below are extracts from the recent financial statements and other financial information for Oden Co and the T&L sector.

Oden Co**Year ending 31 May (all amounts in \$m)**

	2013	2014	2015
Total non-current assets	972	990	980
Total current assets	128	142	126
Total assets	1,100	1,132	1,106
Equity			
Ordinary shares (\$0.50)	300	300	300
Reserves	305	329	311
Total equity	605	629	611
Non-current liabilities			
Bank loans	115	118	100
Bonds	250	250	260
Total non-current liabilities	365	368	360
Current liabilities			
Trade and other payables	42	45	37
Bank overdraft	88	90	98
Total current liabilities	130	135	135
Total equity and liabilities	1,100	1,132	1,106

Oden Co**Year ending 31 May (all amounts in \$m)**

	2013	2014	2015
Sales revenue	1,342	1,335	1,185
Operating profit	218	203	123
Finance costs	(23)	(27)	(35)
Profit before tax	195	176	88
Taxation	(35)	(32)	(16)
Profit for the year	160	144	72

Other financial information (Based on annual figures till 31 May of each year)

	2012	2013	2014	2015
Oden Co average share price (\$)	2.10	2.50	2.40	2.20
Oden Co dividend per share (\$)	0.15	0.18	0.20	0.15
T&L sector average share price (\$)	3.80	4.40	4.30	4.82
T&L sector average earnings per share (\$)	0.32	0.36	0.33	0.35
T&L sector average dividend per share (\$)	0.25	0.29	0.29	0.31
Oden Co's equity beta	1.5	1.5	1.6	2.0
T&L sector average equity beta	1.5	1.4	1.5	1.6

The risk-free rate and the market return have remained fairly constant over the last ten years at 4% and 10% respectively.

Required:

- (a) Explain what a dark pool network is and why Chawan Co may want to dispose of its equity stake in Oden Co through one, instead of through the stock exchange where Oden Co's shares are listed. (5 marks)
- (b) Discuss whether or not Chawan Co should dispose of its equity stake in Oden Co. Provide relevant calculations to support the discussion.

Note: Up to 10 marks are available for the calculations.

(20 marks)

(25 marks)

- 3** In order to raise funds for future projects, the management of Bento Co, a large manufacturing company, is considering disposing of one of its subsidiary companies, Okazu Co, which is involved in manufacturing rubber tubing. They are considering undertaking the disposal through a management buy-out (MBO) or a management buy-in (MBI). Bento Co wants \$60 million from the sale of Okazu Co.

Given below are extracts from the most recent financial statements for Okazu Co:

Year ending 30 April (all amounts in \$000)

	2015
Total non-current assets	40,800
Total current assets	12,300
Total assets	<u>53,100</u>
Equity	24,600
Non-current liabilities	16,600
Current liabilities	
Trade and other payables	7,900
Bank overdraft	4,000
Total current liabilities	<u>11,900</u>
Total equity and liabilities	<u>53,100</u>

Year ending 30 April (all amounts in \$000)

	2015
Sales revenue	54,900
Operating profit	12,200
Finance costs	1,600
Profit before tax	10,600
Taxation	2,120
Profit for the year	<u>8,480</u>

Notes relating to the financial statements above:

- (i) Current assets, non-current assets and the trade and other payables will be transferred to the new company when Okazu Co is sold. The bank overdraft will be repaid by Bento Co prior to the sale of Okazu Co.
- (ii) With the exception of the bank overdraft, Bento Co has provided all the financing to Okazu Co. No liabilities, except the trade and other payables specified above, will be transferred to the new company when Okazu Co is sold.
- (iii) It is estimated that the market value of the non-current assets is 30% higher than the book value and the market value of the current assets is equivalent to the book value.
- (iv) The group finance costs and taxation are allocated by Bento Co to all its subsidiaries in pre-agreed proportions.

Okazu Co's senior management team has approached Dofu Co, a venture capital company, about the proposed MBO. Dofu Co has agreed to provide leveraged finance for a 50% equity stake in the new company on the following basis:

- (i) \$30 million loan in the form of an 8% bond on which interest is payable annually, based on the loan amount outstanding at the start of each year. The bond will be repaid on the basis of fixed equal annual payments (constituting of interest and principal) over the next four years;
- (ii) \$20 million loan in the form of a 6% convertible bond on which interest is payable annually. Conversion may be undertaken on the basis of 50 equity shares for every \$100 from the beginning of year five onwards;

(iii) 5,000,000 \$1 equity shares for \$5,000,000.

Okazu Co's senior management will contribute \$5,000,000 for 5,000,000 \$1 equity shares and own the remaining 50% of the equity stake.

As a condition for providing the finance, Dofu Co will impose a restrictive covenant that the new company's gearing ratio will be no higher than 75% at the end of its first year of operations, and then fall to no higher than 60%, 50% and 40% at the end of year two to year four respectively. The gearing ratio is determined by the book value of debt divided by the combined book values of debt and equity.

After the MBO, it is expected that earnings before interest and tax will increase by 11% per year and annual dividends of 25% on the available earnings will be paid for the next four years. It is expected that the annual growth rate of dividends will reduce by 60% from year five onwards following the MBO. The new company will pay tax at a rate of 20% per year. The new company's cost of equity has been estimated at 12%.

Required:

- (a) **Distinguish between a management buy-out (MBO) and a management buy-in (MBI). Discuss the relative benefits and drawbacks to Okazu Co if it is disposed through a MBO instead of a MBI.** (5 marks)
- (b) **Estimate, showing all relevant calculations, whether the restrictive covenant imposed by Dofu Co is likely to be met.** (12 marks)
- (c) **Discuss, with supporting calculations, whether or not an MBO would be beneficial for Dofu Co and Okazu Co's senior management team.** (8 marks)

(25 marks)

- 4 For a number of years Daikon Co has been using forward rate agreements to manage its exposure to interest rate fluctuations. Recently its chief executive officer (CEO) attended a talk on using exchange-traded derivative products to manage risks. She wants to find out by how much the extra cost of the borrowing detailed below can be reduced, when using interest rate futures, options on interest rate futures, and a collar on the options, to manage the interest rate risk. She asks that detailed calculations for each of the three derivative products be provided and a reasoned recommendation to be made.

Daikon Co is expecting to borrow \$34,000,000 in five months' time. It expects to make a full repayment of the borrowed amount in 11 months' time. Assume it is 1 June 2015 today. Daikon Co can borrow funds at LIBOR plus 70 basis points. LIBOR is currently 3·6%, but Daikon Co expects that interest rates may increase by as much as 80 basis points in five months' time.

The following information and quotes from an appropriate exchange are provided on LIBOR-based \$ futures and options.

Three-month \$ December futures are currently quoted at 95·84. The contract size is \$1,000,000, the tick size is 0·01% and the tick value is \$25.

Options on three-month \$ futures, \$1,000,000 contract, tick size 0·01% and tick value \$25. Option premiums are in annual %.

December calls	Strike price	December puts
0·541	95·50	0·304
0·223	96·00	0·508

Initial assumptions

It can be assumed that settlement for both the futures and options contracts is at the end of the month; that basis diminishes to zero at a constant rate until the contract matures and time intervals can be counted in months; that margin requirements may be ignored; and that if the options are in-the-money, they will be exercised at the end of the hedge instead of being sold.

Further issues

In the talk, the CEO was informed of the following issues:

- (i) Futures contracts will be marked-to-market daily. The CEO wondered what the impact of this would be if 50 futures contracts were bought at 95·84 on 1 June and 30 futures contracts were sold at 95·61 on 3 June, based on the \$ December futures contract given above. The closing settlement prices are given below for four days:

Date	Settlement price
1 June	95·84
2 June	95·76
3 June	95·66
4 June	95·74

- (ii) Daikon Co will need to deposit funds into a margin account with a broker for each contract they have opened, and this margin will need to be adjusted when the contracts are marked-to-market daily.
- (iii) It is unlikely that option contracts will be exercised at the end of the hedge period unless they have reached expiry. Instead, they are more likely to be sold and the positions closed.

Required:

- (a) Based on the three hedging choices available to Daikon Co and the initial assumptions given above, draft a response to the chief executive officer's (CEO) request made in the first paragraph of the question. (15 marks)
- (b) Discuss the impact on Daikon Co of each of the three further issues given above. As part of the discussion, include the calculations of the daily impact of the mark-to-market closing prices on the transactions specified by the CEO. (10 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper

Professional Level – Options Module

Advanced Financial Management

September/December 2015



Time allowed

Reading and planning: 15 minutes

Writing: 3 hours

This question paper is divided into two sections:

Section A – This ONE question is compulsory and MUST be attempted

Section B – TWO questions ONLY to be attempted

Formulae and tables are on pages 11–15.

Do NOT open this question paper until instructed by the supervisor.

During reading and planning time only the question paper may be annotated. You must NOT write in your answer booklet until instructed by the supervisor.

This question paper must not be removed from the examination hall.

The Association of Chartered Certified Accountants

P4
Paper

ACCA

This is a blank page.
The question paper begins on page 3.

Section A – This ONE question is compulsory and MUST be attempted

- 1 Cigno Co is a large pharmaceutical company, involved in the research and development (R&D) of medicines and other healthcare products. Over the past few years, Cigno Co has been finding it increasingly difficult to develop new medical products. In response to this, it has followed a strategy of acquiring smaller pharmaceutical companies which already have successful products in the market and/or have products in development which look very promising for the future. It has mainly done this without having to resort to major cost-cutting and has therefore avoided large-scale redundancies. This has meant that not only has Cigno Co performed reasonably well in the stock market, but it has also maintained a high level of corporate reputation.

Anatra Co is involved in two business areas: the first area involves the R&D of medical products, and the second area involves the manufacture of medical and dental equipment. Until recently, Anatra Co's financial performance was falling, but about three years ago a new chief executive officer (CEO) was appointed and she started to turn the company around. Recently, the company has developed and marketed a range of new medical products, and is in the process of developing a range of cancer-fighting medicines. This has resulted in a good performance in the stock market, but many analysts believe that its shares are still trading below their true value. Anatra Co's CEO is of the opinion that the turnaround in the company's fortunes makes it particularly vulnerable to a takeover threat, and she is thinking of defence strategies that the company could undertake to prevent such a threat. In particular, she was thinking of disposing some of the company's assets and focussing on its core business.

Cigno Co is of the opinion that Anatra Co is being held back from achieving its true potential by its equipment manufacturing business and that by separating the two business areas, corporate value can be increased. As a result, it is considering the possibility of acquiring Anatra Co, unbundling the manufacturing business, and then absorbing Anatra Co's R&D of medical products business. Cigno Co estimates that it would need to pay a premium of 35% to Anatra Co's shareholders to buy the company.

Financial information: Anatra Co

Given below are extracts from Anatra Co's latest statement of profit or loss and statement of financial position for the year ended 30 November 2015.

	2015 \$ million
Sales revenue	21,400
Profit before interest and tax (PBIT)	3,210
Interest	720
Pre-tax profit	2,490

	2015 \$ million
Non-current liabilities	9,000
Share capital (50c/share)	3,500
Reserves	4,520

Anatra Co's share of revenue and profits between the two business areas are as follows:

	Medical products R&D	Equipment manufacturing
Share of revenue and profit	70%	30%

Post-acquisition benefits from acquiring Anatra Co

Cigno Co estimates that following the acquisition and unbundling of the manufacturing business, Anatra Co's future sales revenue and profitability of the medical R&D business will be boosted. The annual sales growth rate is expected to be 5% and the profit margin before interest and tax is expected to be 17.25% of sales revenue, for the next four years. It can be assumed that the current tax allowable depreciation will remain equivalent to the amount of investment needed to maintain the current level of operations, but that the company will require an additional investment in assets of 40c for every \$1 increase in sales revenue.

After the four years, the annual growth rate of the company's free cash flows is expected to be 3% for the foreseeable future.

Anatra Co's unbundled equipment manufacturing business is expected to be divested through a sell-off, although other options such as a management buy-in were also considered. The value of the sell-off will be based on the medical and dental equipment manufacturing industry. Cigno Co has estimated that Anatra Co's manufacturing business should be valued at a factor of 1.2 times higher than the industry's average price-to-earnings ratio. Currently the industry's average earnings-per-share is 30c and the average share price is \$2.40.

Possible additional post-acquisition benefits

Cigno Co estimates that it could achieve further cash flow benefits following the acquisition of Anatra Co, if it undertakes a limited business re-organisation. There is some duplication of the R&D work conducted by Cigno Co and Anatra Co, and the costs related to this duplication could be saved if Cigno Co closes some of its own operations. However, it would mean that many redundancies would have to be made including employees who have worked in Cigno Co for many years. Anatra Co's employees are considered to be better qualified and more able in these areas of duplication, and would therefore not be made redundant.

Cigno Co could also move its headquarters to the country where Anatra Co is based and thereby potentially save a significant amount of tax, other than corporation tax. However, this would mean a loss of revenue for the government where Cigno Co is based.

The company is concerned about how the government and the people of the country where it is based might react to these issues. It has had a long and beneficial relationship with the country and with the country's people.

Cigno Co has estimated that it would save \$1,600 million after-tax free cash flows to the firm at the end of the first year as a result of these post-acquisition benefits. These cash flows would increase by 4% every year for the next three years.

Estimating the combined company's weighted average cost of capital

Cigno Co is of the opinion that as a result of acquiring Anatra Co, the cost of capital will be based on the equity beta and the cost of debt of the combined company. The asset beta of the combined company is the individual companies' asset betas weighted in proportion of the individual companies' market value of equity. Cigno Co has a market debt to equity ratio of 40:60 and an equity beta of 1.10.

It can be assumed that the proportion of market value of debt to market value of equity will be maintained after the two companies combine.

Currently, Cigno Co's total firm value (market values of debt and equity combined) is \$60,000 million and Anatra Co's asset beta is 0.68.

Additional information

- The estimate of the risk free rate of return is 4.3% and of the market risk premium is 7%.
- The corporation tax rate applicable to all companies is 22%.
- Anatra Co's current share price is \$3 per share, and it can be assumed that the book value and the market value of its debt are equivalent.
- The pre-tax cost of debt of the combined company is expected to be 6.0%.

Important note

Cigno Co's board of directors (BoD) does not require any discussion or computations of currency movements or exposure in this report. All calculations are to be presented in \$ millions. Currency movements and their management will be considered in a separate report. The BoD also does not expect any discussion or computations relating to the financing of acquisition in this report, other than the information provided above on the estimation of the cost of capital.

Required:

- (a) Distinguish between a divestment through a sell-off and a management buy-in as forms of unbundling. (4 marks)
- (b) Prepare a report for the board of directors (BoD) of Cigno Co which:
- (i) Estimates the value attributable to Cigno Co's shareholders from the acquisition of Anatra Co before taking into account the cash benefits of potential tax savings and redundancies, and then after taking these into account; (18 marks)
 - (ii) Assesses the value created from (b)(i) above, including a discussion of the estimations made and methods used; (8 marks)
 - (iii) Advises the BoD on the key factors it should consider in relation to the redundancies and potential tax savings. (4 marks)
- Professional marks will be awarded in part (b) for the format, structure and presentation of the report. (4 marks)
- (c) Discuss whether the defence strategy suggested by Anatra Co's CEO of disposing assets is feasible. (6 marks)
- (d) Takeover regulation, where Anatra Co is based, offers the following conditions aimed at protecting shareholders: the mandatory-bid condition through sell out rights, the principle of equal treatment, and squeeze-out rights.

Required:

Explain the main purpose of each of the three conditions. (6 marks)

(50 marks)

Section B – TWO questions ONLY to be attempted

- 2 The Armstrong Group is a multinational group of companies. Today is 1 September. The treasury manager at Massie Co, one of Armstrong Group's subsidiaries based in Europe, has just received notification from the group's head office that it intends to introduce a system of netting to settle balances owed within the group every six months. Previously inter-group indebtedness was settled between the two companies concerned.

The predicted balances owing to, and owed by, the group companies at the end of February are as follows:

Owed by	Owed to	Local currency million (m)
Armstrong (USA)	Horan (South Africa)	US \$12.17
Horan (South Africa)	Massie (Europe)	SA R42.65
Giffen (Denmark)	Armstrong (USA)	D Kr21.29
Massie (Europe)	Armstrong (USA)	US \$19.78
Armstrong (USA)	Massie (Europe)	€1.57
Horan (South Africa)	Giffen (Denmark)	D Kr16.35
Giffen (Denmark)	Massie (Europe)	€1.55

The predicted exchange rates, used in the calculations of the balances to be settled, are as follows:

	D Kr	US\$	SA R	€
1 D Kr =	1.0000	0.1823	1.9554	0.1341
1 US \$ =	5.4855	1.0000	10.7296	0.7358
1 SA R =	0.5114	0.0932	1.0000	0.0686
1 € =	7.4571	1.3591	14.5773	1.0000

Settlement will be made in dollars, the currency of Armstrong Group, the parent company. Settlement will be made in the order that the company owing the largest net amount in dollars will first settle with the company owed the smallest net amount in dollars.

Note: D Kr is Danish Krone, SA R is South African Rand, US \$ is United States dollar and € is Euro.

Required:

- (a) (i) **Calculate the inter-group transfers which are forecast to occur for the next period.** (8 marks)
- (ii) **Discuss the problems which may arise with the new arrangement.** (3 marks)

The most significant transaction which Massie Co is due to undertake with a company outside the Armstrong Group in the next six months is that it is due to receive €25 million from Bardsley Co on 30 November. Massie Co's treasury manager intends to invest this money for the six months until 31 May, when it will be used to fund some major capital expenditure. However, the treasury manager is concerned about changes in interest rates. Predictions in the media range from a 0.5% rise in interest rates to a 0.5% fall.

Because of the uncertainty, the treasury manager has decided to protect Massie Co by using derivatives. The treasury manager wishes to take advantage of favourable interest rate movements. Therefore she is considering options on interest rate futures or interest rate collars as possible methods of hedging, but not interest rate futures. Massie Co can invest at LIBOR minus 40 basis points and LIBOR is currently 3.6%.

The treasury manager has obtained the following information on Euro futures and options. She is ignoring margin requirements.

Three-month Euro futures, €1,000,000 contract, tick size 0.01% and tick value €25.

September	95.94
December	95.76
March	95.44

Options on three-month Euro futures, €1,000,000 contract, tick size 0.01% and tick value €25. Option premiums are in annual %.

Calls			Strike	Puts		
September	December	March		September	December	March
0.113	0.182	0.245	96.50	0.002	0.123	0.198
0.017	0.032	0.141	97.00	0.139	0.347	0.481

It can be assumed that settlement for the contracts is at the end of the month. It can also be assumed that basis diminishes to zero at contract maturity at a constant rate and that time intervals can be counted in months.

Required:

- (b) Based on the choice of options on futures or collars which Massie Co is considering and assuming the company does not face any basis risk, recommend a hedging strategy for the €25 million receipt. Support your recommendations with appropriate comments and relevant calculations. (14 marks)

(25 marks)

- 3 Five years ago the Patel family invested in a new business, Flufftort Co, which manufactures furniture. Some family members became directors of Flufftort Co, others have not been actively involved in management. A venture capital firm, Gupte VC, also made a 20% investment in Flufftort Co. A representative of Gupte VC was appointed to Flufftort Co's board. Flufftort Co also took out a long-term 8.5% bank loan.

Sales have generally been disappointing. As a result, members of the Patel family have been reluctant to invest further in Flufftort Co. Over the last year Gupte VC has taken a tougher attitude towards Flufftort Co. Gupte VC pressurised Flufftort Co to pay a dividend of \$2 million for the year ended 30 June 2015. Gupte VC has also said that if Flufftort Co's financial results do not improve, Gupte VC may exercise its right to compel Flufftort Co to buy back its shares at par on 30 June 2016.

However, Flufftort Co's most recent product, the Easicushion chair, has been a much bigger success than expected. In order to produce enough Easicushion chairs to affect its results substantially, Flufftort Co will need to make significant expenditure on manufacturing facilities and additional working capital.

Extracts from statement of profit or loss for year ended 30 June 2015 and forecast statement of profit or loss for year ended 30 June 2016

	2015	2016 Forecast
	\$m	\$m
Operating profit	8.0	6.0
Finance cost	(3.0)	(3.0)
Profit before tax	5.0	3.0
Tax on profits (20%)	(1.0)	(0.6)
Profit for the period	4.0	2.4
Dividends	(2.0)	–
Retained earnings	2.0	2.4

Note

The forecast statement of profit or loss for the year ended 30 June 2016 is not affected by the proposed investment. This can be assumed only to affect results after 30 June 2016. The figure shown for retained earnings in the 2016 forecast can be assumed to be the net increase in cash for the year ended 30 June 2016.

Summarised statement of financial position as at 30 June 2015

	\$m
Assets	
Non-current assets	69.0
Current assets excluding cash	18.0
Cash	7.6
Total assets	94.6
Equity and liabilities	
Share capital (\$1 shares)	50.0
Retained earnings	2.6
Total equity	52.6
Long-term liabilities	
8.5% Bank loan	30.0
9% Loan note	5.0
Total long-term liabilities	35.0
Current liabilities	7.0
Total liabilities	42.0
Total equity and liabilities	94.6

Notes

1. 55% of shares are owned by the members of the Patel family who are directors, 25% by other members of the Patel family and 20% by Gupte VC.
2. The bank loan is secured on the non-current assets of Flufftort and is due for repayment on 31 December 2019. The loan is subject to a covenant that the ratio of equity to non-current liabilities should be greater than 1:3 on a book value basis. Flufftort has also been granted an overdraft facility of up to \$5 million by its bank.
3. The loan note is held by Rajiv Patel, a member of the Patel family who is not a director. The loan note is unsecured, is subordinated to the bank loan and has no fixed date for repayment.
4. If no finance is available for investment in manufacturing facilities, non-current assets, current assets excluding cash, the bank loan, loan note and current liabilities can be assumed to be the same at 30 June 2016 as at 30 June 2015.

However, the chief executive and finance director of Flufftort Co intend to propose that the company should be refinanced to fund the expanded production of the Easicushion chair. They have not yet consulted anyone else about their proposals.

Details of the proposed refinancing are as follows:

1. The members of the Patel family who are directors would subscribe to an additional 15 million \$1 shares at par.
2. Gupte VC would subscribe to an additional 20 million \$1 shares at par.
3. The 8.5% bank loan would be renegotiated with the bank and the borrowing increased to \$65 million, to be repaid on 30 June 2022. The expected finance cost of the loan would be 10% per annum.
4. Rajiv Patel's loan note would be replaced by 5 million \$1 shares.
5. The refinancing would mean non-current assets would increase to \$125 million, current assets other than cash would increase to \$42 million and current liabilities would increase to \$12 million.
6. Operating profits would be expected to increase to \$20 million in the first full year after the facilities are constructed (year ended 30 June 2017) and \$25 million in the second year (year ended 30 June 2018). No dividends would be paid for these two years, as cash surpluses would be used for further investment as required. Tax on company profits can be assumed to remain at 20%.

Required:

- (a) (i) **Prepare a projected statement of financial position as at 30 June 2016, on the assumption that Gupte VC exercises its rights and Gupte VC's shares are repurchased and cancelled by Flufftort Co.**
(4 marks)
- (ii) **Prepare a projected statement of financial position as at 30 June 2016 on the assumption that the proposed refinancing and investment take place.**
(4 marks)
- (iii) **Prepare projected statements of profit or loss for the years ended 30 June 2017 and 30 June 2018 on the basis that the profit forecasts are correct.**
(4 marks)
- (b) **Evaluate whether the suggested refinancing scheme is likely to be agreed by all finance providers. State clearly any assumptions which you make.**
(13 marks)

(25 marks)

- 4 Moonstar Co is a property development company which is planning to undertake a \$200 million commercial property development. Moonstar Co has had some difficulties over the last few years, with some developments not generating the expected returns and the company has at times struggled to pay its finance costs. As a result Moonstar Co's credit rating has been lowered, affecting the terms it can obtain for bank finance. Although Moonstar Co is listed on its local stock exchange, 75% of the share capital is held by members of the family who founded the company. The family members who are shareholders do not wish to subscribe for a rights issue and are unwilling to dilute their control over the company by authorising a new issue of equity shares. Moonstar Co's board is therefore considering other methods of financing the development, which the directors believe will generate higher returns than other recent investments, as the country where Moonstar Co is based appears to be emerging from recession.

Securitisation proposals

One of the non-executive directors of Moonstar Co has proposed that it should raise funds by means of a securitisation process, transferring the rights to the rental income from the commercial property development to a special purpose vehicle. Her proposals assume that the leases will generate an income of 11% per annum to Moonstar Co over a ten-year period. She proposes that Moonstar Co should use 90% of the value of the investment for a collateralised loan obligation which should be structured as follows:

- 60% of the collateral value to support a tranche of A-rated floating rate loan notes offering investors LIBOR plus 150 basis points
- 15% of the collateral value to support a tranche of B-rated fixed rate loan notes offering investors 12%
- 15% of the collateral value to support a tranche of C-rated fixed rate loan notes offering investors 13%
- 10% of the collateral value to support a tranche as subordinated certificates, with the return being the excess of receipts over payments from the securitisation process

The non-executive director believes that there will be sufficient demand for all tranches of the loan notes from investors. Investors will expect that the income stream from the development to be low risk, as they will expect the property market to improve with the recession coming to an end and enough potential lessees to be attracted by the new development.

The non-executive director predicts that there would be annual costs of \$200,000 in administering the loan. She acknowledges that there would be interest rate risks associated with the proposal, and proposes a fixed for variable interest rate swap on the A-rated floating rate notes, exchanging LIBOR for 9.5%.

However the finance director believes that the prediction of the income from the development that the non-executive director has made is over-optimistic. He believes that it is most likely that the total value of the rental income will be 5% lower than the non-executive director has forecast. He believes that there is some risk that the returns could be so low as to jeopardise the income for the C-rated fixed rate loan note holders.

Islamic finance

Moonstar Co's chief executive has wondered whether Sukuk finance would be a better way of funding the development than the securitisation.

Moonstar Co's chairman has pointed out that a major bank in the country where Moonstar Co is located has begun to offer a range of Islamic financial products. The chairman has suggested that a Mudaraba contract would be the most appropriate method of providing the funds required for the investment.

Required:

- (a) Calculate the amounts in \$ which each of the tranches can expect to receive from the securitisation arrangement proposed by the non-executive director and discuss how the variability in rental income affects the returns from the securitisation. (11 marks)
- (b) Discuss the benefits and risks for Moonstar Co associated with the securitisation arrangement that the non-executive director has proposed. (6 marks)
- (c) (i) Discuss the suitability of Sukuk finance to fund the investment, including an assessment of its appeal to potential investors. (4 marks)
- (ii) Discuss whether a Mudaraba contract would be an appropriate method of financing the investment and discuss why the bank may have concerns about providing finance by this method. (4 marks)

(25 marks)

Formulae

Modigliani and Miller Proposition 2 (with tax)

$$k_e = k_e^i + (1 - T)(k_e^i - k_d) \frac{V_d}{V_e}$$

The Capital Asset Pricing Model

$$E(r_i) = R_f + \beta_i(E(r_m) - R_f)$$

The asset beta formula

$$\beta_a = \left[\frac{V_e}{(V_e + V_d(1 - T))} \beta_e \right] + \left[\frac{V_d(1 - T)}{(V_e + V_d(1 - T))} \beta_d \right]$$

The Growth Model

$$P_0 = \frac{D_0(1 + g)}{(r_e - g)}$$

Gordon's growth approximation

$$g = b r_e$$

The weighted average cost of capital

$$WACC = \left[\frac{V_e}{V_e + V_d} \right] k_e + \left[\frac{V_d}{V_e + V_d} \right] k_d(1 - T)$$

The Fisher formula

$$(1 + i) = (1 + r)(1 + h)$$

Purchasing power parity and interest rate parity

$$S_1 = S_0 \times \frac{(1 + h_c)}{(1 + h_b)} \quad F_0 = S_0 \times \frac{(1 + i_c)}{(1 + i_b)}$$

Modified Internal Rate of Return

$$MIRR = \left[\frac{PV_R}{PV_I} \right]^{\frac{1}{n}} (1 + r_e) - 1$$

The Black-Scholes option pricing model

$$c = P_a N(d_1) - P_e N(d_2) e^{-rt}$$

Where:

$$d_1 = \frac{\ln(P_a / P_e) + (r + 0.5s^2)t}{s\sqrt{t}}$$

$$d_2 = d_1 - s\sqrt{t}$$

The Put Call Parity relationship

$$p = c - P_a + P_e e^{-rt}$$

Present Value Table

Present value of 1 i.e. $(1 + r)^{-n}$

Where r = discount rate

n = number of periods until payment

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	0.980	0.961	0.943	0.925	0.907	0.890	0.873	0.857	0.842	0.826	2
3	0.971	0.942	0.915	0.889	0.864	0.840	0.816	0.794	0.772	0.751	3
4	0.961	0.924	0.888	0.855	0.823	0.792	0.763	0.735	0.708	0.683	4
5	0.951	0.906	0.863	0.822	0.784	0.747	0.713	0.681	0.650	0.621	5
6	0.942	0.888	0.837	0.790	0.746	0.705	0.666	0.630	0.596	0.564	6
7	0.933	0.871	0.813	0.760	0.711	0.665	0.623	0.583	0.547	0.513	7
8	0.923	0.853	0.789	0.731	0.677	0.627	0.582	0.540	0.502	0.467	8
9	0.914	0.837	0.766	0.703	0.645	0.592	0.544	0.500	0.460	0.424	9
10	0.905	0.820	0.744	0.676	0.614	0.558	0.508	0.463	0.422	0.386	10
11	0.896	0.804	0.722	0.650	0.585	0.527	0.475	0.429	0.388	0.350	11
12	0.887	0.788	0.701	0.625	0.557	0.497	0.444	0.397	0.356	0.319	12
13	0.879	0.773	0.681	0.601	0.530	0.469	0.415	0.368	0.326	0.290	13
14	0.870	0.758	0.661	0.577	0.505	0.442	0.388	0.340	0.299	0.263	14
15	0.861	0.743	0.642	0.555	0.481	0.417	0.362	0.315	0.275	0.239	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	0.812	0.797	0.783	0.769	0.756	0.743	0.731	0.718	0.706	0.694	2
3	0.731	0.712	0.693	0.675	0.658	0.641	0.624	0.609	0.593	0.579	3
4	0.659	0.636	0.613	0.592	0.572	0.552	0.534	0.516	0.499	0.482	4
5	0.593	0.567	0.543	0.519	0.497	0.476	0.456	0.437	0.419	0.402	5
6	0.535	0.507	0.480	0.456	0.432	0.410	0.390	0.370	0.352	0.335	6
7	0.482	0.452	0.425	0.400	0.376	0.354	0.333	0.314	0.296	0.279	7
8	0.434	0.404	0.376	0.351	0.327	0.305	0.285	0.266	0.249	0.233	8
9	0.391	0.361	0.333	0.308	0.284	0.263	0.243	0.225	0.209	0.194	9
10	0.352	0.322	0.295	0.270	0.247	0.227	0.208	0.191	0.176	0.162	10
11	0.317	0.287	0.261	0.237	0.215	0.195	0.178	0.162	0.148	0.135	11
12	0.286	0.257	0.231	0.208	0.187	0.168	0.152	0.137	0.124	0.112	12
13	0.258	0.229	0.204	0.182	0.163	0.145	0.130	0.116	0.104	0.093	13
14	0.232	0.205	0.181	0.160	0.141	0.125	0.111	0.099	0.088	0.078	14
15	0.209	0.183	0.160	0.140	0.123	0.108	0.095	0.084	0.074	0.065	15

Annuity Table

Present value of an annuity of 1 i.e. $\frac{1 - (1 + r)^{-n}}{r}$

Where r = discount rate
 n = number of periods

<i>Discount rate (r)</i>											
<i>Periods</i> (n)	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%	
1	0.990	0.980	0.971	0.962	0.952	0.943	0.935	0.926	0.917	0.909	1
2	1.970	1.942	1.913	1.886	1.859	1.833	1.808	1.783	1.759	1.736	2
3	2.941	2.884	2.829	2.775	2.723	2.673	2.624	2.577	2.531	2.487	3
4	3.902	3.808	3.717	3.630	3.546	3.465	3.387	3.312	3.240	3.170	4
5	4.853	4.713	4.580	4.452	4.329	4.212	4.100	3.993	3.890	3.791	5
6	5.795	5.601	5.417	5.242	5.076	4.917	4.767	4.623	4.486	4.355	6
7	6.728	6.472	6.230	6.002	5.786	5.582	5.389	5.206	5.033	4.868	7
8	7.652	7.325	7.020	6.733	6.463	6.210	5.971	5.747	5.535	5.335	8
9	8.566	8.162	7.786	7.435	7.108	6.802	6.515	6.247	5.995	5.759	9
10	9.471	8.983	8.530	8.111	7.722	7.360	7.024	6.710	6.418	6.145	10
11	10.368	9.787	9.253	8.760	8.306	7.887	7.499	7.139	6.805	6.495	11
12	11.255	10.575	9.954	9.385	8.863	8.384	7.943	7.536	7.161	6.814	12
13	12.134	11.348	10.635	9.986	9.394	8.853	8.358	7.904	7.487	7.103	13
14	13.004	12.106	11.296	10.563	9.899	9.295	8.745	8.244	7.786	7.367	14
15	13.865	12.849	11.938	11.118	10.380	9.712	9.108	8.559	8.061	7.606	15
(n)	11%	12%	13%	14%	15%	16%	17%	18%	19%	20%	
1	0.901	0.893	0.885	0.877	0.870	0.862	0.855	0.847	0.840	0.833	1
2	1.713	1.690	1.668	1.647	1.626	1.605	1.585	1.566	1.547	1.528	2
3	2.444	2.402	2.361	2.322	2.283	2.246	2.210	2.174	2.140	2.106	3
4	3.102	3.037	2.974	2.914	2.855	2.798	2.743	2.690	2.639	2.589	4
5	3.696	3.605	3.517	3.433	3.352	3.274	3.199	3.127	3.058	2.991	5
6	4.231	4.111	3.998	3.889	3.784	3.685	3.589	3.498	3.410	3.326	6
7	4.712	4.564	4.423	4.288	4.160	4.039	3.922	3.812	3.706	3.605	7
8	5.146	4.968	4.799	4.639	4.487	4.344	4.207	4.078	3.954	3.837	8
9	5.537	5.328	5.132	4.946	4.772	4.607	4.451	4.303	4.163	4.031	9
10	5.889	5.650	5.426	5.216	5.019	4.833	4.659	4.494	4.339	4.192	10
11	6.207	5.938	5.687	5.453	5.234	5.029	4.836	4.656	4.486	4.327	11
12	6.492	6.194	5.918	5.660	5.421	5.197	4.988	4.793	4.611	4.439	12
13	6.750	6.424	6.122	5.842	5.583	5.342	5.118	4.910	4.715	4.533	13
14	6.982	6.628	6.302	6.002	5.724	5.468	5.229	5.008	4.802	4.611	14
15	7.191	6.811	6.462	6.142	5.847	5.575	5.324	5.092	4.876	4.675	15

Standard normal distribution table

	0·00	0·01	0·02	0·03	0·04	0·05	0·06	0·07	0·08	0·09
0·0	0·0000	0·0040	0·0080	0·0120	0·0160	0·0199	0·0239	0·0279	0·0319	0·0359
0·1	0·0398	0·0438	0·0478	0·0517	0·0557	0·0596	0·0636	0·0675	0·0714	0·0753
0·2	0·0793	0·0832	0·0871	0·0910	0·0948	0·0987	0·1026	0·1064	0·1103	0·1141
0·3	0·1179	0·1217	0·1255	0·1293	0·1331	0·1368	0·1406	0·1443	0·1480	0·1517
0·4	0·1554	0·1591	0·1628	0·1664	0·1700	0·1736	0·1772	0·1808	0·1844	0·1879
0·5	0·1915	0·1950	0·1985	0·2019	0·2054	0·2088	0·2123	0·2157	0·2190	0·2224
0·6	0·2257	0·2291	0·2324	0·2357	0·2389	0·2422	0·2454	0·2486	0·2517	0·2549
0·7	0·2580	0·2611	0·2642	0·2673	0·2704	0·2734	0·2764	0·2794	0·2823	0·2852
0·8	0·2881	0·2910	0·2939	0·2967	0·2995	0·3023	0·3051	0·3078	0·3106	0·3133
0·9	0·3159	0·3186	0·3212	0·3238	0·3264	0·3289	0·3315	0·3340	0·3365	0·3389
1·0	0·3413	0·3438	0·3461	0·3485	0·3508	0·3531	0·3554	0·3577	0·3599	0·3621
1·1	0·3643	0·3665	0·3686	0·3708	0·3729	0·3749	0·3770	0·3790	0·3810	0·3830
1·2	0·3849	0·3869	0·3888	0·3907	0·3925	0·3944	0·3962	0·3980	0·3997	0·4015
1·3	0·4032	0·4049	0·4066	0·4082	0·4099	0·4115	0·4131	0·4147	0·4162	0·4177
1·4	0·4192	0·4207	0·4222	0·4236	0·4251	0·4265	0·4279	0·4292	0·4306	0·4319
1·5	0·4332	0·4345	0·4357	0·4370	0·4382	0·4394	0·4406	0·4418	0·4429	0·4441
1·6	0·4452	0·4463	0·4474	0·4484	0·4495	0·4505	0·4515	0·4525	0·4535	0·4545
1·7	0·4554	0·4564	0·4573	0·4582	0·4591	0·4599	0·4608	0·4616	0·4625	0·4633
1·8	0·4641	0·4649	0·4656	0·4664	0·4671	0·4678	0·4686	0·4693	0·4699	0·4706
1·9	0·4713	0·4719	0·4726	0·4732	0·4738	0·4744	0·4750	0·4756	0·4761	0·4767
2·0	0·4772	0·4778	0·4783	0·4788	0·4793	0·4798	0·4803	0·4808	0·4812	0·4817
2·1	0·4821	0·4826	0·4830	0·4834	0·4838	0·4842	0·4846	0·4850	0·4854	0·4857
2·2	0·4861	0·4864	0·4868	0·4871	0·4875	0·4878	0·4881	0·4884	0·4887	0·4890
2·3	0·4893	0·4896	0·4898	0·4901	0·4904	0·4906	0·4909	0·4911	0·4913	0·4916
2·4	0·4918	0·4920	0·4922	0·4925	0·4927	0·4929	0·4931	0·4932	0·4934	0·4936
2·5	0·4938	0·4940	0·4941	0·4943	0·4945	0·4946	0·4948	0·4949	0·4951	0·4952
2·6	0·4953	0·4955	0·4956	0·4957	0·4959	0·4960	0·4961	0·4962	0·4963	0·4964
2·7	0·4965	0·4966	0·4967	0·4968	0·4969	0·4970	0·4971	0·4972	0·4973	0·4974
2·8	0·4974	0·4975	0·4976	0·4977	0·4977	0·4978	0·4979	0·4979	0·4980	0·4981
2·9	0·4981	0·4982	0·4982	0·4983	0·4984	0·4984	0·4985	0·4985	0·4986	0·4986
3·0	0·4987	0·4987	0·4987	0·4988	0·4988	0·4989	0·4989	0·4989	0·4990	0·4990

This table can be used to calculate $N(d)$, the cumulative normal distribution functions needed for the Black-Scholes model of option pricing. If $d_i > 0$, add 0·5 to the relevant number above. If $d_i < 0$, subtract the relevant number above from 0·5.

End of Question Paper